

THE CHAUTAUQUAN

*A Monthly Magazine Devoted to
the Promotion of True Culture.*

*Organ of the
Chautauqua Literary and
Scientific Circle.*

Vol. IV
December, 1883
No. 3.

*Theodore L. Flood, D.D., Editor
The Chautauqua Press*

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THE CHAUTAUQUAN.

*A MONTHLY MAGAZINE DEVOTED TO THE PROMOTION OF TRUE
CULTURE. ORGAN OF THE CHAUTAUQUA LITERARY AND
SCIENTIFIC CIRCLE.*

VOL. IV.

DECEMBER, 1883.

No. 3.

Chautauqua Literary and Scientific Circle.

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REQUIRED READING
FOR THE
Chautauqua Literary and Scientific Circle for 1883-4.
DECEMBER.



GERMAN HISTORY.

By REV. W. G. WILLIAMS, A. M.

III. THE FRANKS AND MEROVINGIANS.

After the fall of the Western Empire the Franks step into the foreground and show themselves of all the German tribes the most capable of founding a stable government. From the first they were distinguished from the others by their superior military discipline, and by their pride and ambition. They had always been looked upon as formidable warriors. Few of them wore helmets and mail; their breasts and backs were covered only by the shield. From the hips downward they wrapped themselves in close-fitting linen or leather, so as to display each man's tall, upright form. Their principal weapon was the two-edged battle-axe, which served for throwing as well as striking. They also carried frightful javelins with barbed points. Their own laws describe them as brave warriors, profound in their plans, manly and healthy in body, handsome, bold, impetuous, and hardy. But their enemies, perhaps with some justice, denounced them as the most faithless and cruel of men. The distinguishing ornament of the kings was their hair, which was left uncut, flowing freely over the shoulder. The people were still heathen, untamed and uncivilized, yet in constant intercourse with the Romans in Gaul.[\[A\]](#)

CLOVIS, THE FIRST FRANKISH KING.

The name of Clovis is not alone to be remembered as that of the founder of the kingdom of the Franks, but for the remarkable so-called conversion which he experienced during a hard-fought battle with the Alemanni. While the result was yet in doubt, Clovis, in the face of his army, called upon the new God, Christ, and vowed to serve him, if he would help him now. He was victorious; received instruction from St. Remigius, and was then

baptized, with three thousand of his noblest Franks, in the cathedral at Rheims. "Bow thy head in silence, Sigambrian," said the saint; "worship what thou hast hitherto destroyed; war against what thou hast worshiped." This was by no means the only instance of wholesale conversions to Christianity in consequence of a victory. The heathen, when defeated by Christians, commonly ascribed the result to the superior strength of the Christian God, and often resolved to seek his protection for themselves. It was the Catholic, not the Arian faith, which Clovis adopted. He was straightway recognized by the Pope as "the most Christian king," the appointed protector and propagator of the true faith against Arian Germany.

Clovis built up his kingdom with many a deed of blood, but with great vigor. His empire comprised German as well as Roman territory; but struck root firmly in the old native soil, from which it drew ever new strength: and therefore it was that its duration was not merely momentary, like that of the Gothic kingdoms, but it proved the beginning of the monarchy of the Middle Ages, the beginning of a new national life, in which Roman form was animated with fresh German strength. Clovis ruled his wide realm from Paris, a city which had existed even before the days of Cæsar and the Romans in Gaul. He died in Paris at the early age of forty-five.[\[B\]](#)

From Clovis to *Karl der Grosse* (French, *Charlemagne*; Latin, *Carolus Magnus*), a period of two hundred and fifty years, we witness not only the vicissitudes incident to the establishment of a new social and political order upon the ruins of the old, with all the ferocity of manner and barbarity of action to be expected in such an age; but also there is the gradual displacement of the old pagan religions by the newer one called Christianity. It is a period of strifes, of jealousies, and blood. It was toward the last of this period that occurred the memorable battle of Poitiers, between the Franks under Karl, afterward surnamed *Martel*, and the Saracens, who having crossed from Africa and possessed themselves of entire Spain, next collected a large army, and under command of Abderrahman, Viceroy of the Caliph of Damascus, set out for the conquest of France and Germany, as yet an undivided nationality. Thus the new Christian faith of Europe, still engaged in quelling the last strength of the ancient paganism, was suddenly called upon to meet the newer faith of Mohammed, which had determined to subdue the world.

Not only France, but the Eastern Empire, Italy and England looked to Karl, in this emergency. The Saracens crossed the Pyrenees with 350,000 warriors, accompanied by their wives and children, as if they were sure of victory and meant to possess the land. Karl called the military strength of the whole broad kingdom into the field, collected an army nearly equal in numbers, and finally, in October, 732, the two hosts stood face to face, near the city of Poitiers. It was a struggle almost as grand, and as fraught with important consequences to the world, as that of Aëtius and Attila, nearly 300 years before. Six days were spent in preparations, and on the seventh the battle began. The Saracens attacked with that daring and impetuosity which had gained them so many victories; but, as the old chronicle says, “the Franks, with their strong hearts and powerful bodies, stood like a wall, and hewed down the Arabs with iron hands.” When night fell, 200,000 dead and wounded lay upon the field. Karl made preparations for resuming the battle on the following morning, but he found no enemy. The Saracens had retired during the night, leaving their camps and stores behind them, and their leader, Abderrahman, among the slain. This was the first great check the cause of Islam received, after a series of victories more wonderful than those of Rome. From that day the people bestowed upon Karl the surname of *Martel*, the Hammer, and as Charles Martel he is best known in history.

[C]

CHARACTER OF THE GERMAN CHRISTIANITY AT THIS TIME.

The Christianity of the Germans, and even that of the Roman provinces, for many generations after the date of their “conversion,” was a very different kind of religion from that which is now held by enlightened Christians. Constantine and several of his successors were actually worshiped after death by multitudes of the Christians of those days. The apostolic doctrines were not conceived as a system of belief by the people, nor even by their teachers; the personal sovereignty of Christ as a king and warrior, and the future heaven or hell to be awarded by him, were apprehended as practical truths, but were overlaid with a dense mass of superstitious notions and observances, many of them legacies from heathenism. Above all, the Germans indulged without stint their passion for the wonderful; and the power of Christianity over them depended largely on the supply of miracles and of potent relics which it could furnish them. The

workers of miraculous cures were numerous; they were esteemed as the favorites of heaven, and cities and princes contended with one another for their bones. Some of the popes were wise enough to discourage the zeal for miracles; and as late as A. D. 590, Pope Gregory I. wrote to St. Augustine, of England, cautioning him against spiritual pride as a worker of them. But it was not long before the papacy became the great center from which relics of the saints were distributed throughout the Church. The Roman catacombs were ransacked, and bones of saints found in an abundance sufficient to supply Christendom for ages. The Pope's guaranty of genuineness was final; and this resource contributed immeasurably to increase the wealth and power of the Holy See. The legends of the saints, as circulated and preserved, mainly by tradition, were for centuries the intellectual food of the Church at large; and were filled with idle and monotonous tales of wonderful cures in mind and body, wrought by the holy men and women in their lives, or by their corpses or their tombs. No doubt was entertained, even by the most intelligent, of the truth of these miracles. The modern conception of nature, as the work of a divine will which is unchangeable, and which therefore expresses itself in fixed, uniform laws, was then unknown. The spiritual conception of Christianity, as life by a personal trust in a pure, holy, and loving God, was set forth, indeed, by a few writers and preachers, and was doubtless verified in the experience of many a humble heart; but it was far above the thoughts of the people, or even of the clergy at large. To them no religion was of any value which was not magical in its methods and powers, and a charm to secure good fortune or to avert danger. In short, the Church was one thing, Christianity another; and the priestly ambition of the great organization to rule over men's lives and estates entirely eclipsed and obscured the spiritual work of the kingdom which is not of this world. Nothing in the early German character is more attractive than the habitual and general chastity of the people, and their reverence for the marriage tie. But the great migrations corrupted them; and the degradation of marriage in the succeeding centuries was promoted and completed by the influence of the Church. Hardly any agency can be traced in history which has wrought greater social and moral evil than the contempt for human love and for the marriage tie, which was sedulously cultivated by the Roman Church from the beginning of the fourth century. Yet, there are indications enough to satisfy us that the doctrines of the New Testament had not lost their power; and that truth, purity, divine charity, and

Christian heroism were yet kept alive in many hearts. Thousands of men and women, whose minds and lives were darkened by the teachings and practices of asceticism, monasticism and gross superstitions, still cherished a devout, self-sacrificing love for their unseen Master and Lord and stood ready to die for him. Even the idea of Christian brotherhood was not entirely lost; and the common worship of the same Redeemer by master and slave did much to mitigate the horrors that grew out of their relation.[\[D\]](#)

CHARLES THE GREAT.

The history of Germany may now for half a century be ranged about the central figure, Charles the Great, more commonly called Charlemagne. Indeed, so conspicuous a figure is he that it is impossible for all subsequent history to lose sight of him. The decayed Merovingian scepter when it fell into his hands was swayed with such unprecedented vigor and ability that its old name soon disappeared, and henceforth it is the Carolingian, and Charles becomes the head and founder of a new dynasty. The first years of his rule are marked by continuous wars of conquest. The brave and savage Saxons resisted him and the Christianity which he championed until compelled by his all-conquering arms to yield. Saxony emerged from his hands subdued and Christian, divided into eight bishoprics, studded with new cities and abbeys which proved centers of civilization; and that wild country, until then barbarous and pagan, entered into communion with the rest of the empire.

He next turned his attention to Italy, where his career of victory was uninterrupted. He visited Rome, and, dismounting at a thousand paces from the walls, walked in procession to the church of St. Peter on the Vatican Hill, kissing the steps as he ascended in honor of the saints by whom they had been trodden. In the vestibule of the church he was received by the Pope, who embraced him with great affection, the choir chanting the psalm, "Blessed is he who cometh in the name of the Lord." Then they descended into the vaults, and offered up their prayers together at the shrine of St. Peter.[\[E\]](#)

EXTENT OF HIS EMPIRE—HIS CORONATION.

In the course of a reign of forty-five years, Charlemagne extended the limits of his empire beyond the Danube; subdued Dacia, Dalmatia, and Istria, conquered and subjected all the barbarous tribes to the banks of the Vistula, and successfully encountered the arms of the Saracens, the Huns, the Bulgarians, and the Saxons. His war with the Saxons was of more than thirty years duration, and their final conquest was not achieved without an inhuman waste of blood, through what has been considered a mistaken zeal for the propagation of Christianity, by measures which that religion can not be said to sanction or approve. All these wars were very nearly finished in the year 800. Charlemagne then found himself master of France, of Germany, of three-quarters of Italy, and a part of Spain. He had increased by more than a third the extent of territory which his father had left him. These vast possessions were no longer a kingdom, but an empire. He thought he had done enough to be authorized to seat himself on the throne of the West; and, as his father had required at the hands of the Pope his regal crown, so it was from the Pope that he demanded his imperial diadem. He was, therefore, with great ceremony, created Emperor of the West in St. Peter's, at Rome, by Pope Leo III., on Christmas day 800. It was a great event, for that imperial title which had remained buried under the ruins wrought by the barbarians, was drawn thence by the Roman pontiff, and shown to scattered nations and enemies as a rallying sign.

The crown which he received was destined to be for one thousand and six years the symbol of German unity, whilst the assembled people shouted, "Long life and victory to Carolus Augustus, the great and peace-bringing Roman Emperor, whom God hath crowned!" Thus, 324 years after the imperial dignity had disappeared, it was renewed by Charles. In this coronation act Pope Leo III. had fulfilled a function like St. Remy did in consecrating Clovis. His successors constituted it a privilege, and the pontiffs considered themselves the dispensers of crowns. During the whole of the middle ages the imperial consecration could only be given at Rome, and from the hands of the Holy Father. More than one war arose out of this prerogative.^[F]

THE CHARACTER AND ACHIEVEMENTS OF CHARLEMAGNE —HIS PLACE IN HISTORY.

Charlemagne, or Charles the Great, is the name which history has agreed to give to the founder of the German empire—incorporating the epithet with the name itself. We have recited in outline the facts of his wonderful career, as they are recorded in the meagre records of contemporary historians, and must rely upon the same authentic testimony in attempting to estimate his mind, character, and work. But the Charles of history is one; the Charles of heroic legend and popular fame is another. The former is a powerful conqueror and politic statesman, whom some eminent writers regard as the greatest of all monarchs; the latter is a Christian saint, superhuman in strength, beauty, and wisdom, incapable of defeat in war, of error in judgment, or of infirmity or corruption in his own will. Thus the song of Roland says: “His eyes shone like the morning star; his glance was dazzling as the noonday sun. Terrible to his foes, kind to the poor, victorious in war, merciful to offenders, devoted to God, he was an upright judge, who knew all the laws, and taught them to his people as he learned them from the angels. In short, he bore the sword as God’s own servant.” As Theodoric had been the center of the ancient popular minstrelsy, so Charles the Great became the central figure in that more cultivated heroic poetry, chiefly the work of the clergy, in which were celebrated the deeds of the twelve paladins, with Roland and the fight of Roncesvalles:

“When Charlemain with all his peerage fell
By Fontarabia.”

When we consider the profound impression made on the popular mind by this person, as represented in legend and song, we are almost ready to inquire whether its influence upon later German history was not greater than that of his authentic achievements. But it is true that the entire German race owes to him its first political organization. It was the purpose of his life, which never wavered, to unite all the German tribes under the control of one imperial government and of one Christian Church. In the greater part of this work he succeeded, and thus left the stamp of his mind upon the following centuries, through all the Middle Ages. The national consciousness of the collective German tribes dates from his reign, and it is at the beginning of the ninth century that “the Germans” are first spoken of in contrast with the Roman peoples of the empire, although the national

name did not come into general use until four generations later, in the reign of Otto the Great. When Charles mounted the throne, he was twenty-four years of age, in the strength and prime of his youth. His person was huge and strong, combining the presence and muscular power of the heroes of song; so that he found it sport to fight with the gigantic wild bulls in the forest of Ardennes. His passion for labor, war, and danger was that of the adventurous warriors of the great migration. In the momentous affairs of state, he exhibited the want of feeling and the unscrupulousness which have been common to nearly all great warriors; but in daily intercourse with those around him, he had the mildness, cheerfulness, and freshness of spirit which add so much grace to true greatness. These characteristics were those of his people; but that which specially distinguished him was the far-seeing mind, which had caught from ancient Rome the conception of a universal state, and was wise enough, without slavish copying, to adapt this conception to the peculiar requirements of the widely different race he ruled. This lofty intellect appears the more wonderful, that no one can tell how he obtained his mental growth, or who were his instructors; he seems to shine out of the darkness of his age like a sun.

Charlemagne's active mind gave attention to all matters, great and small. His untiring diligence, and his surprising swiftness in apprehension and decision, enabled him to dispatch an amount of business perhaps never undertaken by another monarch, unless by Frederick II., of Prussia, or by Napoleon Bonaparte. He was simple in his own attire, usually wearing a linen coat, woven at home by the women of his own family, and over it the large, warm Frisian mantle; and he demanded simplicity in his followers, and scoffed at his courtiers when their gorgeous silks and tinsel, brought from the East, were torn to rags in the rough work of the chase. Hunting in his favorite forest of Ardennes was the chief delight and recreation of his court. Next to this, he enjoyed swimming in the warm baths at Aix, which became his favorite residence. At his meals he listened to reading; and even condescended to join the monks, detailed for the purpose, in reading exercises. He founded schools in all the convents, and visited them in person, encouraging the diligent pupils, and reproving the negligent. He also introduced Roman teachers of music, to improve the church-singing of the Franks; while he required that sermons should be preached in the language of the people. Thus he diligently promoted popular education,

while he strove to make up by study what he had lost by the neglect of his own culture in youth. He gathered men of learning—poets, historians, and copyists—around him, the most prominent of them being Anglo-Saxons, of whom the wise and pious Alcuin was chief. Even when an old man, he found time, though often only at night, to practice in writing his hand so accustomed to the sword; and having long been familiar with the Latin language, which he tried to diffuse among the people, undertook to learn the Greek also. He highly esteemed his native language, too. He gave German names to the months and the winds; caused a German grammar to be compiled; and took pains to collect the ancient heroic songs of the German minstrels, though his son, in his monkish zeal, destroyed them. He revered the clergy highly: granted them tithes throughout the empire, and everywhere watched over the increasing endowments and estates of the Church, in whose possessions at that time both agriculture and morality were better cared for than elsewhere. Most of the bishops and abbots were selected by the king himself.

Charlemagne's personal character must not be judged by the standards of a time so remote from him as ours. He has been called dissolute; and it is true that he utterly disregarded the marriage tie, when it would limit either his pleasures or his ambition. He married five wives, only to dishonor them. He even encouraged, as it seems, his own daughters to live loose lives at home; refusing to give them in marriage to princes, lest their husbands might become competitors for a share of the kingdom. But he was never controlled by his favorite women, nor did he neglect state business for indulgence. Charlemagne has been censured as cruel; and, indeed, there are few acts recorded in history of more wanton cruelty than his slaughter in cold blood of thousands of Saxons at Verden. Yet this was not done in the exercise of passion or hatred, but as a measure of policy, a means deliberately devised to secure a definite end, in which it was successful. Charlemagne was never cruel upon impulse; but his inclinations were to gentleness and kindness. The key to his character is his unbounded ambition. In the pursuit of power he knew no scruple; the most direct and efficient means were always the right means to him. There is no doubt of his earnest attachment to the Christian Church and to the orthodox doctrines, as he understood them. But this was not associated with an appreciation of Christian morality, or a sense of human brotherhood. His

passion for conquest was in large part a fanatical zeal for the propagation of a religion which he regarded as inseparable from his empire.

Charlemagne was held in high honor by foreign nations. The Caliph of Bagdad, Haroun-al-Raschid, wielded in the East a power comparable with his own. To Charlemagne he sent a friendly embassy, with precious gifts, and it was reciprocated in the same spirit. The kings of the Normans expressed their respect for him in a similar way. But his own taste esteemed the ring of a good sword more than gold. His person and his private life have been vividly depicted to us by Einhard (Eginhard), a youth educated at his court, to whom, according to legend, the emperor gave one of his daughters for a wife. Charlemagne was tall and strongly formed, measuring from crown to sole seven times the length of his own foot. He had an open brow, very large, quick eyes, an abundance of fine hair, which was white in his last years, and a cheerful countenance.[\[G\]](#)

RESULTS OF HIS WARS AND RULE.

Some writers have sought to represent Charlemagne as a royal sage, a pacific prince, who only took up arms in self-defense. Truth compels a more faithful though less flattering portraiture. He had no invasion to dread. The Saracens were scattered, the Avars (Bavarians) weakened, and the Saxons impotent to carry on any serious war beyond their forests and marshes. If he led the Franks beyond their own frontiers, it was that he had, like so many other monarchs, the ambition of reigning over more nations, and of leaving a high-sounding name to posterity. All that he attempted beyond the Pyrenees proved abortive. It would have been of greater value had he subdued the Bretons, so far as to have made them sooner enter French nationality, instead of contenting himself with a precarious submission. The conquest of the Lombard kingdom profited neither France nor Italy, but only the Pope, whose political position it raised, and whose independence it secured for the future. The country for which those long wars had the happiest result, was that one which had suffered most from them, Germany. Before Charlemagne, Almayne was still Germany—that is to say, a shapeless chaos of pagan or Christian tribes, but all barbarian, enemies of one another, united by no single tie. There were Franks, Saxons, Thuringians, and Bavarians. After him there was a German people, and there will be a kingdom of Germany. It was great glory for him to have

created a people—a glory which few conquerors have acquired; for they destroy much more than they found. His reign lasted forty-four years, and may be summed up as an immense and glorious effort to bring under subjection the barbarian world and all that which survived the Roman civilization; to put an end to the chaos born of invasion, and to found a settled state of society in which the authority of the emperor, closely united to that of the Pope, should maintain order alike in Church and State—a very difficult problem, which it was given Charlemagne to solve, but of which all the difficulties did not become apparent until after his death. The work of Charlemagne, in fact, did not last. The name of this powerful though rude genius is not the less surrounded with a lasting glory; and it has remained in the memory of nations with that of three or four other great men who have done, if not always the greatest amount of good, at least have made the most noise in the world. As to Charlemagne, the amount of good accomplished very far surpasses that which was only vain renown and sterile ambition. He created modern Germany; and if that chain of nations, the links of which he had sought to rivet, broke, his great image loomed over the feudal times as the genius of order, continually inviting the dispersed races to emerge from chaos, and seek union and peace under the sway of a strong and renowned chief.

Charlemagne died, January 28, 814, in his seventy-second year, and was buried at Aix-la-Chapelle, in a church which he had built there after his Italian conquests, in the Lombard style. Eginhard, his secretary and friend, who wrote his life, tells us that he was considerably above six feet in height, and well proportioned in all respects, excepting that his neck was somewhat too short and thick. His imperial crown, which is still preserved at Vienna, would fit only the head of a giant. His air was dignified, but at the same time his manners were social. Charlemagne had no fewer than five wives; of his four sons, only one survived him, Louis, the youngest and most incapable, who succeeded him on the imperial throne.^[H]

[To be continued.]

^[A] Lewis.

^[B] Lewis.

^[C] Taylor.

^[D] Lewis.

[E] Menzies.

[F] Menzies.

[G] Lewis.

[H] Menzies.



EXTRACTS FROM GERMAN LITERATURE.



WALTHER VON DER VOGELWEIDE.

As an introduction to a brief extract upon Walther von der Vogelweide, we give Longfellow's beautiful little poem:

WALTHER VON DER VOGELWEIDE.

Vogelweide the Minnesinger,
When he left this world of ours,
Laid his body in the cloister,
Under Würtzburg's minster towers.

And he gave the monks his treasures,
Gave them all with this behest:
They should feed the birds at noontide
Daily on his place of rest;

Saying, "From these wandering minstrels
I have learned the art of song;
Let me now repay the lessons
They have taught so well and long."

Thus the bard of love departed;
And, fulfilling his desire,
On his tomb the birds were feasted
By the children of the choir.

Day by day, o'er tower and turret,
In foul weather and in fair,
Day by day, in vaster numbers,
Flocked the poets of the air.

On the tree whose heavy branches
Overshadowed all the place,
On the pavement, on the tombstone,
On the poet's sculptured face,

On the cross-bars of each window,
On the lintel of each door,
They renewed the War of Wartburg,
Which the bard had fought before.

There they sang their merry songs

There they sang their merry carols,
Sang their lauds on every side;
And the name their voices uttered
Was the name of Vogelweide.

Till at length the portly abbot
Murmured, "Why this waste of food?
Be it changed to loaves henceforward
For our fasting brotherhood."

Then in vain o'er tower and turret,
From the walls and woodland nests,
When the minster bells rang noontide,
Gathered the unwelcome guests.

Then in vain, with cries discordant,
Clamorous round the Gothic spire,
Screamed the feathered Minnesingers
For the children of the choir.

Time has long effaced the inscriptions
On the cloister's funeral stones,
And tradition only tells us
Where repose the poet's bones.

But around the vast cathedral,
By sweet echoes multiplied,
Still the birds repeat the legend,
And the name of Vogelweide.

Walther's lyrical poems are distinguished from those of most of his contemporaries by a strong impress of sincerity and a wide range of thought.

When he hails the coming of the spring after a long winter, he imitates in the gladness of his heart the carols of the birds, and goes on in melodious verses to speak of the beauty of the lady to whom he dedicates his song, but whom he never names. In the next song the reader, to his surprise, will find the minstrel changed into a satirist, who denounces the political and

religious corruptions of his time, rebukes the Pope for his worldly ambition and predicts a speedy ruin of the world. These are not all the notes of the scale on which his songs are constructed. As a specimen of his lighter and more popular style, the following strophe in praise of German women may serve:

In many foreign lands I've been
And knights and ladies there have seen;
But here alone I find my rest—
Old Germany is still the best;
Some other lands have pleased me well;
But here—'tis here I choose to dwell.
German men have virtues rare,
And German maids are angels fair.

He rises to a higher strain than this in other lyrics, where he places domestic virtue above external beauty, and speaks of *minne* in the higher interpretation of the word. "Even where it can not be returned," he says, "if devoted to one worthy of it, it ennobles a man's life. His affection for one teaches him to be kind and generous to all." Walther pleasantly describes himself as by no means good-looking, and censures all praise bestowed on men for their merely exterior advantages. And he is no fanatical worshiper of feminine beauty, affirming that it may sometimes be a thin mask worn over bad passions.

With regard to their moral and social purport the verses of Walther have a considerable historical interest. They show us how insecurely the Church held the faith and loyalty of German men in the thirteenth century.

Walther is bold and violent in his defiance and contempt of the Pope's usurpation of temporal authority. Referring in one place to a fable commonly believed in his times, he says: "When Constantine gave the spear of temporal power, as well as the spear and the crown to the See of Rome, the angels in heaven lamented, and well they might; for that power is now abused to annoy the emperor and to stir up the princes, his vassals against him." The poet was as earnest in dissuading the people from contributing money to support the Crusades. "Very little of it," he says, "will ever find its way into the Holy Land. The Pope is now filling his Italian coffers with our German silver." This saying seems to have been very popular for a tame moralist who lived in Walther's time complains that, by making such statements, the poet was perverting the faith of many people. "All his fine verses," the moralist adds, "will not atone for that bad libel on Rome." Yet the author of it was quite orthodox in doctrine, and was enthusiastic in his zeal for rescuing the Holy Sepulcher from the Saracens.

Many of his verses express earnestly his love for his native land, and his grief for social and political disorders of his times. He believes that the world is falling a prey to anarchy. "I hear the rushing of the water," he says, "and I watch the movements of the fish that swim in its depth. I explore the habits of the creatures of this world in the forest and in the field, from the beast of the field down to the insect, and I find that there is nowhere any life that is not vexed by anarchy and strife. Warfare is found everywhere, and yet some order is preserved even among animals; but in my own native

land, where the petty princes are lifting themselves up against the emperor, we are hastening on to anarchy.” The course of events proved that he was too true in this prediction. Resignation and despair, rather than any hope of a reconciliation of religion with practical life, characterize other meditative poems. The following is one of the best of this class:

I sat one day upon a stone,
And meditated long, alone,
While resting on my hand my head,
In silence to myself I said:
“How, in these days of care and strife,
Shall I employ my fleeting life?
Three precious jewels I require
To satisfy my heart’s desire:
The first is honor, bright and clear,
The next is wealth, and far more dear,
The third is heaven’s approving smile;”
Then, after I had mused a while
I saw that it was vain to pine
For these three pearls in one small shrine;
To find within one heart a place
For honor, wealth, and heavenly grace;
For how can one in days like these
Heaven and the world together please?

—From “*Outlines of German Literature*”
—Gostwick and Harrison.

HANS SACHS.

Riches of Poverty.

Why art thou cast down, my heart?
Why trouble, why dost mourn apart,
O'er naught but earthly wealth?
Trust in thy God, be not afraid,
He is thy friend, who all things made!

Dost think thy prayers he doth not heed?
He knows full well what thou dost need;
And heaven and earth are his!
My Father and my God, who still
Is with my soul in every ill.

The rich man in his wealth confides;
But in my God my trust abides.
Laugh as ye will, I hold
This one thing fast, that He hath taught:
Who trusts in God shall want for naught.

Yes, Lord: thou art as rich to-day
As thou hast been, and shall be aye:
I rest on thee alone;
Thy riches to my soul be given,
And 'tis enough for earth and heaven.

The legends of Hans Sachs are all pointed with satire. Readers now-a-days find in them a coarseness which jars their ideas of reverence and refinement, but which in the sixteenth century was in perfect keeping with the popular taste. One of the best of his legends is that of "St. Peter and the Goat." "We are told that once upon a time St. Peter was perplexed by an apparent prevalence of injustice in the world; and ventured to think that he could arrange matters better if he held the reins of government. He frankly confesses these thoughts to his Master. Meanwhile a peasant girl comes to him and complains that she has to do a hard day's work, and at the same time to keep in order a frolicsome young goat. 'Now,' says the Lord to

Peter, ‘you must have pity on this girl, and must take charge of the goat. That will serve as an introduction to your managing the affairs of the universe.’”

The legend goes on:

“The young goat had a playful mind
And never liked to be confined;
The Apostle at a killing pace,
Followed the goat, in a desperate chase;
Over the hills and among the briers
The goat runs on and never tires,
While Peter, behind, on the grassy plain,
Runs on, panting and sighing in vain.
All day, beneath a scorching sun,
The good Apostle had to run
Till evening came; the goat was caught
And safely to the Master brought,
Then, with a smile, to Peter said
The Lord: ‘Well, friend, how have you sped?
If such a task your powers has tried
How could you keep the world so wide?’
Then Peter, with his toil distressed,
His folly, with a sigh, confessed;
‘No, Master, ’tis for me no play
To rule one goat for one short day;
It must be infinitely worse
To regulate the universe.’”

MARTIN LUTHER.

The Book of Psalms.

The heart of man is like a ship out on a wild sea, and driven by storm-winds blowing from all the four quarters of the world; now impelled by fear

and care for coming evil, now disturbed by vexation and grief for present misfortune, now urged along by hope and a confidence of future good, now wafted by joy and contentment. These storm-winds of the soul teach us how to speak in good earnest, to open our hearts and to utter their contents. The man actually in want and fear does not express himself quietly, like a man who only talks about fear and want; a heart filled with joy utters itself and sings in a way not to be imitated by one who is all the time in fear; “It does not come from the heart,” men say, when a sorrowful man tries to laugh, or a merry man would weep.... Now of what does this book of Psalms mostly consist but of earnest expressions of the heart’s emotions—the storm-winds, as I have called them? Where are finer expressions of joy than the Psalms of praise and thanksgiving? There you look into the hearts of the saints, as if you looked into a fair and delightful garden, aye, or into heaven itself—and you see how lovely and pleasant flowers are springing up there out of manifold happy and beautiful thoughts of God and all His mercies.... But again, where will you find deeper, more mournful and pitiful words of sorrow than in the Psalms devoted to lamentation? I conclude then that the Psalter is a hand-book for religious men, wherein every one, whatever may be his condition, may find words that will rhyme with it; and Psalms as exactly fitted to suit his wants as if they had been written solely for his benefit.—*From the Preface to Luther’s Book of Psalms.*

Light in Despondency.

When the sky is black and lowering, when thy path in life is drear,
Upward lift thy steadfast glances, ’mid the maze of sorrow here.
From the beaming Fount of Gladness shall descend a radiance bright,
And the grave shall be a garden, and the house of darkness light,
For the Lord will hear and answer when in faith his people pray;
Whatsoever he hath appointed shall but work thee good alway.
E’en thy very hairs are numbered, God commands when one shall fall;
And the Lord is with his people, helping each and blessing all.

Our Defense.

A strong tower is the Lord our God,
To shelter and defend us;
Our shield his arm, our sword his rod
Against our foes befriend us.
That ancient enemy—
His gathering powers we see,
His terror and his toils;
Yet victory with its spoils
Not earth but heaven shall send us!

Though wrestling with the wrath of hell,
No might of man avail us,
Our captain is Immanuel,
And angel comrades hail us!
Still challenge ye his name?
“Christ in the flesh who came”—
The Lord, the Lord of Hosts!
Our cause his succor boasts;
And God shall ne’er fail us!

While mighty truth with us remain,
Hell’s arts shall move us never;
Nor parting friendship, honors, gains,
Our love from Jesus sever:
They leave us when they part
With him a peaceful heart;
And when from dust we rise,
Death yields us as he dies,
The crown of life forever!



GOTTHOLD EPHRAIM LESSING.

The Parable of “The Three Kings,” from “Nathan the Wise.”

In the oldest times, and in an eastern land,
There lived a man who had a precious ring.
This gem—an opal of a hundred tints—
Had such a virtue as would make the wearer
Who trusted it, beloved by God and man.
What wonder, if the man who had this ring
Preserved it well, and, by his will, declared
It should forever in his house remain?
At last when death came near, he called the son
Whom he loved best, and gave to him the ring,
With one strict charge:—“My son, when you must die,
Let this be given to your own darling child—
The son whom you love best, without regard
To any rights of birth.”—’Twas thus the ring
Was always passed on to the best-beloved.
Sultaun! you understand me?

Saladin.

Yea. Go on!—

Nathan. A father, who, at last possessed this ring
Had three dear sons—all dutiful and true—
All three alike beloved.—But, at one time,
This son, and then another, seemed most dear—
Most worthy of the ring; and it was given,
By promise, first to this son, then to that,
Until it might be claimed by all the three.
At last, when death drew nigh, the father felt
His heart distracted by the doubt to whom
The ring was due. He could not favor one
And leave two sons in grief! How did he act?
He called a goldsmith in, gave him the gem,
And bade him make exactly of that form,
Two other rings, and spare nor cost nor pains
To make all three alike. And this was done
So well. the owner of the first. true ring.

Could find no shade of difference in the three.
And now he called his sons—one at a time—
He gave to each a blessing and a ring—
One of the three—and died—

Saladin.

Well, well. Go on.

Nathan. My tale is ended. You may guess the sequel:—
The father dies; immediately each son
Comes forward with his ring, and asks to be
Proclaimed as head and ruler of the house;
All three assert one claim, and show their rings—
All made alike. To find the first—the true—
It was as great a puzzle as for us—
To find the one true faith.

Saladin. Is that, then, all the answer I must have?

Nathan. 'Tis my apology, if I decline
To act as judge, or to select the ring—
The one, true gem, of three all made alike;
All given by one—

Saladin. There! talk no more of “rings.”
The three religions, that, at first, were named,
Are all distinct—aye, down to dress—food—drink—

Nathan. Just so! and yet their claims are all alike,
As founded upon history, on facts
Believed, and handed down from sire to son,
Uniting them in faith. Can we—the Jews—
Distrust the testimony of our race?
Distrust the men who gave us birth, whose love
Did ne'er deceive us; but when we were babes,
Taught us, by means of fables, for our good?
Must *you* distrust your own true ancestors,
To flatter mine?—or must a Christian doubt
His father's words, and so agree with ours?—

Saladin. Allah!—the Israelite is speaking truth,
And I am silenced—

Nathan. Let me name the rings
Once more!—The sons at last, in bitter strife,
Appeared before a judge, and each declared
He had the one true gem, given by his father;
All said the same, and all three spoke the truth;
Each, rather than suspect his father's word,
Accused his brethren of a fraud—.

Saladin. What then?
What sentence could the judge pronounce? Go on.

Nathan. Thus said the judge:—"Go, bring your father here;
Let him come forth! or I dismiss the case.
Must I sit guessing riddles? Must I wait
Till the true ring shall speak out for itself?—
But stay!—'twas said that the authentic gem
Had virtue that could make its wearer loved
By God and man. That shall decide the case.
Tell me who of the three is best beloved
By his two brethren. Silent?—Then the ring
Hath lost its charm!—Each claimant loves himself,
But wins no love. The rings are forgeries;
'Tis plain, the first, authentic gem was lost;
To keep his word with you, and hide his loss,
Your father had these three rings made—these three,
Instead of one—"

Saladin. Well spoken, judge, at last!

Nathan. "But stay," the judge continued; "hear one word—
The best advice I have to give; then go.—
Let each still trust the ring given by his father!—
It might be, he would show no partial love;
He loved all three, and, therefore, would not give
The ring to one and grieve the other two

the ring to one and grieve the other two.
Go, emulate your father's equal love.
Let each first test his ring and show its power;
But aid it, while you test; be merciful,
Forbearing, kind to all men, and submit
Your will to God. Such virtues shall increase
Whatever powers the rings themselves may have;
When these, among your late posterity,
Have shown their virtue—in some future time,
A thousand thousand years away from now—
Then hither come again!—A wiser man
Than one now sitting here will hear you then,
And will pronounce the sentence.”

Saladin.

Allah! Allah!

Nathan. Now, Saladin, art thou that “wiser man?”
Art thou the judge who will, at last, pronounce
The sentence?

*[Saladin grasps Nathan's hand, and holds
to the end of the conversation.]*

Saladin. I the judge?—I'm dust! I'm nothing!
'Tis Allah!—Nathan, now I understand;
The thousand thousand years have not yet passed;
The judge is not yet come; I must not place
Myself upon his throne! I understand—
Farewell, dear Nathan! Go.—Be still my friend.



READINGS IN PHYSICAL SCIENCE.

Abridged from Science Primer on Physical Science, by Prof. Geikie.

III.—RIVERS AND GLACIERS.

We have found that the water of the river is largely derived from springs, and that all spring-water contains more or less mineral materials dissolved out of the brooks. Every river, therefore, is carrying not merely water, but large quantities of mineral matters into the sea. It has been calculated, for instance, that the Rhine in one year carries into the North Sea lime enough to make three hundred and thirty-two thousand millions of oyster shells. This chemically-dissolved material is not visible to the eye, and in no way affects the color of the water. At all times of the year, as long as the water flows, this invisible transport of some of the materials of rocks must be going on.

But let us now again watch the same river in flood. The water is no longer clear, but dull and dirty. You ascertained that this discoloration arises from mud and sand suspended in the water. You see that over and above the mineral matter in chemical solution, the river is hurrying seaward with vast quantities of other and visible materials. And thus it is clear that at least one great part of the work of rivers must be to transport the mouldered parts of the land which are carried into them by springs or by rain.

But the rivers, too, help in the general destruction of the surface of the land. Of this you may readily be assured, by looking at the sides or bed of a stream when the water is low. Where the stream flows over hard rock, you find the rock all smoothed and ground away; and the stones lying in the water-course are all more or less rounded and smoothed. When these stones were originally broken by frosts or otherwise, from crags and cliffs, they were sharp-edged, as you can prove by looking at the heaps of blocks lying at the foot of any precipice, or steep bank of rock. But when they fell, or

were washed into the river, they began to get rolled and rubbed, until their sharp edges were ground away, and they came to wear the smooth rounded forms which we see in the ordinary gravel.

While the stones are ground down, they, at the same time, grind down the rocks which form the sides and bottom of the river-channel over which they are driven. You can even see in some of the eddies of the stream how the stones are kept moving round until they actually excavate deep round cavities, called pot-holes, in the solid rock.

Now, it is clear that two results must follow from this ceaseless wear and tear of rocks and stones in the channel of a stream. In the first place, a great deal of mud and sand must be produced; and, in the second place, the bed of the river must be ground down so as to become deeper and wider. The sand and mud are added to the other similar material washed into the streams by rain from the mouldering surface of the land. By the deepening and widening of the water-courses, such picturesque features as gorges and ravines are excavated out of the solid rock.

Look, again, at the channel of a river in summer. You see it covered with sheets of gravel in one place, beds of sand in another, while here and there a piece of hard rock sticks up through these different kinds of river-stuff. Note some portion of the loose materials, and you find it to be continually shifting. A patch of gravel or sand may remain for a time, but the little stones and grains of which it is made up are always changing as the water covers and moves them. In fact, the loose materials over which the river flows are somewhat like the river itself. You come back to its banks after many years, and you find the river there still, with the same ripples, and eddies, and gentle murmuring sound. But though the river has been there constantly all the time, its water has been changing every minute, as you can watch it changing still. So, although the channel is always more or less covered with loose materials, these are not always the same. They are perpetually being pushed onward, and others, from higher up the stream, come behind to take their place.

It is not in the bottoms of the rivers, then, that the material worn away from the surface of the land can find any lasting rest. And yet the rivers do get rid of a good deal of this material as they roll along. You have, perhaps,

noticed that a river is often bordered with a strip of flat plain, the surface of which is only a few feet above the level of the water. Most of our rivers have such margins, and, indeed, seem each to wind to and fro through a long, level, meadow-like plain. Now this plain is really made up from the finer particles of decomposed rocks which the river has carried along. During floods, the river, swollen and muddy, rises above its banks, and spreads over the low ground on either side. Whenever this takes place, the overflowing water moves more slowly over the flats; and, as its current is thus checked, it can not hold so much mud and sand, but allows some of these materials to settle down to the bottom. In this way the overflowed tracts get a coating of soil laid over them by the river, and when the waters retire this coating adds a little to the height of the plain. The same thing takes place year after year, until by degrees the plain gets so far raised that the river, which all this while is also busy deepening its channel, can not overflow it even at the highest floods. In course of time the river, as it winds from side to side, cuts away slices of the plain and forms a newer one at a lower level. And thus a series of terraces is gradually made, rising step by step above the river.

Still the laying down of its sand and mud by a river to form one or more such river-terraces is, after all, only a temporary disposal of these materials. They are still liable to be carried away, and in truth they are carried off continually as the river eats away its banks.

When the current of a river is checked as it enters the sea or a lake, the feebler flow of the water allows the sand and mud to sink to the bottom. By degrees some portions of the bottom come in this way to be filled up to the surface of the river, and wide flat marshy spaces are formed on either side of the main stream. During floods these spaces are overflowed with muddy water, in the same way as in the case of the valley plains just described, and a coating of mud or sand is laid down on them until they slowly rise above the ordinary level of the river, which winds about among them in endless branching streams. Vegetation springs up on these flat swampy lands; animals, too, find food and shelter there; and thus a new territory is made by the work of the river.

These flat river-formed tracts are called deltas, because the one which was best known to the ancients, that of the Nile, had the shape of the Greek

letter Δ (*delta*). This is the general form which is taken by accumulations at the mouths of rivers; the flat delta gets narrow toward the inland, and broader toward the sea. Some of them are of enormous size; the delta of the Mississippi, for example.

Each delta, then, is made of materials worn from the surface of the land, and brought down by the river. And yet vast though some of these deltas are, they do not show all the materials which have been so worn away. A great deal is carried far out and deposited on the sea-bottom; for the sea is the great basin into which the spoils of the land are continually borne.

Having now followed the course taken by the water which falls on the land as rain, we come to that taken by snow.

On the tops of some of the highest mountains in Britain snow lies for great part of the year. On some of them, indeed, there are shady clefts wherein you may meet with deep snow-wreaths even in the heat of summer.

But in other parts of Europe, where the mountains are more lofty, the peaks and higher shoulders of the hills gleam white all the year with unmelted snow.

Let us see why it is that perpetual snow should occur in such regions, and what part this snow plays in the general machinery of the world.

You have learned that the higher parts of the atmosphere are extremely cold. You know also that in the far north and the far south, around those two opposite parts of the earth's surface called the Poles, the climate is extremely cold—so cold as to give rise to dreary expanses of ice and snow, where sea and land are frozen, and where the heat of summer is not enough to thaw all the ice and drive away all the snow. Between these two polar tracts of cold, wherever mountains are lofty enough to get into the high parts of the atmosphere where the temperature is usually below the freezing-point, the vapor condensed from the air falls upon them, not as rain, but as snow. Their heads and upper heights are thus covered with perpetual snow. In such high mountainous regions the heat of the summer always melts the snow from the lower hills, though it leaves the higher parts still covered. From year to year it is noticed that there is a line or limit below which the ground gets freed of its snow, and above which the snow

remains. This limit is called the snow-line, or the limit of perpetual snow. Its height varies in different parts of the world. It is highest in the warmer regions on either side of the equator, where it reaches to 15,000 feet above the sea. In the cold polar tracts, on the other hand, it approaches the sea-level. In other words, while in the polar tracts the climate is so cold that perpetual snow is found even close to the sea-level, the equatorial regions are so warm that you must climb many thousand feet before you can reach the cold layers of the air where snow can remain all the year.

There is, you see, one striking difference between rain and snow. If rain had been falling for the same length of time, the roads and fields would still have been visible, for each drop of rain, instead of remaining where it fell, would either have sunk into the soil, or have flowed off into the nearest brook. But each snowflake, on the contrary, lies where it falls, unless it happens to be caught up and driven on by the wind to some other spot where it can finally rest. Rain disappears from the ground as soon as it can; snow stays still as long as it can.

You will see at once that this marked difference of behavior must give rise to some equally strong differences in the further procedure of these two kinds of moisture. You have followed the progress of the rain; now let us try to find out what becomes of the snow.

In such a country as ours, where there is no perpetual snow, you can without much difficulty answer this question. Each fall of snow in winter-time remains on the ground as long as the air is not warm enough to melt it. Evaporation, indeed, goes on from the surface of snow and ice, as well as from water: so that a layer of snow would in the end disappear, by being absorbed into the air as vapor, even though none of it had previously been melted into running water. But it is by what we call a thaw that our snow is chiefly dissipated; that is, a rise in the temperature, and a consequent melting of the snow. When the snow melts, it sinks into the soil and flows off into brooks in the same way as rain.

In the regions of perpetual snow the heat of summer can not melt all the snow which falls there in the year. What other way of escape, then, can the frozen moisture find?

You will remember that the surplus rainfall flows off by means of rivers. Now the surplus snow-fall above the snow-line has a similar kind of drainage. It flows off by means of what are called glaciers.

When a considerable depth of snow has accumulated, the pressure upon the lower layers from what lies above them squeezes them into a firm mass. The surface of the ground is usually sloped in some direction, seldom quite flat. And among the high mountains the slopes are often, as you know, very steep. When snow gathers deeply on sloping ground, there comes a time when the force of gravity overcomes the tendency of the pressed snow to remain where it is, and then the snow begins to slide slowly down the slope. From one slope it passes on downward to the next, joined continually by other sliding masses from neighboring slopes until they all unite into one long tongue which creeps slowly down some valley to a point where it melts. This tongue from the snow-fields is the glacier. It really drains these snow-fields of their excess of snow as much as a river drains a district of its excess of water.

But the glacier which comes out of the snow-fields is itself made not of snow, but of ice. The snow, as it slides downward, is pressed together into ice. You have learned that each snowflake is made of little crystals of ice. A mass of snow is thus only a mass of minute crystals of ice with air between. Hence when the snow gets pressed together, the air is squeezed out, and the separated crystals of ice freeze together into a solid mass. You know that you can make a snowball very hard by squeezing it firmly between the hands. The more tightly you press it the harder it gets. You are doing to it just what happens when a glacier is formed out of the eternal snows. You are pressing out the air, and allowing the little particles of ice to freeze to each other and form a compact piece of ice. But you can not squeeze nearly all the air out, consequently the ball, even after all your efforts, is still white from the imprisoned air. Among the snowfields, however, the pressure is immensely greater than yours; the air is more and more pressed out, and at last the snow becomes clear transparent ice.

A glacier, then, is a river, not of water, but of ice, coming down from the snow-fields. It descends sometimes a long way below the snow-line, creeping down very slowly along the valley which it covers from side to side. Its surface all the time is melting during the day in summer, and

streams of clear water are gushing along the ice, though, when night comes, these streams freeze. At last it reaches some point in the valley beyond which it can not go, for the warmth of the air there is melting the ice as fast as it advances. So the glacier ends, and from its melting extremity streams of muddy water unite into a foaming river, which bears down the drainage of the snow-fields above.

A river wears down the sides and bottom of its channel, and thus digs out a bed for itself in even the hardest rock, as well as in the softest soil. It sweeps down, too, a vast quantity of mud, sand, and stones from the land to the sea. A glacier performs the same kind of work, but in a very different way.

When stones fall into a river they sink to the bottom, and are pushed along there by the current. When mud enters a river it remains suspended in the water, and is thus carried along. But the ice of a glacier is a solid substance. Stones and mud which fall upon its surface remain there, and are borne onward with the whole mass of the moving glacier. They form long lines of rubbish upon the glacier, and are called moraines. Still the ice often gets broken up into deep cracks, opening into yawning clefts or crevasses, which sometimes receive a good deal of the earth and stones let loose by frost or otherwise from the sides of the valley. In this way loose materials fall to the bottom of the ice, and reach the solid floor of the valley down which the ice is moving; while at the same time similar rubbish tumbles between the edge of the glacier and the side of the valley.

The stones and grains of sand which get jammed between the ice and the rock over which it is moving are made to score and scratch this rock. They form a kind of rough polishing powder, whereby the glacier is continually grinding down the bottom and sides of its channel. If you creep in below the ice, or catch a sight of some part of the side from which the ice has retired a little, you will find the surface of the rock all rubbed away and covered with long scratches made by the sharp points of the stones and sand.

You will now see the reason why the river, which escapes from the end of a glacier, is always muddy. The bottom of the glacier is stuck all over with stones, which are scraping and wearing down the rock underneath. A great deal of fine mud is thus produced, which, carried along by streams of

water flowing in channels under the glacier, emerges at the far end in the discolored torrents which there sweep from under the ice.

A glacier is not only busy grinding out a bed for itself through the mountains; it bears on its back down the valley enormous quantities of fallen rock, earth and stones, which have tumbled from the cliffs on either side. In this way blocks of rock as big as a house may be carried for many miles, and dropped where the ice melts. Thousands of tons of loose stones and mud are every year moved on the ice from the far snowy mountains away down into the valleys to which the glaciers reach.

The largest glaciers in the world are those of the polar regions. North Greenland, in truth, lies buried under one great glacier, which pushes long tongues of ice down the valleys and away out to sea. When a glacier advances into the sea, portions of it break off and float away as icebergs. So enormous are the glaciers in these cold tracts that the icebergs derived from them often rise several hundred feet above the waves which beat against their sides. And yet, in all such cases, about seven times more of the ice is immersed under water than the portion, large as it is, which appears above. You can realize how this happens if you take a piece of ice, put it in a tumbler of water, and watch how much of it rises out of the water. Sunk deep in the sea, therefore, the icebergs float to and fro until they melt, sometimes many hundreds of miles away from the glaciers which supplied them.

You will come to learn afterward that, once upon a time, there were glaciers in Britain. You will be able with your own eyes to see rocks which have been ground down and scratched by the ice, and big blocks of rock and piles of loose stones which the ice carried upon its surface. So that, in learning about glaciers, you are not merely learning what takes place in other and distant lands, you are gaining knowledge which you will be able by and by to make good use of, even in your own country.



SUNDAY READINGS.

SELECTED BY THE REV. J. H. VINCENT, D.D.

[*December 2.*]
FROM THE “CHRISTIAN’S PATTERN.”
By THOMAS À KEMPIS.

“He that followeth me walketh not in darkness, saith the Lord.” These are the words of Christ, by which we are admonished that we ought to imitate his life and manners, if we would be truly enlightened and delivered from all blindness of heart.

Let therefore our chief endeavor be to meditate upon the life of Jesus Christ.

What will it avail thee to dispute sublimely of the Trinity, if thou be void of humility, and art thereby displeasing to the Trinity?

Truly, sublime words do not make a man holy and just; but a virtuous life maketh him dear to God.

I had rather feel compunction, than know the definition thereof.

If thou didst know the whole Bible, and the sayings of all the philosophers, by heart, what would all that profit thee without the love of God?

Vanity of vanities! all is vanity, but to love God and serve him only.

It is therefore vanity to seek after perishing riches.

It is also vanity to seek honors.

It is vanity to follow the desires of the flesh, and to labor for that for which thou must afterward suffer grievous punishment.

It is vanity to wish to live long, and to be careless to live well.

It is vanity to mind this present life, and not those things which are to come.

It is vanity to set thy love on that which speedily passeth away, and not to hasten thither, where everlasting joys remain.

All men naturally desire to know; but what availeth knowledge without the fear of God?

Surely an humble husbandman that serveth God is better than a proud philosopher, that, neglecting himself, studies the course of the heavens.

He that knoweth himself is vile in his own eyes, and is not pleased with the praises of men.

If I understood all things in the world, and had not charity, what would that help me in the sight of God, who will judge me according to my deeds.

There are many things, to know which doth little profit the soul.

And he is very unwise, that minds any other things than those that tend to the welfare of his soul.

Many words do not satisfy the soul; but a pure conscience giveth confidence toward God.

The more thou knowest, and the better thou understandest, the more grievously shalt thou be judged, unless thy life be the more holy.

Be not therefore lifted up; but rather let the knowledge given thee make thee afraid.

If thou thinkest that thou knowest much: yet there are many more things which thou knowest not.

Be not over wise, but rather acknowledge thine own ignorance.

The highest and most profitable lesson is, the true knowledge of ourselves.

It is great wisdom to esteem ourselves nothing, and to think always well and highly of others.

We are all frail, but remember, none more frail than thyself.

[*December 9.*]

It is good that we be sometimes contradicted; and that men think ill of us, and this, although we do not intend well.

For then we more diligently seek God for our inward witness, when outwardly we are contemned by men.

Wherefore a man should settle himself so fully in God, that he need not seek comforts of men.

When a man is afflicted, tempted, or troubled with evil thoughts; then he understandeth better the great need he hath of God.

So long as we live in this world, we can not be without temptation.

Hence it is written in Job, “The life of man is a warfare upon earth.”

Temptations are often very profitable to men, though they be troublesome and grievous; for in them a man is humbled, purified, and instructed.

All the saints have passed through, and profited by, many tribulations, and temptations:

And they that could not bear temptations, became reprobates and fell away.

There is no place so secret, where there are no temptations.

There is no man that is altogether secure from temptations while he liveth.

When one temptation goeth away, another cometh; and we shall ever have something to suffer.

Many seek to fly temptations, and fall more grievously into them.

By flight alone we can not overcome, but by patience and humility we conquer all our enemies.

He that only avoideth them outwardly, and doth not pluck them up by the roots, shall profit little: yea, temptations will soon return unto him, and he shall feel them worse than before.

By patience (through God's help) thou shalt more easily overcome, than by harsh and disquieting efforts in thy own strength.

Often take counsel in temptations; and deal not roughly with him that is tempted.

The beginning of temptation is inconstancy of mind, and little confidence in God.

For as a ship without a rudder is tossed to and fro with the waves, so the man that is negligent is many ways tempted.

Fire trieth iron, and temptation a just man.

We know not often what we are able to do: but temptations show us what we are.

We must be watchful, especially in the beginning of the temptation; for the enemy is then more easily overcome, if he be not suffered to enter the door of your hearts, but be resisted without the gate at his first knock.

Wherefore one said, "Withstand the beginning: for an after remedy comes too late."

First, there occurreth to the mind a simple evil thought; then a strong imagination; afterward delight; and lastly consent.

And so by little and little our malicious enemy getteth entrance, while he is not resisted in the beginning.

And the longer one is slack in resisting, the weaker he becomes daily, and the enemy stronger against him.

Some suffer the greatest temptation in the beginning of their conversion; others in the latter end.

Others again are much troubled almost throughout their life.

Some are but slightly tempted, according to the wisdom which weigheth the states of men, and ordereth all things for the good of his elect.

We ought therefore, when we are tempted, so much the more fervently to pray unto God; who surely will give with the temptation, a way to escape, that we may be able to bear it.

Let us therefore humble ourselves under the hand of God, in all temptations and tribulations; for he will exalt the humble in spirit.

In temptations and afflictions a man is proved how much he hath profited.

Neither is it any such great thing if a man be devout and fervent, when he feeleth no affliction; but if in time of adversity he bear himself patiently, there is hope then of great proficiency.

Some are kept from great temptations, and are overcome in small ones; that being humbled, they may never trust themselves in great matters, who are baffled in so small things.

[*December 16.*]

Turn thine eyes unto thyself, and beware thou judge not the deeds of other men.

In judging others a man laboreth in vain, often erreth, and easily sinneth; but in judging and examining himself, he always laboreth fruitfully.

We often judge of things according as we fancy them: for affection bereaves us easily of a right judgment.

If God were always our desire, we should not be so much troubled when our inclinations were opposed.

But oftentimes something lurks within, which draweth us after it.

Many secretly seek themselves in their actions, but know it not.

They live in peace of mind when things are done according to their will: but if things succeed otherwise than they desire, they are straightway troubled.

Diversity of inclinations and opinions often causes dissensions between religious persons, between friends and countrymen.

An old custom is hardly broken, and no man is willing to be led farther than himself can see.

If thou dost more rely upon thine own reason, than upon Jesus Christ, late, if ever, shalt thou become illuminated.

The outward work without charity, profiteth nothing; but whatsoever is done out of charity, be it ever so little and contemptible in the sight of the world, is wholly fruitful.

For God weigheth more with how much love one worketh, than how much he doeth.

He doth much that loveth much.

He doth much that doth a thing well.

He doth well that serveth his neighbor, and not his own will.

Often it seemeth to be charity, and it is rather carnality; because natural inclinations, self-will, hope of reward, and desire of our own interest, are motives that men are rarely free from.

He that hath true and perfect charity seeketh himself in nothing; but only desireth in all things that God should be exalted.

He envieth none, because he seeketh not his own satisfaction; neither rejoiceth in himself, but chooses God only for his portion.

He attributes nothing that is good to any man, but wholly referreth it unto God, from whom, as from the fountain, all things proceed: in whom finally all the saints rest.

O that he had but one spark of true charity, he would certainly discern that all earthly things are full of vanity!

[*December 23.*]

When one that was in great anxiety of mind, often wavering between fear and hope, did once humbly prostrate himself in prayer, and said, O, if I knew that I should persevere! he presently heard within him an answer from God which said, If thou didst know it, what wouldst thou do? Do what thou wouldst do then, and thou shalt be safe.

And being herewith comforted and strengthened, he committed himself wholly to the will of God, and his anxiety ceased:

Neither had he any mind to search curiously farther what should befall him; but rather labored to understand what was the perfect and acceptable will of God, for the beginning and accomplishing every good work.

Hope in the Lord, and do good, saith the prophet, and inhabit the land, and thou shalt be fed.

One thing there is that draweth many back from a spiritual progress, and diligent amendment; the horror of the difficulty, or the labor of the combat.

But they improve most in virtue, that endeavor most to overcome those things which are grievous and contrary to them.

For there a man improveth more, and obtaineth greater grace, where he more overcometh himself and mortifieth himself in spirit.

Gather some profit to thy soul wheresoever thou art; so if thou seest or hearest of any good examples, stir up thyself to the imitation thereof.

But if thou seest anything worthy of reproof, beware thou doest not the same.—And if at any time thou hast done it, labor quickly to amend it.

Be mindful of the profession thou hast made, and have always before thine eyes the remembrance of thy Savior crucified.

Thou hast good cause to be ashamed, looking upon the life of Jesus Christ, seeing thou hast as yet no more endeavored to conform thyself unto him, though thou hast walked a long time in the way of God.

A religious person that exerciseth himself seriously and devoutly in the most holy life and passion of our Lord shall there abundantly find whatsoever is necessary and profitable for him; neither shall he need seek any better thing out of Jesus.

A CHRISTMAS PRAYER.

Come thou O Lord, and dwell within me, giving me light, and love, and liberty. May the spirit of the sweet Christmas Child possess me! May the Star of Bethlehem abide above my dwelling place! May the angels who seek thee be drawn toward me, and surround my path! May their song fill my life. Glory to God in the highest. On earth peace, good will to men.

[December 30.]

This life will soon be at an end; consider therefore how thy affairs stand as to the next.

Man is here to-day; to-morrow he is gone.

When he is out of sight, he is soon forgotten.

Thou shouldst so order thyself in all thy thoughts and all thy actions, as if thou wert to die to-day.

Hadst thou a clear conscience, thou wouldst not fear death.

It were better to avoid sin than to fly death.

If thou art not prepared to-day, how wilt thou be to-morrow?

To-morrow is uncertain, and how knowest thou that thou shalt live till to-morrow?

What availeth to live long, when we are so little the better?

Alas! long life doth not always mend us; but often increased guilt.

O, that we had spent but one day well in this world!

When it is morning, think thou mayst die before night.

When evening comes, dare not to promise thyself the next morning.

Be therefore always in readiness; and so live that death may never take thee unprepared.

Many die suddenly, and when they look not for it; for “in such an hour as you think not, the Son of man cometh.” Matt. xxiv: 44.

When that last hour shall come, thou wilt have a far different opinion of thy whole life.

How wise and happy is he, that laboreth to be such in his life as he would wish to be found at the hour of his death.

Whilst thou art in health, thou mayst do much good, but when thou art sick, I know not what thou wilt be able to do.

Few by sickness grow better; and they who travel much are seldom sanctified.

Trust not in friends and kindred, neither put off the care of thy soul till hereafter, for man will sooner forget thee than thou art aware of.

If thou art not careful for thyself now, who will be careful for thee hereafter?

The time present is very precious; now are the days of salvation, now is the acceptable time.

But alas! that thou shouldst spend thy time no better here, where thou mightest purchase life eternal. The time will come when thou shalt desire

one day or hour to amend in, and I can not say it will be granted thee.

Ah fool! why dost thou think to live long, when thou canst not promise thyself one day!

How many have been deceived, and suddenly snatched away!

How often dost thou hear, such a man is slain, another is drowned, a third has broken his neck with a fall; this man died eating, and that playing?

One perished by fire, another by sword, another of the plague, another was slain by thieves! Thus death is the end of all, and man's life suddenly passeth away like a shadow.

Who shall remember thee when thou art dead? Do, do now, my beloved, whatsoever thou art able to do: for thou knowest not when thou shalt die, nor yet what shall be after thy death.

Now, while thou hast time, lay up for thyself everlasting riches.

Keep thy heart free, and lifted up to God, because thou hast here no abiding city.

Send thither thy daily prayers, and sighs, and tears, that after death thy spirit may happily pass to the Lord. *Amen.*



POLITICAL ECONOMY.

By G. M. STEELE, D.D.

III. EXCHANGE.

1. Exchange is the mutual and voluntary transfer of the right of property held by different persons. This implies, (*a*) the existence of the *right of property*; (*b*) that the transfer must be *mutual*, otherwise there is no exchange; (*c*) that it be *voluntary*, otherwise it would be robbery.

2. The principles that form the basis of exchange are the same as those implied in the great law of association and individuality; namely, those which give rise to the combination and division of labor. There is usually some one kind of labor, or at most a few kinds, for which each individual is competent. But the variety of occupations so nearly corresponds with the variety of aptitudes in every well-ordered community, that each may, with little effort, find the calling to which he is suited.

But while each individual is thus limited in his productive capabilities, his claims and wants are nearly limitless. He is in need of a thousand commodities, only a very few of which he can produce. He depends for the remainder of these upon his fellow-men. On the other hand, he can produce a thousand times as much of the few kinds of commodities to which he devotes himself, as he himself needs. These he transfers to his fellow-men, taking in return the surplus of their several products. This is exchange, or commerce. It is implied in the very constitution of man. Association is an imperative condition of humanity.

3. A distinction is sometimes made between *commerce* and *trade*—a wise distinction, as it seems to me, though observed by but few writers. The former is the *object* to be accomplished; the latter is the *agency* through

which it is accomplished. Thus, a farmer has wheat, butter, eggs, poultry, wool, etc., which he wishes to exchange for cloth, sugar, agricultural implements, boots and shoes, and a hundred other articles. He can not go to the several producers of these, carrying his own products to exchange for them, except at immense disadvantage. Hence arises the necessity for the trader, or merchant. Trade and commerce have sometimes been represented as mutually antagonistic. This is true only to a certain extent. The great economical point to be guarded is to have no more traders than are necessary to make the exchanges. When the industrial and commercial conditions of a country are such that the producers and consumers, who are the real exchangers, are placed and kept at a great distance from each other, so that they can not combine with each other except through the agency of a great number of middle-men, the conditions are highly detrimental to the interests of the parties chiefly concerned. Beyond a certain point, the greater the power of trade, the worse it is for commerce. It is nevertheless true that there are certain natural obstacles to direct commerce which can be surmounted only by some kind of intermediate agency; and this makes the trader necessary. In this respect, and to this extent, trade is an aid to commerce. Yet commerce should be as direct as possible. To this end it is desirable that the greatest number of commodities for which productive facilities exist, should be produced in the same community.

4. The general law of exchange is *value for value*. This will be obvious if we recur to one of our statements concerning the nature of value, namely, that is the quantity of one commodity that may be equitably exchanged for a given quantity of another. It will be still more obvious if we recall the complete definition: value is our estimate of the sacrifice requisite to secure possession of a desired object. Thus, if it require the labor of one day to produce a pair of shoes, and the labor also of a day to produce three bushels of oats, then the rule of exchange would be three bushels of oats for a pair of shoes, because the required labor in the one case is precisely equal to that in the other.

This is the fundamental law, but it is modified in its operation by certain other facts and principles. Chief among these is the law of *supply and demand*. By supply is meant the quantity of any commodity which is in the market. Demand signifies the quantity which is desired at a given price. The definitions are sometimes erroneously given of supply as the quantity which

exists, and demand as the quantity desired. But a man may offer for sale a load of wheat, provided the price is a dollar a bushel, but withdraw it from the market if the price is but ninety cents. A thousand people in a certain town may desire diamond necklaces, but not half a dozen may be able to purchase them. Hence supply is all that is offered in the market; and demand is desire with ability to purchase.

Demand and supply affect prices in this way. Suppose a community has been exclusively using wood for fuel, and their wood can be had at a certain price. After a time a coal mine is discovered in the vicinity, and coal can be furnished much cheaper than wood. This would lessen the demand for wood. As there would be the same amount for sale as before, the seller would be in competition, and the price would fall. So if for any reason before the discovery of the coal the supply of wood had been diminished one half, the demand being the same, the price would rise. Thus we have the general principle that other things being equal, the greater the supply, the less the price; the smaller the supply, the greater the price; the greater the demand, the greater the price; and the smaller the demand, the less the price. In other words, the price varies directly as the demand, and inversely as the supply. In general price varies as the cost of production plus or minus the effect of supply and demand. These principles are affected again in many ways which we can not here explain. Yet the variations are always temporary, and the price or market value always tends to seek the level of cost of production.

5. Trade has been spoken of as an agent of exchange. An *instrument* also is needed. The primitive method of exchange was by barter. That is, by giving the commodity one produces for that which one desires to possess. But this was early found inconvenient. The man who made shoes and wished to exchange some of them for a coat, would not readily find a coat-maker in want of shoes; or if he should, the latter very likely would not want just so many pairs of shoes as would be equal in value to the coat. All other exchanges might be at a similar disadvantage. What is needed is a commodity which will be a *medium* of exchange—which every one will be willing to receive for any commodity which he has for sale, and which will command anything which he wishes to buy. Such a commodity is usually the main element in the machinery of exchange, and is what constitutes *money*.

This instrument in order to meet the want, it is generally believed, must have the following characteristics: 1. Value in the material of which it is made. 2. Uniformity of value throughout the world. 3. Much value in small bulk. 4. Approximate constancy of value. 5. Not readily destructible. 6. Divisibility into small portions which are capable of being reunited. 7. Of universal use. 8. Capable of receiving stamps and marks. Most of these properties are found in gold and silver, if not to such an extent as has been claimed for them, at least so far that they have been the basis of the money of the civilized world.

6. But supplementing in a certain way, and representing these, the instrument of exchange comprises also the large element of *credit*. This consists chiefly of book accounts, promissory notes, bank notes, government notes, bank deposits, checks, drafts, bills of exchange, stocks and bonds. One of the great agencies in modern commerce by which credit is made effectual as a part of the mechanism of exchange is that of *banks*. Banks are institutions which serve to abbreviate and facilitate the business of exchange and to extend and render available the credit of the community.

There are four kinds of banks, namely: savings banks, banks of deposit, banks of circulation and issue, and banks of discount. In our modern banking system the last three are generally found in combination, that is, each bank exercises all the functions implied.

A savings bank is an institution in which small sums of money are deposited from time to time as they accumulate in the hands of persons of moderate incomes. The depositors are credited with these amounts, and receive a certain, usually not very large, rate of interest in any case, and an additional amount contingently. The bank loans the money thus deposited in large sums to trustworthy persons who can furnish good security, the rate of interest being somewhat higher than that paid to the depositor.

The benefit of such an institution is two fold. In the first place there are many persons who have small sums of money which they desire to be earning something in some safe place. The amount is too small to be loaned to advantage. Such persons are not likely to know how, even if the sums at their disposal were sufficient, to find the best investment, or to determine concerning the security offered. But put into the hands of men who make

this their business, under rules devised by the best financial talent of the community, and who can combine these small sums and invest them to the best advantage, it is made both safe and profitable for the small capitalists.

In the second place there are many persons who wish to unite their labor and skill with capital in some productive enterprise, and having no capital of their own, desire to borrow. They do not know the persons who have money to loan. The savings bank affords them an opportunity and gives them an advantage which they would not otherwise have. It is a benefit first to those who have some surplus, but are unable to loan it to advantage; secondly to those who are in want of capital, but do not know where to find it.

A *bank of deposit* grows out of the necessities of commerce in a community where much business is transacted. All persons engaged in trade will find from time to time large or smaller accumulations of money in their hands which it is not safe without considerable expense, to keep by them. Hence the custom of depositing these for safe keeping in the bank. Usually no interest is paid as the money may be withdrawn any time at the will of the depositor. It was early found that only a small proportion of these deposits were likely to be withdrawn at any one time; hence a considerable proportion of them could be loaned on short time, and thus the bank would in this way receive compensation for its care, without expense to the depositors. In this way, too, the capital of the community could be kept more fully employed.

But the credit factor in the deposit system soon came to have a much wider scope than is here indicated. Instead of each depositor going to the bank and drawing his money as he needs it, he now gives an order or *check* on the bank to any man to whom he may have occasion to make a payment. In many cases the receiver of such a check also has deposits at the same bank. In such a case he sends in the check to be deposited with his cash for the day. The amount is debited to the drawer of the check, and credited to the depositor of it, and thus by a simple *transfer of credit* much business is done without the intervention of any money. This expands into a great and complicated system of exchange between individuals doing business at different banks, by banks in different cities, and by traders in remote nations. Goods are sold in one locality and paid for in the goods of another

locality by means of drafts, bills of exchange, etc., meeting and canceling one another, so that very little money is transferred from point to point.

The function of *discount and loan*, as has been intimated, is in modern banking usually combined with that of *deposit*, as also that of *circulation or issue*. When the capital of a bank is paid in by the stockholders, and the officers elected, it is then ready for business under regulations imposed by its charter. There are two ways in which the public is accommodated. First, when a wholesale city merchant sells a bill of goods to a country retail merchant, it is frequently the case that the former makes out his bill, which the latter accepts, promising to pay in thirty, sixty or ninety days. This accepted bill the wholesale merchant carries to his bank, where it is received with his endorsement, and the cash, less the interest for the given time, is paid him or placed to his credit. This is *discounting* a bill. A loan is sometimes made by a borrower's giving his own note endorsed by some reliable person, and payable in some brief time as above. Sometimes the note is discounted; at other times the interest is paid when the note is taken up.

The function of *circulation* is exercised by the issuing of bank-notes to be circulated as money. When a bank is instituted the stockholders are required to pay in their respective shares in metallic or lawful money. But as the borrower would find coin most inconvenient to carry about, the device arose of substituting notes of the bank, payable on demand, thus leaving the specie in the bank. It was further soon observed that only a very small proportion of these notes were likely to be called for at any one time. Hence a large part of the specie could be used for other purposes instead of being kept idle in the vaults. Under the national bank system now in operation the capital of the bank may be largely invested in United States bonds which are retained in the government treasury, but on which the bank draws the usual interest. The bills of the bank are then guaranteed by the government, so that there is never any loss to the holder of the bills, even if the bank fails.

PROTECTION AND FREE TRADE.

7. We have space but for a very brief outline of this important question. It is one which has for a long time agitated the public mind, and one on

which honest and highly intelligent men widely differ. A *protective tariff* so called, is a system of duties levied by the government of a country on certain commodities produced in other countries to prevent their coming into unequal competition with similar commodities of domestic production in such a way as to cripple or destroy the industries implied in the latter.

Free trade is opposed to all those duties, the design of which is to afford any advantage to domestic industry. It implies the same freedom between producers in different nations as between those in the same community.

The main arguments in favor of protection are as follows:

(1) It is the only sure defense of new and feeble industries against the unequal competition of those long established in other or older communities. Freedom of competition is admitted as desirable, but it is denied that this exists under the conditions referred to. A community which has long experience, skilled labor, and accumulated capital, possesses great advantages in the contest with a nation destitute of them.

(2) It is urged that a restrictive system gives a steady and uniform market at an expense less than the benefit accruing.

(3) It is also supposed to be essential to societal completeness; that is, to such a diversification of industry as will most profitably meet the diversity of ability and aptitude in the community.

(4) It is thought to be necessary to the highest prosperity of the unprotected interests. Among these agriculture is the most prominent. It is for its advantage that the tax of transportation be saved by having manufacturing communities in the midst of agricultural areas. Also, a community compelled to confine itself to agriculture mainly, must virtually transport its soil, the land constantly diminishing in fertility.

The advocates of free trade, on the other hand, present the following arguments in its favor, and objections against protection:

(1) Free trade is said to be the method of nature.

(2) It is objected that protection violates the right of every man to do what he will with his own.

(3) It is said to be of the nature of a tax on all the other industries for the support of those protected.

(4) It is objected that the restrictive system causes a diminution of exports from the protected country, on the principle that if the latter does not buy of the former, then the former can not pay for the goods of the latter.

(5) Another argument is that “infant industries” under protection never come to maturity.

(6) Finally, the case of the United States is cited as an instance of free trade on a large scale between widely remote sections, with the most satisfactory results.



READINGS IN ART.

III.—MODERN SCULPTURE.

The ten centuries following the second have no sculptural remains of value. The dark ages threw their shadow over art, as over literature and society. No doubt the feeling prevalent in the early Church that the “graven image” might become an idol, hindered the progress of the plastic art quite as much as the general decay that pervaded every form of human undertaking.

In the first half of the thirteenth century lived Nicola Pisano, the founder, one might say, of modern sculpture. Nicola is supposed to have been influenced by his study of the remains of Greek sculpture to be seen at Pisa, his home. Applying the principles of the Greek work to the modern subjects, his sculpture inaugurated the Italian renaissance. Church decoration was the field of labor to which all artists of those centuries betook themselves, and Pisano executed his best work, bas-reliefs, on the façades and pulpits of the churches of Pisa, Siena, and other Italian cities. A marble urn of St. Dominic, now at Bologna, is among his celebrated works. Pisano had many followers, among whom were his son (more famous, however, as an architect), and Andrea Orcagna. The latter belonged to Florence, to whose churches he devoted his genius. His masterpiece in sculpture is the tabernacle of the Virgin in the church of San Michele, at Florence. It is a pyramid-shaped altar in white marble; the profusion of reliefs which cover it represent the life of the Virgin. A little before the time of Orcagna lived Giotto, at one time a leader of artistic activity in Florence. He is known well by his beautiful campanile, or bell-tower, and the bas-reliefs with which it is decorated are his best-known sculptures. The basement story is decorated, and, says a writer, speaking of these ornamentations, “This rich cycle of works represents with perfect clearness, and in simple and truly artistic treatment, the whole progress, from the creation of the first man, through the successful conflict with the forces of

nature, up to the climax of a life illumined by learning and art, and secured under the maternal shelter of the Church.”

It was in the fifteenth century that sculpture attained its highest standpoint. Foremost among the artists of this “golden age,” as it has been called, is Lorenzo Ghiberti, the Florentine. The latter was first brought into prominence in 1401, when leading men of Florence offered a prize for the best design for a bronze folding door to be used in the baptistery of San Giovanni. Each artist was allowed a year to complete the test panel, the subject of the design of which was to be the “Sacrifice of Isaac,” and the work was to be a bas-relief. Ghiberti was declared the victor, even by his most famous rivals, Donatello and Brunelleschi. For twenty-one years he labored at his doors, and at the end of that time was entrusted with another. The latter occupied him nearly as long as the first, and was even superior, Michael Angelo declaring it worthy to be the gate of paradise. While busy at the gate of the baptistery, Ghiberti executed three bronze statues of St. John the Baptist, St. Matthew, and St. Stephen, and a bronze sarcophagus of St. Zenobius. Donatello has been mentioned as a rival of Ghiberti in the contest for the door: he deserves mention as one of the most faithful followers of nature during this period. He even carried his naturalism to excess, copying the deformed, the horrible, and the grotesque. There are, however, several fine statues by him in San Michele. Among these are the statues of St. Peter and St. Mark, in niches on the outside, and a fine statue of St. George. The first equestrian statue of modern art was by Donatello, and is at Padua.

Lucca del Robbia lived at the same time, and his name is associated with the beautiful terra-cottas found in such quantities in the churches of Florence. These works are in white, on a pale-blue ground, and were glazed by a process now unknown. The subjects used on them were almost invariably the Madonna and Child. But Robbia did much in marble and bronze. In the Uffizi is to be seen a frieze for the front of an organ, by him. “It represents boys and girls of different ages, dancing, singing, and playing on various musical instruments, and is full of charming simplicity and childlike grace, and rich and varied in action. Some of the figures are almost wholly detached from the background, particularly in the representation of the dance.” There are many more names which might be added to this Tuscan or Florentine school of sculpture. Andrea Verocchio is

the only one we will mention, and his strongest influence was exerted as the teacher of that master-artist of the sixteenth century, Leonardo da Vinci.

The works of the fifteenth century are very numerous; they crowd the churches of Rome, Florence, and the neighboring cities. Not only in Tuscany, but in Upper and Lower Italy these artists were employed, and many native artists, imitators of the school, have left sculptures on the tombs and in the churches of Venice, Naples, and Como. The subjects of artistic effort, it will be noticed, are nearly always religious. Lübke says of this period: "It was chiefly devoted to the ornamentation of tomb-monuments and altars, which, with few exceptions, were built up against the wall in the shape of a triumphal arch, and required much plastic decoration in the way of reliefs and detached figures. Pulpits, founts, holy-water basins, singing-galleries, and choir-screens were also adorned with rich carvings. This abundant supply of work necessarily called forth a corresponding amount of skill, and the nature of the subject helped the artistic and realistic taste of the time to express itself. There was a decided effort to attain a correct likeness in portrait-statues of the dead, and in the numerous reliefs there was a tendency to portray the varied scenes of life."

But a new form of plastic art was to appear in the coming century. To quote from the same author: "Italian plastic art had during the fifteenth century gained a new form from the study of the antique, and had made considerable advances in the unceasing effort after truth and life.... But hitherto, the expression of an often severe and tasteless realism was predominant, and now, under the influence of a profound and repeated study of the antique, an inspiration toward the ideal, the beautiful, and the sublime, was to assert itself; and this gave rise to a higher and freer style.... Plastic art gained a freer and nobler comprehension, a broad, bold treatment of forms, and a style simplified so as to bring out what was fundamental and essential, which might, for a moment, compete with the antique." Leonardo da Vinci was one of the first in the list of masters of the fifteenth century, but, unfortunately, we have lost his best work. Andrea Contucci, better known as Sansovino, executed many sculptures which are unparalleled in beauty of treatment and form. In the baptistery at Florence is one of the noblest of these—the baptism of Christ. The figures of John the Baptist and Christ are life-like, free, and perfectly developed. There is nothing more interesting among what Sansovino has left than the

decorations of the Holy House of Loreto. "Taken as a whole, this work is probably the most important collective creation in the sculpture of this golden age." There are a great number of reliefs employed in the ornamentation, and the niches are filled by single statues; of the former the Annunciation and the Nativity are the most important.

But by far the ablest of the sculptors was Michael Angelo Buonarroti, of Florence. It was as a sculptor that he chose to regard himself, although, as in the case of so many of the Italian artists, he was both a painter and architect beside. Numerous works attributed to him are in existence. Mythological subjects, as well as religious, are to be seen among them. Thus there are bas-reliefs at Florence representing Hercules in his contest with the centaurs, and a statue of Bacchus in the Uffizi. The colossal marble statue of David in the academy at Florence, is said to have been carved out of a rejected block. The most ambitious undertaking of Michael Angelo was the mausoleum of Pope Julius II. The designs were drawn on a grand scale, and the master had gone to Carrara to get out the marble, when a misunderstanding between him and the Pope stopped the work. It was afterward re-attempted, but never finished. Some of the detached figures intended for the tomb are still seen. Among them the famous Moses, in the church of San Pietro, at Vincolo. Two groups at Florence were executed for the sarcophagi of Giuliano and Lorenzo de Medici. The statues of the princes are seated in niches in the wall: at their feet, on the lids of the coffins, are the groups: on that of the former the design is Day and Night; on the latter Dawn and Evening. We can mention no more of his designs, but will add the fine criticism of a German critic: "If we compare Michael Angelo with those who went before, we see at once that art reached one of those turning-points at which it enters on a new period with an undreamed-of future opening before. His deeply emotional soul was content neither with the contemplative realism of the fifteenth century, which was based on its truth to nature, nor with the quiet, harmonious beauty of contemporaneous masters. Each of his works exists for its own sake only, and here we see a kinship with the antique. But again: each of them is also the product of the stormy inward struggles of a man who is ever aiming at the highest ideal, and untiringly striving after a new expression of his thoughts—a man to whom achievement gave but little satisfaction, so that often he left his works unfinished. Here we see the strongest contrast to

antique art. Nearly all his sculptured works are in one sense or another incomplete, and many he had to drop, because under the mighty stress of his ideas, and in his eagerness to liberate from the marble the slumbering soul within, he had made a false stroke and spoiled the block.”

The influence of Michael Angelo was predominant. The productions of almost every sculptor of the times were marked by both his strong and weak points. The Michelangesque manner, as it has been called, was evident in the sculptures of the following century.

Outside of this Tuscan school there were during the sixteenth century several prominent artists; at Modena, Antonio Begarelli, who worked mainly in terra-cotta, and who left many works in the churches of his native city.

At Padua lived Riccio, who executed a bronze candelabrum which has become famous for both its size and its excessive ornamentation. It was eleven feet in height and laden with innumerable fantastic reliefs and figures mostly taken from mythology. A pupil of Sansovino, Jacopo Tatti, was the leader in Upper Italy. He worked mainly at Venice. The bronze of the sacristy of St. Mark in that city, the choir-screen in the same church, and several figures of evangelists in bronze are among his religious works. In the Doge's palace are two large statues of Mars and Neptune which are particularly fine. He also did portrait-sculptures of much merit. But during this century art was by no means confined to Italy, though Italy then, as always, took the lead. In the North there was a steady work in the plastic art. The influence of the antique was wanting, and the materials in which the works were executed were different. Wood carving was very popular; invariably much gilding and brilliant coloring was used. The work was mainly on the altars of the churches, on shrines, figures for niches in the church walls and choir stalls. Michael Pacher, of Austria, was eminent in this art; Veit Stoss, of Cracow, and Jörg Syrlin, of Ulm. In nearly all of the old churches of Germany are these highly colored carvings in wood.

But stone was used as extensively, and in a somewhat wider variety of works. Many monuments, the buttresses of churches, lecterns, doors, and choir-piers, were made in stone and decorated in the usual manner by reliefs

and figures. Nearly all the German cities boast more or less of stone work in their churches.

The leading artist of the time was Adam Krafft, who worked mainly in Nuremberg. A very fine and powerful work by him is the Seven Stations, as it is called. It represents the repeated fainting of Christ beneath the burden of the cross. The work is done in relief. The face and expression of the Savior is noble and expressive in every case. This work was followed by Christ on the Cross. In 1492 he executed the history of the Passion for a monument on the exterior of St. Sebald's church.

The monuments of the time are mainly very superior. Among them may be mentioned that of Emperor Henry II. and his consort by Riemenschneider, the marble monument of Bishop Rudolph von Schrenburg in the Würzburg cathedral, and the marble memorial to the Emperor Frederic III. in Vienna. The celebrated school of metal works of Nuremberg flourished during this period. The best known representatives belonged to the family of Vischer, and in Peter Vischer the most complete artistic development was reached. The earliest work, by Hermann Vischer in 1457, was the bronze baptismal font in Wittenberg. Peter, his son, began his work on the tomb of Archbishop Ernst in Magdeburg cathedral, but his *chef d'œuvre* was the tomb of St. Sebald in the church of that saint at Nuremberg. Vischer and his five sons were engaged on this for eleven years. The sarcophagus rests on a base elaborately wrought in relief, and the whole is enclosed; the cover is composed of three arched canopies supported on eight slender columns. The base, pillars and canopies are wrought exquisitely; although the ornaments are profuse, yet a perfect simplicity and purity of style is preserved. There are very many other productions attributed to Vischer—a fine relief in the cathedral at Regensburg, several tombs, and, as examples of his treatment of antique designs, an Apollo at Nuremberg, and a relieve of Orpheus and Eurydice in the Berlin Museum.

One of the most magnificent tombs of this period was that of the Emperor Maximilian at Innsbrück; several of its figures were from Peter Vischer's hands. Twenty-eight colossal bronze statues of the ancestors of the imperial house and of heroes surrounded the monument. Besides these there were a large number of gracefully poised female figures, and twenty-

three figures of the patron saint of the House of Austria. The whole was surmounted by a marble cenotaph on which a figure of the Emperor knelt. Several artists were engaged on this monument. The sculptures of this period in other countries are not very prominent. In France there was considerable attention given to plastic art. Many fine choir-screens have been preserved, and some exceedingly rich tombs. Among the latter are the monuments of Louis XII. and his wife (1530), of Francis I. (1552), and of Henry II. (1583), all in the church of St. Denis in Paris. A set of artists who were engaged on the decorations of the palace of Fontainebleau was known as "the Fontainebleau school." The leader of this group was Jean Goujon. The sculpture of Spain during this period followed largely the Italian schools. The most lavish treatment is visible in the decorations of the churches, particularly in the altars. The high altar of the cathedral at Toledo is one of the most costly and ornate of its time (about 1500).

"The seventeenth and eighteenth centuries were marked by a decadence of sculpture. Plastic art sought to become striking, rejected everything that could limit her art and gave herself up freely to her longing after what was striking. Henceforth it was decreed that every plastic work must be spirited. The most striking effects must be aimed at in the expression of inward emotion through mien, attitude and position.... Besides the drapery must be arranged in all sorts of ways conducive to effect.... Thus all dignity, simplicity and distinctness in sculpture, all plastic style was lost, and was succeeded by a senseless striving after outward effect and mere decoration." The best Italian artists of these years were Lorenzo Bernini (1598-1680), who showed well the perversion of the principles of art, and Alessandro Algardi. The French claimed as their most celebrated masters in the seventeenth century, Pierre Puget, who worked chiefly at Genoa, and François Girardon, both of whom are noted for their exaggerations; in the eighteenth century were Houdon and Pigalle.

Franz Duquesnoy, the Fleming, worked at Rome in the seventeenth century and gained a fine reputation by his life-like figures of children. In Berlin, Andrew Schlüter executed superior works. Among these are the masks of dying warriors carved above the windows of the court of the Arsenal. An equestrian statue of the Great Elector is his best work.

In the latter half of the eighteenth century a revival of sculpture took place; this has been attributed to the efforts of Popes Clement XIV. and Pius VI., to the publications of Winckelmann, and to the unearthing of the treasures of Pompeii and Herculaneum. The first sculptor to initiate works of purer taste was Canova (1757-1822); he came of a race of stone cutters, and while at work at his trade executed the figures which attracted the attention of a Venetian, who educated him for an artist. Canova's early works were mythological in subject. He had studied sculptures unearthed at Pompeii and Herculaneum, and under their influence executed his "Apollo crowning himself with laurel" and "Theseus vanquishing the Minotaur." In 1802 Canova was invited by Napoleon to Paris where he executed a colossal statue of the emperor. His figures of women were his most pleasing works. Of the many monuments he executed, the best is that of Christina in the church of the Augustines at Vienna. But few artists escaped the influence of Canova. Among his best known followers were Dannecker, of Stuttgart; Chaudet, a French artist, and Flaxman, an English sculptor.

For a brief outline of the sculptor of the nineteenth century we can do nothing better than quote from Lübke:

The Danish artist, Bertel Thorvaldsen (1770-1844), penetrated farther than all these masters into the spirit and the beauty of classical art; and created, with inexhaustible fertility of imagination, and with the noblest feeling for form, an array of works which are conceived with a pure, chaste, and noble appreciation of the Greek spirit. In his celebrated frieze of the triumph of Alexander in the Villa Carlotta, on the lake of Como, the genuine Grecian relief style is revived in all its perfect purity and severity. He also treats with the versatility of genius and with charming simplicity the subjects of ancient mythology, in numerous statues, groups, and smaller reliefs; and even introduces into the domain of Christian representation a novel, beautiful, and dignified treatment, in the sculptures executed by him for the Church of Our Lady in Copenhagen. Among his monumental works we may mention the statues of Gutenberg at Mayence, and of Schiller at Stuttgart, the Dying Lion at Lucerne, the equestrian statue of the Elector Maximilian at Munich, and the tombs of the Duke of Leuchtenberg in St. Michael's Church at Munich, and of Pope Pius VII. in St. Peter's Church at Rome.

While the wide domain of idealistic sculpture was thus again cultivated with such versatility of inspiration, the Berlin artist, Johann Gottfried Schadow (1764-1850), adopted a more realistic style, especially directed toward lifelike composition and distinct characterization of individual peculiarities. His monument of the Count von der Mark in the Church of Ste. Dorothy in Berlin, the statue of Frederic the Great at Stettin, and, in a less degree, the Blücher monument at Rostock, and that of Luther at Wittenberg, as well as many others, are vigorous protests against the mannerism of the hitherto prevailing tendency, and re-open to sculpture a field which had now been almost lost to her for two hundred years.

Thus a new path was opened to modern sculpture, in pursuing which it has of late years accomplished great results, and which assures to it still greater beauty, and diversity of attainment, if only it hold fast to the principles already secured, and go on with true dignity toward its goal. Even if the world of ideal forms should never again acquire that importance for us which it possessed for the Greeks, nevertheless the daily life of humanity still contains a wealth of exquisite motives, full of beauty and *naïveté*, which give to the sculptor's fancy ample incitement to ideal creations. There is, moreover, in the chaste grace and pure dignity of the antique conceptions, an imperishable charm, which appeals to every human sentiment, and secures for all productions conceived in a similar spirit the warm interest of those who delight to refresh themselves with the simple beauty that belongs to every true manifestation of nature. Hence the idealistic style of this art of Greece, as it has been recognized by the present and endowed with new activity, becomes forever the most priceless and precious possession of modern sculpture.

The new-born historic feeling of the several nations demands to-day that their heroes, the defenders of their liberties, the representatives of their intellect, their warriors in the battles both of the sword and of thought, shall be preserved to fame in the true likeness of their actual forms. As a consequence, sculpture is compelled to probe the depths of the individual consciousness; to investigate the characteristics of each individual intellect as expressed in the figure, the physiognomy, and even in the externals of attitude and garb; and even to give utterance to the mysterious life of the soul, as far as it lies within her power. Without losing sight of the great importance which the study of the sculptures of the fifteenth century has

upon this tendency, the influence of the antique should not be undervalued; since, without the sense of beauty so secured, a realistic degeneracy and exaggeration would be very sure to follow.

Among the German schools of sculpture of to-day, that of Berlin takes the lead. Frederick Tieck of this school adopted the antique style in a series of admirable productions, and especially in the decorative sculpture designed by him for the theater; while the path which Schadow had taken was followed up nobly and rationally during the long and influential labors of Christian Rauch (1777-1857). This artist's important position is due less to his wealth of creative ideas than to his delicate feeling for nature, his fine appreciation of the genuine plastic style, and his incomparable care in execution. His importance, however, does not consist merely in his numerous works, but also in the influence he exercised on his large circle of talented scholars. While he shows a true classical beauty in his ideal works, like his victories and his many admirable reliefs, his statues of Prince Blücher, of Generals Bülow and Scharnhorst, his colossal equestrian statue of Frederic the Great at Berlin, his superb statues of Queen Louise, and of Frederic William III. in the mausoleum at Charlottenburg, his bronze statues of Dürer at Nuremberg, of Kant at Königsberg, of King Max I. at Munich, and many others, prove him a sculptor of the first rank for delicate characterization, and life-like suggestiveness of composition. Many excellent scholars have gone from his studio into careers of independent importance and masterly ability; and these form, with their vigorous activity, which is never at a loss for employment in important undertakings, the nucleus of the present school of Berlin.

Among the most conspicuous of the Berlin artists should be reckoned Friedrich Drake, whose reliefs on the statue of Frederic William III. in the Thiergarten at Berlin are full of simple grace. Another of this school is Schievelbein (died in 1867), who showed a great deal of imagination, especially in the composition of reliefs; as in the great frieze representing the destruction of Pompeii, in the new museum, and also in the relief on the bridge at Dirschau.

Ernst Rietschel (1804-61) claims indisputably one of the first places among the sculptors of his century, as regards versatility of endowment, delicate feeling for form, and depth of sentiment. He derived from Rauch

his faithful and characteristic representation of life, and his painstaking execution. His double monument of Schiller and Goethe at Weimar, his monument of Lessing in Brunswick (in a still purer and happier style), and the statue of Luther executed for a monument at Worms, are good examples of these traits. In the group of the Virgin with the body of Christ, which he executed for the Friedenskirche near Potsdam, he produced a work full of striking expression, and of the deepest religious feeling; while the subjects of his numerous representations in relief for the pediment of the opera house at Berlin, and the theater and museum at Dresden, represent him with equal dignity and merit in the department of the ideal antique subjects. Ernst Hähnel is a Dresden artist, whose powerful compositions for the Dresden theater and museum are antique in treatment, but who also produced monumental statues, works of the most delicate characterization, such as the Beethoven at Bonn, the Emperor Charles IV. at Prague, and the statues designed for the Dresden Museum, especially the noble Raphael. Recently, also, Schilling has distinguished himself by his ideal groups of the divisions of the day,—Morning, Noon, Evening, Night,—designed for the Brühl Terrace.

In Munich, the talented Ludwig Schwanthaler (1802-48) was the chief representative of a more romantic style, which opened a new field of fresh ideas to modern sculpture. This master, who was endowed with an almost inexhaustible imagination, carried out a great number of extensive works during his short life, in supplying the plastic decorations for most of the buildings erected by King Louis. While these are distinguished by fertility of invention, and an excellent decorative taste, the artist, spurred on to ceaseless labor, and hindered by bodily infirmities, did not succeed in giving his monumental creations that thorough development of form which is an essential of sculpture. It can not be denied, however, that a grand monumental conception is visible in these productions, as is especially proved in the colossal statue of Bavaria in Munich. A numerous school had its origin in this artist's studio.

In France, sculpture early endeavored to free herself from the rigid rule of the antique, and carried the prevailing effort after dramatic effect, expression and passion, even to an extreme point of realism. Individual artists have kept to a noble and more moderate style; as Bosio, and the admirable sculptors Rude and Duret; but, on the other hand, P. J. David



SELECTIONS FROM AMERICAN LITERATURE.

DR. HORACE BUSHNELL.

Dr. Bushnell's mind was one of the rarest. What it was in his books, that it was in private, with certain very piquant and unforgettable flavors added.—*Dr. Burton.*

I think he had no capacity, with all his eminent powers, for enmity. Goodness and wisdom were the powers that amounted to genius in him by being so great.—*Rev. C. A. Bartol.*

WRONG RESISTED.—As it is said that ferocious animals are disarmed by the eye of man, and will dare no violence if he but steadily look at them, so it is when right looks upon wrong. Resist the devil, and he will flee from you; offer him a bold front, and he runs away. He goes, it may be, uttering threats of rage; but yet he goes.

GREAT MEN.—The great and successful men of history, are, commonly, made such by the great occasions they fill. They are the men who had faith to meet such occasions; and therefore the occasions marked them, called them to come and be what the successes of their faith would make them. The boy is but a shepherd, but he hears from his panic-stricken countrymen of the giant champion of their enemies. A fire seizes him, and he goes down to the army, with nothing but his sling, and his heart of faith, to lay that champion in the dust. Next he is a great military leader, then the king of his country. As with David, so with Nehemiah; as with him, so with Paul, and Luther. A Socrates, a Tully, a Cromwell, a Washington—all the great master-spirits—the founders and law-givers of empires, and defenders of the rights of men, are made by the same law. These did not shrink despairingly within the compass of their poor abilities, but in their heart of faith embraced each one his cause, and went forth under the inspiring force of their call to apprehend that for which they were apprehended.

FAMILY RELIGION—WHY A FAILURE.—The father prays, in the morning, that his children may grow up in the Lord, and calls it the principal good of their life, that they are to be Christians, living to God and for the world to come. Then he goes out into the field, or shop, or house of trade, and, delving there all day in his gains, keeps praying from morning to night, without knowing it, that his family may be rich. His plans and works, faithfully seconded by an affectionate wife, pull exactly contrary to the pull of his prayers, and to all their common teaching in religion. Their tempers are worldly, and make a worldly atmosphere in the home. Pride, the ambition of show, and social standing, envy to what is above, and jealousy of what is below, follies of dress and fashion, and the more foolish elation, when a son is praised, or a daughter admired in the matter of personal appearance, or, what is no better, a manifest preparing and foretasting of this folly, when the son or daughter is so young as to be more certainly poisoned by the infection of it. Oh, these unspoken, damning prayers! how many they are, and how they fill up all the days! The mornings open with a reverent, fervent-sounding prayer of words; and then the days come after piling up petitions of ends, aims, tempers, passions and works, that ask for anything and everything but what accords with genuine religion. The prayer of the morning is that the son, the daughter—all the sons and daughters—may be Christians; and then the prayers that follow are for anything but that—in fact, for things most contrary to that. Is it any wonder, when we consider this common disagreement between the prayers of the family, and all other concerns, ends, and enjoyments of the common life beside, that so many fine shows of family piety are yet followed by so much of godless, and even reprobate, character in the children?

DR. NOAH PORTER.

How to Read History.

Whately pertinently observes, in his annotations upon Lord Bacon's "Essay on Studies:" "In reference to the study of history I have elsewhere remarked upon the importance, among the intellectual qualifications for such a study, of a vivid imagination. The practical importance of such an

exercise of imagination to a full and clear, and consequently, profitable view of the transactions related in history can hardly be over-estimated.”

To stimulate and aid the imagination in its efforts to reproduce the past, historical plays and poems, and, more recently, historical novels have been abundantly employed. Their usefulness has been the subject of frequent discussion, and of various opinions. It has been forcibly, and perhaps not untruly said, that the majority of the present generation of English readers have learned more of English history from Shakspeare and Walter Scott than from the entire library of professed historians. Of course no man would contend that either Shakspeare or Scott could be substituted for the usual historical authorities, but only that they may supplement them in certain important particulars. Many other historical plays and novels are invaluable as enabling the reader to enter more fully into the spirit of past times. They are of especial service in helping him to appreciate the feelings and motives of prominent personages, and vividly to reproduce the manners and institutions of another age. It is not often that an historical writer is endowed with the painstaking zeal of the antiquarian, and the creative power of the poet. If we can not have the two gifts in a single writer, we must seek for them apart in the historian and the novelist.

Thackeray’s “Henry Esmond” is an admirable example of a good historical novel, when carefully and conscientiously written by a man of rare gifts and of a rarer honesty. No reader of this tale of the times of Queen Anne could fail to derive from it such impressions of the state of manners and of morals in the higher circles, as well as of the political jealousies and the religious feuds which divided men of all classes, as no formal history could possibly convey—such as even the most abundant and painstaking research into the less accessible resources of historical knowledge would fail to impart to a man of feeble capacity to picture and recombine. The service is not a slight one which is rendered to the world when such a painstaking explorer of historical truth as Thackeray gathers his materials with faithful and laborious research, and weaves them together into so fascinating and instructive a story. But this tale, marvelous as it is for its elaborated truthfulness and picturesque effects, strikingly illustrates the possible dangers and disadvantages to which the historical novel may be abused. Thackeray was not without his prejudices. These, with his desire for producing striking effects, are manifest in the occasional *overdrawing* of

this generally well-balanced representation of one of the most interesting periods of English history. It is notorious that Walter Scott gave very serious offense to multitudes of his admiring readers by some of his portraiture of the representative characters of the great historical parties of Scotland and England. With all the good sense and candor which he had at command, his sympathies were too intense and his prejudices too tenacious to allow him to write otherwise than he did, though he knew he should excite the indignation of thousands of his fervid countrymen. Mrs. H. B. Stowe says in the preface to her recent historical romance, "Oldtown Folks:" "I have tried to make my mind as still as a looking-glass or a mountain lake, and thus to give you merely the images reflected therein." But a fervid and sympathetic nature like hers can no more free itself from a theological or personal bias in representing the New England of the past, over which she has laughed, and wept, and speculated, and struggled all her life, than the "mountain lake" can hold itself in glassy smoothness against the gusts and breezes that sweep upon it from the heights above.

The fact deserves notice that of late professed historians have indulged somewhat freely in romancing, and so in a sense turned their histories into quasi-historical novels, especially when they attempt to give elaborate and eloquent portraiture of the leading personages, in which the most lavish use is made of effective epithets and pointed antitheses. Macaulay, among recent historians, has set the fashion very decidedly in this direction. In his efforts to make history minute, vivid, and effective, he has often described like an impassioned advocate, and painted, like a retained attorney, with the most unsparing expenditure of contrasts and epithets. Carlyle gives sketches, alternately in chalk and charcoal, that exhibit his saints and demons, now in ghastliest white, and then in the most appalling blackness. But though he draws caricatures he draws them with the hand of an artist. Froude, by research, eloquence and audacity combined, attempts to reverse the settled historic judgments of all mankind in respect to characters that had been "damned to everlasting fame." Bancroft and Motley abound in examples of this tendency to paint historical characters so much to the life that the impression is made that the result is only a painting to which there never was reality.

WASHINGTON IRVING.

To a true poet-heart add the fun of Dick Steele—
Throw in all of Addison, *minus* the chill,
With the whole of that partnership's stock and good will,
Mix well, and while stirring him o'er as a spell,
The fine old English gentleman, simmer it well,
Sweeten just to your own private liking, then strain,
That only the finest and clearest remain;
Let it stand out of doors till a soul it receives,
From the warm, lazy sun loitering down through green leaves.
And you'll find a choice nature, not wholly deserving
A name either English or Yankee—just Irving.—*Lowell.*

... Washington Irving, one of the best and pleasantest acquaintances I have made this many a day.—*Sir Walter Scott.*

The Style of Mr. Irving is always pleasing.—*Macaulay.*

Throughout his polished pages no thought shocks by its extravagance, no word offends by vulgarity or affectation.—
Edinburgh Review.

A Rainy Sunday in an Inn.

It was a rainy Sunday in the gloomy month of November. I had been detained in the course of a journey by a slight indisposition, from which I was recovering; but I was still feverish, and was obliged to keep within doors all day, in an inn of the small town of Derby. A wet Sunday in a country inn; whoever has had the luck to experience one, can alone judge of my situation. The rain pattered against the casements, the bells tolled for church with a melancholy sound. I went to the windows in quest of something to amuse the eye, but it seemed as if I had been placed completely out of the reach of all amusement. The windows of my bedroom looked out among tiled roofs and stacks of chimneys, while those of my sitting-room commanded a full view of the stable-yard. I know of

nothing more calculated to make a man sick of this world than a stable-yard on a rainy day. The place was littered with wet straw that had been kicked about by travelers and stable-boys. In one corner was a stagnant pool of water surrounding an island of muck; there were several half-drowned fowls crowded together under a cart, among which was a miserable crest-fallen cock, drenched out of all life and spirit, his drooping tail matted, as it were, into a single feather, along which the water trickled from his back; near the cart was a half-dozing cow, chewing the cud, and standing patiently to be rained on, with wreaths of vapor rising from her reeking hide; a wall-eyed horse, tired of the loneliness of the stable, was poking his spectral head out of a window, with the rain dripping on it from the eaves; an unhappy cur, chained to a dog-house hard by, uttered something every now and then between a bark and a yelp; a drab of a kitchen wench tramped backward and forward through the yards in pattens, looking as sulky as the weather itself; everything, in short, was comfortless and forlorn, excepting a crew of hard-drinking ducks, assembled like boon companions round a puddle, and making a riotous noise over their liquor.

I sauntered to the window, and stood gazing at the people picking their way to church, with petticoats hoisted mid-leg high, and dripping umbrellas. The bells ceased to toll, and the streets became silent. I then amused myself with watching the daughters of a tradesman opposite, who, being confined to the house for fear of wetting their Sunday finery, played off their charms at the front windows, to fascinate the chance tenants of the inn. They at length were summoned away by a vigilant vinegar-faced mother, and I had nothing further without to amuse me.

The day continued lowering and gloomy; the slovenly, ragged, spongy clouds drifted heavily along; there was no variety even in the rain; it was one dull, continued, monotonous patter, patter, patter, excepting that now and then I was enlivened by the idea of a brisk shower, from the rattling of the drops upon a passing umbrella. It was quite refreshing (if I may be allowed a hackneyed phrase of the day) when in the course of the morning a horn blew, and a stage-coach whirled through the street with outside passengers stuck all over it, cowering under cotton umbrellas, and seethed together, and reeking with the steams of wet box-coats and upper benjamins. The sound brought out from their lurking-places a crew of vagabond boys and vagabond dogs, and the carrot-headed hostler, and that

nondescript animal yclept Boots, and all the other vagabond race that infest the purlieu of an inn; but the bustle was transient: the coach again whirled on its way; and boy and dog, and hostler and Boots, all slunk back again to their holes; the street again became silent, and the rain continued to rain on.

The evening gradually wore away. The travelers read the papers two or three times over. Some drew round the fire and told long stories about their horses, about their adventures, their overturns and breakings-down. They discussed the credits of different merchants and different inns, and the two wags told several choice anecdotes of pretty chambermaids and kind landladies. All this passed as they were quietly taking what they called their nightcaps; that is to say, strong glasses of brandy and water or sugar, or some other mixture of the kind; after which they one after another rang for Boots and the chambermaid, and walked off to bed in old shoes cut down into marvelously uncomfortable slippers. There was only one man left,—a short-legged, long-bodied plethoric fellow, with a very large sandy head. He sat by himself with a glass of port wine negus and a spoon, sipping and stirring, and meditating and sipping, until nothing was left but the spoon. He gradually fell asleep bolt upright in his chair, with the empty glass standing before him; and the candle seemed to fall asleep too, for the wick grew long and black, and cabbaged at the end, and dimmed the little light that remained in the chamber. The gloom that now prevailed was contagious. Around hung the shapeless and almost spectral box-coats of departed travelers, long since buried in deep sleep. I only heard the ticking of the clock, with the deep-drawn breathings of the sleeping toper, and the drippings of the rain—drop, drop, drop—from the eaves of the house.

Irving's Last Interview with Scott.

It was at Sunnyside, on a glorious afternoon in June, 1855, that surrounded by scenery which Irving has best described, he narrated to me (S. Austin Allibone) the following account of his last interview with Scott:

“I was in London when Scott arrived after his attack of paralysis, on his way to the continent in search of health. I received a note from Lockhart, begging me to come and take dinner with Scott and himself the next day. When I entered the room Scott grasped my hand, and looked me steadfastly in the face. ‘Time has dealt gently with you, my friend, since we parted,’ he

exclaimed:—he referred to the difference in himself since we had met. At dinner, I could see that Scott's mind was failing. He was painfully conscious of it himself. He would talk with much animation, and we would listen with the most respectful attention; but there was an effort and an embarrassment in his manner; he knew all was not right. It was very distressing, and we (Irving, Lockhart, and Anne Scott) tried to keep up the conversation between ourselves, that Sir Walter might talk as little as possible. After dinner he took my arm to walk up-stairs, which he did with difficulty. He turned and looked in my face, and said, 'They need not tell a man his mind is not affected when his body is as much impaired as mine.' This was my last interview with Scott. I heard afterward that he was better; but I never saw him again."

Two years later (in 1857), in narrating the same event, Irving told me that as Scott passed up the stairs with him after dinner, he remarked, "Times are sadly changed since we walked up the Eildon hills together."

JAMES KIRKE PAULDING.

There is no better literary *manner* than the manner of Mr. Paulding. Certainly no American, and possibly no living writer of England has more of those numerous peculiarities which go to the formation of a happy style.—*Edgar A. Poe.*

His works are exclusively and eminently natural, and his descriptions of natural scenery are often eminently beautiful.—*London Athenæum.*

TIME A DESTROYER.—I saw a temple, reared by the hands of man, standing with its high pinnacle in the distant plain. The streams beat about it; the God of nature hurled his thunderbolts against it; yet it stood firm as adamant. Revelry was in the halls; the gay, the young, the beautiful were there. I returned, and lo! the temple was no more. Its high walls lay scattered in ruin; moss and grass grew rankly there; and, at the midnight hour, the owl's long cry added to the solitude. The young, the gay, who had reveled there, had passed away. I saw a child rejoicing in his youth, the idol of his mother, and the pride of his father. I returned and the child had

become old. Trembling with the weight of years, he stood the last of his generation, a stranger amidst all the desolation around him. I saw an old oak standing in all its pride upon the mountain; the birds were caroling in its boughs. I returned and saw the oak was leafless and sapless; the winds were playing at their pastime through the branches. "Who is the destroyer?" said I to my guardian angel. "It is Time," said he. When the morning stars sang together for joy over the new-made world, he commenced his course, and when he has destroyed all that is beautiful on the earth, plucked the sun from his sphere, veiled the moon in blood; yea, when he shall have rolled the heavens and the earth away as a scroll, then shall an angel from the throne of God come forth, and, with one foot upon the land, lift up his hand toward heaven, and swear by heaven's eternal, "time was, but time shall be no more."

[End of Required Reading for December.]



RETURNING.

By MARY HARRISON.

“The spirit shall return to the God who gave it.”

White clouds upon heaven's bosom rest,
Begotten of the sunshine's love,
Now nestled like a fondled dove
Upon a woman's loving breast.

Heaven feeds her baby clouds, they grow,
Then leave her for their manhood's life;
And wail and scramble in the strife
Through which all earth-born children go.

They sink and wander in the gloom
Of winding subterranean ways,
And learn the loss of heavenlier days,
By groping through their chosen tomb.

At length, lights gleam along the distant way,
With eager thoughts of childhood, blest,
And hopes of entering into rest,
They leap to airy, sunny day.

Now rivers slave them to the fields
To fill the cattle-troughs with drink,
And dress the rose-boughs on their brink,
And feed the grass the meadow yields.

For friends and good, they look behind,
Then curse the past, and pray to be
Unborn again within the sea,
For birth has been to them unkind.

All scenes have gone! no good has come!
From bank to bank the waters heave
With tides which only mock and grieve,
Despairs of long-lost, hopeless home.

And looking but for lulling sleep,
The last deep solace of the grave,
They lean to meet the leaning wave

They leap to meet the reaping wave,
And find their lost home in the deep.

So through his day, blind man has striven,
As vapor-clouds, he came to be,
Drawn from, then wandering to the sea,
Invisible, with God in heaven.



EDUCATION OF THE NEGRO POPULATION.

By ATTICUS G. HAYGOOD, D.D.

FACTS ON THE SURFACE.

The records in the Department of Education, in Washington City, show that in the recent slave States of the Union the total school population was, in 1881, 5,814,261. Of these, 3,973,676 were white; 1,840,585 colored children. Counting both races the total school enrollment for 1881 was 3,034,896; of these 2,232,337 were white, 802,559 colored children. Nearly half the white, and more than half the colored school population was, in 1881, out of school. In some of these States the school term is from three to five months; in the cotton States not more than three. Perhaps five months each year is as long a school term as the conditions and needs of the laboring classes in these States will allow.

In 1881 these States expended upon their public schools \$13,359,784; except perhaps in one state this money was expended without distinction of race. The races have schools of their own; doctrinaires would mix them by force of law; those who are actually doing the work of education in these States know that this can not be done, and that only harm would come of it, if the experiment were attempted. For neither race would do so well if taught together; the colored children do not desire mixed schools, and the white children will not attend them. In such conditions law is helpless, and force is folly; also ruin.

OTHER FACTS.

The official figures give the numbers; parole evidence is necessary to complete the statement of the case. In 1881 there were, as the Department of Education reports, in the Southern States 17,248 common schools for colored children. With exceptions so few that they are inappreciable in

these statements, the teachers in these 17,248 common schools were colored—the large majority being women. The majority of these teachers are pitifully incompetent; some of them are well furnished for their work, and are doing it faithfully and successfully. Nearly all of these colored teachers who are of any use have received their preparation in the various schools for higher instruction established by societies and churches in the Northern States. Some of the best work is done in schools established and carried on by individual devotion—I will not say enterprise. Taking them all together there are nearly one hundred and fifty of these schools, called, as fancy or circumstances prompted or allowed, universities, colleges, institutes, seminaries, normal schools, etc., etc. There is hardly an “academy” among them.

OVER-NAMING.

Many will think me wrong in the opinion I now offer; some of the wisest of the teachers in the real work of teaching negroes will agree with me: it is a misfortune that the names given these schools are so out of proportion to their real work and character. None of them, even in catalogues, go beyond the ordinary college course; many of them do not come up to it; in none of them do more than a very small number complete this course. There is not a university, in any proper sense, among them all. It is not in the spirit of censure that I speak of these things, but of deep interest in the great and necessary work, that the good people engaged in these schools are trying, with rare consecration and in the teeth of a thousand discouragements, to accomplish.

The great names for these schools have done harm. They are misleading to begin with, and that is an evil. It is hard enough to get the indifferent or the antagonistic people to understand the subject of the education of the negroes at best; it is harder when new meanings have to be given to old names in order to state acts. I am of the opinion that the names given to most of these schools have done some harm in the North—whence the money has been drawn to support them. Northern men have sometimes spoken to me on these subjects in language that made it plain that they would have helped more but from a conviction that “schools and not universities are what these poor people need.” *Per contra*, it may well

enough be answered, some have given largely to build “universities” that would not give to establish schools. As to the influence on northern sentiment of the *too-great names*, those who know that sentiment better than I do can express themselves more definitely. I know that the big names have done harm in the States where the schools are. At this point let me say, I am only stating what I believe to be facts. Comments, inferences, justifications, do not concern me just now.

First, then, the large names have excited prejudice among the white people who did not know what was back of the names. Most of them, for a long time, did not know what the universities and colleges were really trying to do; the majority do not know at this time.

Some of those who did know something thought the whole business a mere sham; for a long time only a few southern white people really knew that faithful, wise and successful teaching was done in these colleges and universities—most of it not being college or university work at all. The few who really knew what good work was being done could over-look the ambitious names—it being a weakness in the South and West, yielded to by not a few, to give great names to small schools for white youth. The wiser and kinder-hearted ones could condone the offense of over-large names in view of their own example.

The big names did as much as anything else to anger the poor whites against all negro education. People who know human nature will understand this statement without explanation: those who do not know human nature will not understand it anyway.

The worst evil, in the long run, of this big naming of schools for the negroes, fell upon the negroes themselves. It aggravated the tendency—very strong among them—to be satisfied with the name of a thing in the lack of the thing itself, and, what is more, not knowing that they can lack the thing when they have the name. Take, for example, “University,” an admirable school well known to me. Its annual enrollment will average three hundred; its catalogue course reaches from the primary studies through an ordinary college curriculum; one in ten attempts this college course; one in fifty may complete it. The whole three hundred tell their friends: “I was educated in University.” It gives them importance. They

pass as scholars beyond their merits among their own people. In many of them it breeds injurious conceits—of a sort that makes enemies of those who might be friends, and prejudices with the uninformed—who in all countries are the majority—the whole subject of negro education. It is to be feared that only a few colored students know the difference between “University” and a real university.

NO SHAM IN THE WORK DONE.

Let me say with emphasis at this point: there is no sham in the work done in these schools. It is genuine, honest, useful work. This is a general statement; there may be, doubtless are, some schools that do not deserve this praise. But the point I wish to make plain to the readers of THE CHAUTAUQUAN is this: if there be sham it is not in the work done, but in the name given the place where it is done. I asked one of the veterans: “Why did you call this school a university?” He answered: “We hoped it would grow to it some day.” How could I blame the hopefulness of those who did the naming? So many of our white schools had been named under the same sort of prophetic impulse.

TRAINING SCHOOLS.

It is those schools backed by the churches and benevolent societies of the North that are doing the most of the work of preparing teachers among the colored people for the colored people. The very best of the more than seventeen thousand colored teachers have learned whatever they know in these schools. Most of the Southern State governments have recognized the necessity of preparing colored teachers, and make annual appropriations to carry on this work. A few States have established schools of their own; generally they make appropriations to some of the best of the schools established by others.

The great and crying need in the work of education among the people is better teachers in their common schools. They can not be prepared in a day or a year; for it takes much money and much time. The training schools are without endowments, and their patrons are unable to pay more than the lowest tuition fees. If these schools—call them universities, colleges,

institutes, seminaries—what you will, are to keep going at their present rate, to say nothing of improvement, white people must furnish the money, for the best of reasons; the negroes have not money to do this sort of work. Most of this money will have to come from Northern pockets, if it comes at all. The State of New York is worth more in property returned for taxation than all the Southern States together—leaving out Missouri, counted in the census of 1880 among “Western States.”

THE JOHN F. SLATER FUND

Begins to do its blessed work. This fund is dedicated to the work of “Uplifting the lately emancipated population of the Southern States and their posterity, by conferring on them the blessings of Christian education,” and it seeks to accomplish this result by “the training of teachers among the people requiring to be taught.” This fund works through existing institutions; it does not found new schools; there are already more good and deserving schools than it can help. Many times the sum this fund affords could be wisely used.

There is not space in this article to discuss the question, but my opinion may be stated: It is necessary that the United States government should aid the States to make their public schools more efficient. Whatever may be true of other sections, the Southern States, owing to the facts of their history and to conditions now existing, are not able to do the work that is upon them.

As to the sentiment in these States on the subject of negro education, it may be said in brief: The outcry of small village papers does not always even reflect the sentiment of the people, and there are certain facts that indicate that the work of educating the negroes will go on with less and less hindrance. Three such facts I mention in closing this article: (1) The duty and necessity of educating the negro has been recognized by every representative church in the South. (2) This necessity is recognized in the educational system of every Southern State. (3) No man who believes he has any political or educational “future,” any longer opposes, under his proper name, the education of his negro fellow citizens.



Dress changes, but we are not to suppose on that account that the make of the body changes also. Politeness or rudeness, knowledge or ignorance, more or less of a certain degree of guilelessness and simplicity, a serious or playful humor; these are but the outer crust of a man, and may all change; but the heart changes not, and the whole of man is in the heart. One age is ignorant, but the fashion of being learned may come; we are all moved by self-interest, but the fashion of being disinterested will never come. Amidst the countless myriads of creatures born in the space of a hundred years, nature may perhaps produce two or three dozen of rational beings whom she must scatter over the world, and you can readily imagine that they are never found any where in such large numbers as to set the fashion of virtue and uprightness.—*Fontenelle*.



MAN OF LEARNING, TELL ME SOMETHING.

By MARGARET MEREDITH.

I wonder if men could not be persuaded to alter their style of conversation with girls, to talk to us as they talk to men?

We have a feeling that learned young men are the dullest of talkers; not because they talk weightily; Oh, no! because they talk so lightly, and lightness is not their forte.

A diligent student, a very cormorant, perhaps, of knowledge, dons a white necktie and sallies forth, and resolutely leaves behind for the evening every material he has wherewith to make himself agreeable. He is not witty, he is too busy to be a gossip, he is too little in company to learn an easy jog of commonplace or compliment. So he sits on a sofa, and the girl makes some opening remark, to which he replies with studied interest; and at the pause she magnetically feels that it is best to make a longer remark this time. If she were talking to a lad, she might drift into expressing some of her real ideas, and find profit and pleasure in airing them; but for the amusement of this young savant, by no means. Still, at his next turn to speak, or the next, she has come suggestively near some subject worth talking of; if he were with a man he would instantly plunge in, and in five minutes they would be deep in discussion or description, sharpening their wits by every sentence, fixing what they have read, shaping their crude opinions, thoroughly enjoying each other; and for this they need not be equals in cultivation, nor altogether equals in mind.

Why should it be so different when talking with a woman? There is no reason, but habit. One says, "People dislike to talk shop; the busy scholar wants a rest." On the contrary, most people, I think, would rather talk shop than anything else. If it is their life interest and their strong point, they have so much more to say. The truth is, they fear that the listener will object, and so "in company" they avoid it. I wager the listener would be delighted.

I do not write so much to those who can get up at will a brilliant flow of mere scintillation. That is a scarce enough article to be valuable. Yet they might use it occasionally on sense as well as on nonsense, and make themselves all the more notably entertaining.

I once knew a grave professional man who was said to be both clever and cultivated, but for me there seemed no possible way to enjoy him. His visits were the most empty occasions. He was “a desirable person to be visited by,” but he was unendurable; though he did not fail to be politely attentive in more ways than one. I was glad he was going away. Just then a mutual friend came on the scene, who had views on this matter. I know she gave him the benefit of them, as well as if she had told me; for such an amazing change I never saw. The passive sitter waked up, the bore became a charming talker, and all because he had taken his own permission to be agreeable in his natural way. I was so sorry when he left town!

That instance of transformation is what inspires my appeal. The thing would seem grounded and settled, incapable of cure, but what one exhortation can accomplish has been proved.

And it is a case in which the butterfly may well spring full-colored from the chrysalis, for the stuff that talk is made of is all there; not repartee, of course, or always brilliant expression for one’s thoughts and facts; but thoughts and facts very simply used make an evening world-wide different from a succession of laboriously-framed sentences carefully intended to be about something in which the man does not take any interest, and the woman sees he does not. Can we wonder that the sand-man has to be struggled with many a time by both parties? Young boys do not blink with sleep under your very eyes; but full-grown men often do, and largely because they insist on pursuing at thirty-five about the same topics of conversation that they used at eighteen.

Don’t you, Mr. Dry-as-dust, want to turn over a new leaf? My opportunities of learning are limited, perhaps, while yours are constant. If I am to spend an hour, or two or three, with you, will not you give me some advantage from your well-furnished store-house? If I do not respond then possibly you may stand excused, and never again run the risk of talking over my head.

But give me one fair trial, and see if we are not “better company” and better friends ever afterward.



HIBERNATION.

By the REV. J. G. WOOD, M.A.

The hedgehog, like the bat, is carnivorous.

Toward the end of autumn it looks out for some retired spot, a perfectly dry cavity in the ground or in the rock being the favorite resort. Here it gathers together a large quantity of dry moss, leaves, grass, etc., covers itself with them, rolls itself into a ball, and sinks into the hibernating lethargy.

It is rather remarkable that a hibernating animal is much more sensitive to a slight touch than to general handling. If, for example, a single hair of a hibernating bat or a single quill of a hibernating hedgehog be raised, the creature gives a quick start, and takes a few breaths before relapsing into lethargy. Yet a bat may be sunk under water, or have a thermometer tube passed into its stomach, without being awakened.

When a hibernating bat is sunk under water of the same temperature as that of its body, it does not even attempt to breathe. A similar experiment was tried with a hedgehog, and after it had been under water for twenty-one minutes, one tiny bubble of air rose to the surface. I need scarcely say that if the animal had been awake, it would have been drowned in less than a fourth of the time.

For the bat, no food can be found until the warm weather returns, and so the hibernation is unbroken for at least five months. But, though food be almost entirely withdrawn from the hedgehog, some nutriment remains, and therefore the animal is so constituted that it can discover and consume the food which has been provided for it.

This food chiefly consists of snails, which are themselves hibernators, and which during the winter months conceal themselves so effectually that

they are seldom detected except by their two great wintry foes, the thrush and the hedgehog.

The hedgehog, not possessing so wide a range of hibernating temperature as the bat, which actually “hibernates” daily for a short time even during the hottest summers, is roused by an hour or two of warm sunshine such as we often experience about February. Awakened by the warmth, the hedgehog unrolls itself, creeps out of its refuge, and trundles (I know no better word to describe its peculiar pace) away in search of food. Taught by instinct, it is sure to come upon one of the strongholds of the snail, eats as many as it needs, returns to its home, and sleeps until awakened in a similar manner.

Then we have the vegetable-eating squirrel, which is a partial hibernator.

During the later weeks of autumn, the squirrel may be seen in the act of making provision for the winter. In the first place it collects a vast store of fallen leaves, moss, twigs, and similar materials, and with them constructs its winter nest.

Squirrels have two distinct kinds of nest, one for the winter and the other for the summer. Both nests are of considerable size, and both are so well concealed that to detect them is a very difficult task. The summer nest is comparatively light in texture, and is placed near the ends of lofty boughs, where it is hidden by the leaves. Moreover, its position renders it almost unassailable, as the branch on which it is built would not even endure the weight of a small boy. In the winter, when the leaves are off the trees, the nests are very conspicuous, and in the New Forest, where I gave some time to watching the habits of the squirrel, they are exceedingly numerous.

In fact, the squirrels of the New Forest swarm in such numbers, and do so much damage to the young twigs of the trees, that many hundreds must be shot annually, just as is the case with rabbits. They are always shot just before hibernating, because, as they put on new robes for the winter, their skins fetch the best prices. Moreover, the animals become fat, as is the case with all hibernators, and so their flesh is in good condition for the table. Squirrel-pie is a well-known luxury in some parts of England, and is far superior to rabbit-pie, as it is free from the peculiar flavor which attaches itself to the rabbit, and to many persons is exceedingly repulsive.

The winter nest is a very large one, containing at least four or five times as much material as would serve for a summer's nest. Instead of being placed at the end of a bough, it is always set in the hollow caused by the junction of several large branches with the trunk. The exterior is so skilfully formed, that when the tree is viewed from below, even the most practised eyes will often fail to detect the nest, large as it is.

The amount of material which a squirrel employs in this nest is really wonderful. I have taken out of a single nest armful after armful of leaves, until quite a large mound was raised at the foot of the tree, and I should think that there was enough material to fill two large wheelbarrows, even if it were pressed down closely.

I may here mention that the nest of the squirrel is known in some parts of England by the name of "drey," and in others by that of "cage." The latter term is employed in the New Forest.

The house being ready, next comes the task of laying up a store of food. This consists chiefly of nuts, which the animal chooses with marvelous sagacity, or rather, instinct. No one ever yet found an unsound or worm-eaten nut in a squirrel's store. The animal does not rely on a single storehouse, but hides its treasures here and there within easy range of its nest. Many nuts it buries, and owing to this habit, nut-trees are apt to spring up in unexpected places, for, if the weather should be exceptionally severe, the squirrel awakens but seldom from its winter sleep, and so does not need the store which it has hidden. Or, it may die or be killed after it has laid up its food, and so the buried nuts will take root and produce trees.

A remarkable instance of this fact occurred in the grounds of Walton Hall, belonging to the late Charles Waterton.

In former days there had been in the estate an old wooden mill. It had been disused for many years, and at last the only relic of it was the upper millstone which was left on the ground. The reader may be aware that the center of the upper stone is pierced with a tolerably large hole, through which the corn makes its way between the stones.

In the autumn of 1813, some nut-eating, hibernating animal, almost certainly a squirrel, had found this stone, and thought that the hole would

make an admirable hiding-place for a nut. For some reason, the nut was never eaten, and consequently began to germinate. Mr. Waterton, who pervaded his grounds at all hours of day and night, detected the green shoot at once when it appeared in the spring of the following year. Foreseeing that the shoot, if it lived long enough to become a tree, would raise the stone from the ground, he had a fence put round it, and gave special orders for its preservation.

His prevision proved to be perfectly correct. In course of years, the little shoot became a large tree twenty-five feet in height, and bearing fine crops of fruit annually, and Mr. Edmund Waterton told me that in his boyhood he had often climbed it for the purpose of procuring nuts. After the stem was large enough to fill the orifice in which it had been planted it lifted the stone, and raised it some eight or nine inches above the ground.

As might be imagined, in the course of years the pressure of the stone destroyed the bark, and stopped the circulation of the sap, so that the tree died. In order to save it from being blown down, the trunk and branches were cut away some feet above the stone. On my last visit to Walton Hall, shortly before Mr. Waterton's death, the stone was still suspended above the ground, and as a memorial of so remarkable a result of hibernation, I made a careful sketch of it, which was published by Messrs. Macmillan.

It is also noticeable as an example of the slow, silent, and almost irresistible power of vegetation. Even the soft and pulpy mushroom has been known to raise a flat, heavy paving-stone fairly off the ground. Had the mushrooms been allowed to grow, and the paving-stone laid on them, it would have crushed them under its weight. But the vital powers of growth are so tremendous, even when acting upon so feeble a medium, that they performed a feat which would have been thought impossible had it not been witnessed.

In some parts of South America, where the growth of vegetation is surprisingly rapid, there used to be, and may be still, a mode of inflicting capital punishment by the power of vegetation. We all know the sharply-pointed and bayonet-like leaves of certain aloes. The victim was simply fastened to the ground over a spot where an aloe was just starting from the

earth, and before a day had gone by, the leaves would grow completely through the body.

I briefly mention these examples in order to show how all nature is linked together, and that the hibernation of animals and the growth of vegetables are parts of one great system.

Owing to the manner in which the squirrel disperses his treasures, we can not tell the amount of the store required by each animal, but in Northern America we find one which gives the needful information. This is the chipping squirrel, chipmunk, so called from its cry. Its scientific name is *Tamias Lysteri*.

It is a little creature not larger than a two-thirds grown rat, and is very conspicuous on account of the black and yellow stripes which run along its back. Being a creature which leads a subterranean life for the greatest part of its time, it does not possess the bushy tail of the tree-inhabiting squirrels.

Its underground habitation is a most elaborate composition of galleries and chambers, so that there is plenty of space for storage. Audubon once dug up a nest inhabited by four chipping squirrels, and found in it two pecks of acorns, a quart of large nuts, rather more than two quarts of buckwheat, besides about half a pint of grass seeds and ordinary wheat. Considering that the animals would pass the greater portion of the winter months in lethargy, and would only eat at long intervals, the amount of food is really surprising.

In former days, when the red men were supreme and depended solely on hunting for their food, many a tribe has been saved from extermination for want of food in the winter time by digging up the nests of the chipping squirrel, and eating the inhabitants as well as their stores.

In the dormouse we have another instance of hibernation brought into contact with man.

This pretty little creature, which is too familiar to need description, possesses in a great degree the power of becoming fat toward the end of autumn. The ancient Romans were well aware of this fact, and had regular establishments called “gliraria” for the express purpose of fattening dormice for the table.

The dormouse makes a singularly comfortable nest for itself. It is nearly spherical and is composed externally of grass blades woven together in a very ingenious manner. The animal only leaves a small aperture, concealed by grass blades which can be pulled asunder when the inmate enters or leaves the nest, and which resume their position like the folds of a drawn curtain. I once had a remarkably fine specimen of a dormouse nest which was cut out of a hedge. The curtain of grass blades was so admirably formed that it could seldom be detected by any one who did not know the specimen.

Around, but not in this nest, the dormouse places its store of winter food, which is much of the same nature as that of the squirrel, and mostly consists of nuts. For this reason the Germans call the creature by the appropriate name of hazelmaus.

It was made in the fork of a hazel-branch, and was about four feet from the ground, so that the small branches served to strengthen as well as conceal it. The nest was exactly six inches long by three in width, and was made almost entirely of several kinds of grass, the broad-bladed sword-grass being the chief material. Interwoven with the grass-blades were sundry leaves, all hazel and maple, and none of them having been taken from the branch on which the nest was built. It is therefore possible that a dormouse may have placed the nest in Mr. Waterton's mill-stone. I do not, however, think it probable, because there was no bush near the stone, and, as far as is known, the dormouse always stores its food close to its nest. The squirrel, however, ranges farther afield, and may often be seen in the winter-time digging through the snow, at some distance from its tree, so as to disinter the hidden food.

Another vegetable-eating hibernating rodent is the too well-known hamster (*Cricetus frumentarius*) of Northern Europe.

It is about a foot in length, but, on account of its numbers, is a most formidable enemy to the agriculturist. Even when seeking its daily food it is terribly destructive to the crops, but its worst raids are made at the end of the autumn, when it provides a store for the winter. For this purpose it excavates a deep and complicated system of burrows, in which it stores a quantity of grain so enormous that after the harvest the farmers are in the

habit of digging up the hamster's burrows and securing their stolen property.

A single hamster carried off sixty pounds of wheat for its winter store, while another had thought that a hundred weight of beans were necessary for its subsistence. The animal wakes very early from its hibernation, sometimes even in February. It does not, however, come out of its burrow at once, but remains beneath the earth until the warm weather has fairly set in.

Now we come to the bears.

I need not say that intertropical bears do not require to hibernate. Moreover of those bears which inhabit the colder climates the adult males seldom, if ever, hibernate, while the young of both sexes are very uncertain in this respect. For example, with the grizzly bear the young males and females are found at large throughout the whole of winter, and the same is the case with the polar bear. With the brown bear of Northern Europe and the black bear of North America the young animals seem to be rather capricious in hibernating.

In all cases, however, when the adult female bear is about to add to the family she prepares for hibernating. With the exception of the polar bear, who is obliged to form a most remarkable habitation, the female chooses a safe retreat long before it is required, and gradually conveys into it a large quantity of leaves, moss, and small branches, so as to make a comfortable bed.

Shortly before hibernating she becomes enormously fat, and the new fur which she puts on is quite half as long again as that of the summer raiment. Hunters, therefore, are naturally anxious to kill the bear just before hibernating.

In the first place, a fully developed winter fur, taken before it has been injured by use, will sell for twice as much money as the fur of the same animal when taken in summer or after hibernating. In the next place, the fat, which is so well-known as "bear's-grease," always commands a ready sale. Lastly, as bear's meat, prepared either by freezing or smoking, forms the greatest part of winter food in many a family, it is a matter of the greatest consequence to have that meat in the best condition.

How valuable it is under such circumstances may be realized by reading the life of the old American hunter, Daniel Boone, and seeing how, when his wife and children were nearly dying of hunger and cold, he forced his way across the half-frozen river, succeeded in killing a bear, and by almost superhuman exertions transported all the meat across the river to his hut.

Supposing that the bear is not interrupted in her work, she retires to the den just before winter, and closes the entrance as well as she can.

In this place of refuge the young are born. They are at first scarcely larger than rats, but increase in size, drawing the whole of their nourishment from their mother, who takes no food during the whole of the winter and early spring. In consequence, when she and her young emerge, the latter are fat and strong, while the mother is but the shadow of her former self. Here again is a wonderful example of the many ways in which God “giveth meat to all flesh.”

When a male or young female hibernates it comes out of its refuge as fat as it was on entering it. The hibernation is so perfect that there is scarcely any waste of tissue, as is the case with the mother bear, whose young practically subsist on the store of fat which she laid up in the autumn.

The polar bear when about to become a mother is obliged to find a very different kind of refuge, as there are neither caves, hollow trees, or branches, and often there is nothing but ice as a resting-place and snow as a covering. So she depends for shelter upon the snow. After selecting a convenient snow-drift, she scrapes a hole in it, and suffers the snow to fall upon her as it will.

In that country, where even the human inhabitants are obliged to make their houses out of snow or perish, she is soon buried under many feet of snow. Her thick fur keeps the snow from contact with the skin, while the heat of her body gradually melts the snow away from around her, so that she lies in a sort of tent.

Now comes the question, ventilation. Were she alone all the time she would need no communication with the external air, as the hibernation would be perfect, and respiration would not be required. But her young,

who do not hibernate, must breathe continually from the time of their birth, and she, being disturbed by them, is forced to breathe occasionally.

Now, it is found that when animals are buried under snow their warm breath continually ascends, and makes a passage into the air. The aperture is a very small one, but quite sufficient for the purpose; and even in our Scotch Highlands sheep are enabled to breathe in a similar manner when buried in the terrible snow-drifts, which are apt to overwhelm whole flocks at a time.—*London Sunday Magazine*.



ZENOBIA.

By ADA IDDINGS GALE.

Midst clash of arms, she comes, and glittering spear,
Bold, bright and beautiful, her flashing eye;
Crowned, gemmed and robed in cloth of Tyrian dye.
Palmyra's pride, unequaled far or near.
Proudly she moves and with imperious mien
Views with a sweeping glance each column o'er,
While they in rapture kneeling do adore,
And rising, vow allegiance to their queen.
The trumpet's peal, a thousand helmets shine,
The long ranks into perfect order pass,
And at the command move on. Alas!
That fortune's star for such should e'er decline,
That pomp of pride, that dreams of regal sway
Should like the mists of morning melt away.

The man of the least mental powers may be perfect if he move within the limits of his own capacities and abilities, but even the noblest advantages become obscured, annulled, and annihilated, when symmetry, that is so indispensable, is broken through. This mischief will still oftener appear in these present times; for who will be able to satisfy the requirements of a present ever calling for more exertion and in the highest state of excitement?—*Goethe*.

CHARACTER BUILDING.

By JAMES KERR.

Failure in any enterprise often rouses to fresh effort. You fall in order to rise again. You are thrown down that you may rise higher. Failure may thus carry in its bosom a rich harvest of good. In men of spirit, who are not easily cowed, it acts as a spur to exertion. Every time such a man is thrown down, and, like the fabled Titan, touches mother earth, he rises again with renewed strength. Many a great orator has failed ignominiously in his first attempt; but if he has the right stuff in him he is not disheartened. Like the late Lord Beaconsfield, he says indignantly: "The time will come when you will hear me!" He says it, and he keeps his word. We have a similar instance in M. Thiers, the French historian and statesman. When as a young man he made his *debut* in the Chamber of Deputies, his speech was not a success. He felt that he had failed. On returning home he said to his friends, "I have been beaten; but never mind, I am not cast down, I am making my first essay in arms. Beaten to-day, beaten to-morrow; it is the fate of the soldier and the orator. In the tribune, as under fire, defeat is as useful as a victory. We begin again!" Such was the spirit of the man, such his indomitable resolution; and we all know that his efforts were at last crowned with complete success.

Failure, disappointment, and difficulties to be surmounted, doubtless contribute an element of strength to the character. We thus learn to persevere in a difficult task. Speaking of the failures, delays, and obstacles met with at the siege of Troy, Shakspeare puts these words into the mouth of Agamemnon—

"Which are, indeed, naught else
But the protractive trials of the great Jove,
To find persistive constancy in man."

Trials, misfortunes and difficulties of every kind, if properly met, are a means of discipline. In the struggle with them we are made stronger. They brace the mind, and give it firmness. A disposition naturally gentle requires this tonic to prepare it for the rougher duties of life. Many can say that the disappointments and trials they have met with have given a firmness to their temper which was much needed, and have been of the greatest service to them.

I have never known any one who had difficulties to contend with in his youth, and who wrestled with them successfully, who was not thankful for them later in life. They felt that these difficulties, resisted and overcome, helped to mould their character and make them stronger and better men than they would otherwise have been.

We read in the letters of Thomas Erskine, of Linlathen, as follows: “A friend of mine once repeated to me a sentence which he thought utter nonsense, but to me it seemed to have a meaning. *What were rocks made for, my brethren? Even that mariners might avoid them.* There was a gain in having avoided rocks, which there would not be if rocks had never existed.”

In the same manner we may say, What was evil made for? Even that we may avoid it. There is a gain in having avoided and resisted evil, which there would not be if evil had never existed.

The trials and troubles of life afford an education to which no other is equal. We have not the finest type of character in the monk and the nun, who lead a life of seclusion far away from the evil of the world. Their virtues are only negative. It is not among those who are shut up within stone walls and jealously guarded, that you obtain the noblest type of character. On the contrary, it is among those who have had to struggle with evil in all its forms in the strife and conflict of life. In this way virtue is strengthened, and a character formed nobler than a life of mere innocence could impart.

It is seen that in those places where there is the greatest amount of vice, there are also to be found many examples of the greatest virtue. It is said that nowhere are there such good people as in London, and the reason assigned is that nowhere are there so many bad people. The Londoner lives in the midst of temptations which have to be avoided and resisted—thus the habit of virtue and of self-control is formed. Those who are good, in spite of

manifold temptations to evil, are likely to be very good. Their virtue will be of a more robust type than that of those who are immured in nunneries, and who are kept innocent by temptation being removed out of their way.

There are two ways of dealing with mankind. You may remove them from every temptation, and thus keep them innocent in outward act. Or you may place them in the midst of temptations, trusting to their power of resisting them. You wish, for example, to guard a man from the habit of drunkenness. You shut him up within stone walls, where the very smell of drink is unknown; or you place him in a lonely island, where there is no beverage to be had stronger than pure water.

In this way you get rid of the temptation, but you sacrifice the man. You make of him a nonentity. Others, not less wise, would pursue a different course. They would leave him a free agent in the world, with all its trials and temptations. The probability is he would defend himself from the danger; for, after all, even in the most drink-loving nations, it is only a small proportion of the population that give way to this vice. This latter method has the advantage, instead of sacrificing the man, of improving him. It contributes an element of strength to his character, and trains him to be a brave soldier in the battle of life.

There is much in this avoidance of evil and keeping it in check. It is the great means available for the development of our moral nature. What exercise is to the body, resistance to evil is to the mind.



THE RECREATIONS OF THE PARIS WORKMAN.

By R. HEATH.

The recreations of the better class of Paris workmen wear a character of Arcadian simplicity.

On fêtes, and especially during that of the Republic, which, though nominally confined to the fourteenth of July, continues for several Sundays afterward, there is much dancing and all the ordinary amusements of a fair.

The first day of the week, is, however, only a holiday once a month, for the majority of workmen. On the afternoon of pay-Sunday the workman takes his family outside the barrier for a walk into the country. They have a simple dinner at one of the numerous restaurants in the neighborhood, and wander in the woods, plucking the wild flowers, or find a quiet nook, where one of the party reads aloud. These happy afternoons fill the workman's heart with joy, and he begins to recall his childhood and to talk of his old home in some distant province. He takes his wine, is joyously excited, but nothing more; the whole family return by train or tram-car, laden with lilac or wild flowers, and are safe in bed by eleven o'clock.

Saturday evening is the favorite time for the theater. The workman prefers the drama, and if the scene is pathetic, is easily moved to tears.

On Sunday afternoon a few visit the Louvre, the Luxembourg, and the Salon, and other picture galleries when open. They are observed to fix their attention mostly on historical scenes, or pictures which touch the feelings; a scene from the Inquisition, a mother weeping over her children, or an inundation, or a famine.

Compared with the German, the Paris workman can hardly be said to possess any musical faculty whatever. The loud and harsh noises to be

heard night and day in Paris indicate that the popular ear must be in an almost infantine condition. Cracking their whips with the utmost violence is the ceaseless delight of Parisian drivers, and during the fête and for many days after, the urchins on the street render life unsupportable by constant detonations of gunpowder.

To judge from the way the workmen gather round bookstalls, and the avidity with which the young among them may be seen devouring a book while waiting for the tram, reading must be a real enjoyment to the more intelligent. I have seen a young fellow in a blouse reading a book as he sat astride on the back of a heavy cart-horse. A friend, a lady who has made friends with a family at Belleville, finds them not only to possess a good library, but to be well acquainted with French literature. When a workman is a reader his taste will be good. He will despise novels, especially of the vicious order; his favorite books are histories of the Revolution, such as Lamartine's "Girondins;" Louis Blanc's "Dix Ans;" "Histoire de Deux-Décembre," etc.; and for classics, Voltaire, Rousseau, and perhaps Corneille.

If in the present adult population many may be found with literary and artistic tastes, the workmen of the next generation will be educated men, in the vulgar sense of the word; for it would be difficult to give adequate expression to the fury with which the instruction of the people is pressed forward. All classes combine; the Republicans because they sincerely believe that popular instruction is the great panacea for all the ills of the world; Conservatives, because they hope that it will make the people reasonable; Catholics, because they fear to lose even those who still hold to the church.

Primary instruction is now compulsory and gratuitous. The choice of the school rests with the father or guardian, but he can not neglect to have his child instructed by some one and somewhere. The communal schools are excellent, and the greatest pains taken with the instruction. For the present generation there are multitudes of lecture courses, popular and gratuitous. I have no means of exactly knowing the number, but it is said that there are now in Paris during the season as many as 2,000 courses of lectures of one kind or another. A very great number of these are open to the public.

In a speech made last December at the West London School of Art, Mr. Mundella, M.P., stated that he had recently been in France for the purpose of inquiring into the new system of education, which came into operation on the 1st of October last year, and that while there he had spent some time in trying to ascertain the progress the French were making in giving instruction in art. The Vice-President of the Council declared himself “perfectly astounded by the facts that had come to his knowledge on the subject. He had seen in Paris placards, six feet long, offering gratuitous instruction to every person employed in certain trades who would come and accept it. He found schools of art, which were attended by hundreds and thousands of students, in every part of the country. These schools were supported, not only by government aid, but by the different municipalities out of the local rates and taxes. Thus all the artisans of Paris, and a large number of those in the country, were receiving gratuitous art instruction. The Paris municipality expended £32,000 in this way last year, and that sum will be largely exceeded during the present year. He had brought with him the ‘Paris Budget for Education’ for next year (1883), and he found from it that that city with its population of 1,900,000 would spend on education double the amount that was expended for the education of the four millions who lived in London.”

Why then may we not hope to see many Garfields in the French Republic? The first great difficulty is the strong feeling of caste which exists as powerfully in the workman as in any other class.

M. Poulot has related an amusing instance of the way a young lady of the middle class and her mother turned away from him with a kind of horror when they learnt that he actually *worked* in a factory, and helped to make the steam engines. But I have met with an instance quite as startling on the other side. Meeting at the house of a mutual friend, an orator, who, a few days before, I had heard deliver a strong philippic against the government, at a meeting mainly composed of workmen, and on a question of interest to them, I asked him to introduce me to one of his friends. He assured me that he only knew them in the meetings, but that he did not know the address of any. Nothing could give a stronger impression of the immense chasm between the working class and those not actually members of it, than to find one of their prominent advocates—a man who, I believe, has been devoted

for years to their cause—without a single private friend among working-men.—*Good Words*.



A RUSSIAN NOVELIST.

By GABRIEL MONOD.

France has just lost an author who, though he never wrote in French, had made France his adopted country, and had been adopted by her as one of her most illustrious novelists—Ivan Tourgénief. From the time when the petty persecution of the Russian government obliged him to leave his native land, he settled in France with his friends the Viardots, paying only short occasional visits to Russia. It was at Bougival, near Paris, that he died on the third of September, of a painful disease from which he had been suffering for more than two years. His works were often translated into French from the manuscript itself, and appeared simultaneously in French and in Russian; and though he depicted Russian types and manners exclusively, his reputation was as great in Paris as at St. Petersburg, and he passed with the general public for a great French writer. He has contributed, more than any one else, to make Russia understood in France, and to create a sympathy between the two nations. Contemporary Russia lives complete in his works. In his “Memoirs of a Russian Nobleman,” or “Recollections of a Sportsman,” he has given expression to the sufferings, the melancholy, the poetry, of the Russian country-folk, and prepared the way for the emancipation of the peasants; in “A Nest of Nobles” he has depicted the monotonous life of the lesser gentry, living on their small fortunes in the heart of Russia; in “Dimitri Roudine,” in “Smoke,” and in “The Vernal Waters,” we find those Russian types which are met with all over Europe—those nomads whose incoherent brains are seething with all sorts of ideas, social, political, and philosophical; those spirits in search of an ideal and a career, whom the narrow and suffocating social life of Russia has turned into idlers and weaklings; those worldlings, with their eccentric or vulgar frivolity; those women, amongst whom we may find all that is most cruel in coquetry and most sublime in self-devotion. Last of all, in “Fathers and Sons,” he has revealed, with a prophetic touch, the first symptoms of that moral malady of Nihilism which is eating at the heart of modern Russia,

and in “Virgin Soil” he has given us a faithful and impartial description of the society created by the Nihilistic spirit. Tourgénief is a realist; his personages are real, his pictures are drawn from life, his works are full of true facts; but he is at the same time a true artist, not only in virtue of the power with which he reproduces what he has seen, but because he has the faculty of raising his personages to the dignity of human types of lasting truth and universal significance, and because he describes, not all he sees, but only what strikes the imagination and moves the heart. He is wholesomely objective; he does not describe his heroes, he makes them act and speak; the reader sees and hears and knows them as if they were living people—loves them and is sorry for them—hates and despises them. Tourgénief is one of those novelists who have created the greatest number of living types; he is one of those in whom we find the largest, the most sensitive, the most human heart. He has shown, like Dickens, all that warmth of heart can add to genius.—*The Contemporary Review*.



A LAY OF A CRACKED FIDDLE.

By FREDERICK LANGBRIDGE.

When I was quite a tiny mite,
And life a joyful ditty,
I used to know a poor old wight
Who fiddled through the city.
Alas! it's thirty years ago—
Time is so quaint and flighty!
And now I've mites myself, you know,
And not so very mighty.
And he's unvexed by flat and sharp;
He's guessed the awful riddle,
And, haply, got a golden harp
In place of that old fiddle.

And yet, methinks, I see him now—
So clear the memory lingers—
His long grey hair, his puckered brow,
His trembling, grimy fingers,
The comforter that dangled down
Beyond his waist a long way,
The beaver hat with battered crown,
He'd pause to brush—the wrong way,
The brown surtout that still could brag
Its buttons down the middle,
And, crowning all, the greenish bag
That held the sacred fiddle.

Two tunes he played, and only two,
One over, one beginning;
“God Save the Queen's” collapse we knew
Was “Kitty Clover's” inning.
How startlingly the bow behaved—
Curveted, jerked, and bounded—
The while our gracious queen was saved,
And knavish tricks confounded!
And oh! the helpless, hopeless woe,
Brimful and running over,
In (*very slow*) the o—o—oh

Of bothering Kitty Clover!

And so he'd jerk and file and squeak
Like twenty thousand hinges,
While every sympathetic cheek
Was racked with shoots and twinges.
The lawyer left his lease or will,
The workman stopped his hammer,
The druggist ceased to roll the pill,
And ran to calm the clamor.
From doors and windows jingled down
A dancing shower of copper,
Accompanied by many a frown,
And sometimes speech improper.

He gathered up the grudging dole,
And sought a different station,
But always with a bitter soul,
And deep humiliation.
For what though music win you pence,
If praise it fail to win you?
If fees are paid to hurry hence,
And never to continue?
“Bad times for art,” he'd sometimes say
To any youthful scholar;
“They'd rather grub for brass to-day,
Than listen to Apoller.”

And so with quaint, pathetic face,
Aggrieved and disappointed,
The minstrel moved from place to place,
And mourned the times disjointed.
His hat was browner than of yore,
His grizzled head was greyer,
And none had ever cried “Encore,”
Or praised the poor old player.
I came to feel (and was not wrong)—
His day was nearly over—

He'd not be bothered very long
By cruel Kitty Clover.

One day, within a shady square,
Where people lounged or sat round,
He'd played his second woeful air,
And now he took the hat round.
He met with many a gibe and grin,
With coarser disaffection,
The while he tottered out and in,
Receiving the collection.
At length he stopped, with downcast eye,
Beneath a lime tree's cover,
Where sat a maiden, sweet and shy,
Beside her handsome lover.

Half hidden in her leafy place,
The modest little sitter
Just glanced into the fiddler's face,
And read his story bitter.
Unskilled in life and worldly ways,
By womanhood's divining,
She knew the minstrel's soul for praise
And sympathy was pining.
Herself with all a heart could need,
No dearest dream denied her,
She felt her gentle spirit bleed
For that poor wretch beside her.

She hung her head a little while,
Then, growing somewhat bolder,
She rose, and with a blush and smile,
Just touched the minstrel's shoulder.
"How charmingly you play," she said.
"How nice to be so clever!
My friend and I" (her cheeks grew red)
"Could sit entranced for ever.
I've taken lessons—all in vain;

My touch is simply hateful.
Oh! if you'd play those tunes again,
I'd be so very grateful."

He rosined up his rusty bow
(His eyes were brimming over),
Then (o—o—oh!) meandered slow
Through endless "Kitty Clover."
He'd suffered many a cruel wrong
Amid a sordid nation;
He'd waited wearily and long—
At last the compensation!
What cared he now for snub and sneer
From churlish fools around him?
In those sweet eyes he saw a tear,
And felt that fame had crowned him.

And you, my friends, may laugh or frown,
And still I'll risk the saying,
That angels stooped from glory down
To hear the fiddler playing.
And he that holds the golden pen,
That chief of all the bright ones,
Who registers the deeds of men,
The wrong ones and the right ones—
He oped the book, and did record
A sweet and gracious deed there—
A deed performed to Christ the Lord
That he shall smile to read there.



BLUE LAWS.

An interesting and suggestive chapter in our early colonial history is found in the constitution, laws and court records of Connecticut. That some of the enactments and judicial proceedings, to those ignorant of the peculiar condition of the colonists, seem ludicrous, and fit to provoke the unfriendly criticism they have received, is not denied. But an honest, competent critic can not take them thus, and will not hastily discredit the intelligence of the men who, under new and most trying circumstances, made such regulations for their little commonwealth as the exigencies of the situation seemed to demand. We do not approve of all the laws of that olden-time as wise and just; nor do we think the administration always beyond just reproach; but we do venerate the men who for the glory of God and the good of society enacted and rigorously enforced them.

The ancient orthography is retained as a specimen of the English of that day:

CONSTITUTION OF 1638.

“For as much as it hath pleased the Almighty God, by the wise disposition of his divine providence, so to order and dispose of things, that we, the inhabitants, and residents of Windsor, Hartford and Weathersfield, are now dwelling in and upon the river of Conneticut, and the lands thereunto adjoining; and well knowing, when a people are gathered together, the word of God requires, that, to the maintienence of the peace and union of such a people there should bee an orderly and decent government established, according to God, to order and dispose of the affaires of the people at all seasons, as occasions shall require; doe therefore associate and conjoine ourselves to bee as one publique State or Commonwealth; and doe for ourselves and our successors, and such as shall be adjoined to us at any time hereafter, enter into combination, and confederation together, to meinteine and preserve libberty, and the purity of the

gospell of our Lord Jesus, which we now profess; as also the discipline of the churches which, according to the truth of said gospell is now practiced amongst us, as allso in all our civil affaires to be guided, and governed according to such lawes, rules, orders and decrees, as shall bee made, ordered, and decreed, as followeth.”

Then follows the constitution in eleven well considered sections, making provision for the three departments—legislative, judicial and executive. We freely confess our admiration of this wonderful document, but can not, for want of room, print it. This is the less necessary as it evidently formed the basis of the charter of 1662, and its leading provisions have been copied, with some modifications, into the constitutions of the several States, and of the United States. As the first written constitution formed for and adopted by a free people, for their own government, it is a marvel of excellence. Written without a model, it asserts for its authors a more comprehensive and thorough statesmanship than is usually attributed to the leaders in colonial politics at that early day.

The most peculiar feature of their civil polity was that only the righteous were to be in authority, and all power was vested in members of the church; and the conservative influence of religion variously confessed. The church and state were separate, yet, not inconsistently, we find an article headed:

“MAINTENANCE OF MINISTRY.”

“Whereas, the most considerable persons in the land came to these parts of America, that they might enjoye Christe, in his ordinances without disturbance; and whereas, amongst many other precious meanes, the ordinances have beene and are dispensed amongst us with much purity and power, they took it into their serious consideration that a due maintenance might bee provided, and settled, both for the present and the future, for the encouragement of the minister’s worke therein; and doe order that those who are taught in the Word, in the several plantations, bee called together, that evry man voluntarily sett downe what hee is willing to allow to that end and use; and if any man refuse to pay a meete proportion, that then hee bee

rated by authority, in some just and equall way; and if after this any man withhold, or delay due payment, the civil power bee exercised as in other just debts.”

The “Capitall Lawes” were severe, and the executive officers a terror to evil-doers. The death penalty was denounced against criminals convicted of either of fourteen different offenses. The burglar for the third offense lost his life.

1. “If any man after legall conviction shall have or worship any other god but the Lord God, hee shall bee put to death.”—Deut. 13:6, 17:2.

2. “If any man or woman bee a witch, that is, hath or consulteth with a familliar spiritt, they shall bee put to death.”—Exodus 22:18; Levit. 20:27.

3. “If any person shall blaspheme the name of God, the Father, Sonne or Holy Ghost, with direct, express, presumptuous or high-handed blasphemy, or shall curse, in like manner, hee shall bee put to death.”

4. “If any man shall commit any willful murder—which is manslaughter committed from hatred, malice or cruelty—not in a man’s just and necessary defense, nor by mere casualty against his will, hee shall bee put to death.”

8. “If any person committeth adultery with a married or espoused wife, the adulterer and adulteress shall surely bee put to death.”

12. “If any man shall conspire or attempt any invasion, insurrection or rebellion against the Commonwealth hee shall bee put to death.”

The laws were specially severe against the social evil, and the homes of the colonists guarded not only against the crimes, but against all dalliance with evil, and imprudent conduct that might weaken the family bonds. The purity and bliss of the home might not be endangered with impunity, and the wayward were punished with wholesome severity. Here is a court

record: “Martha Malbon, for consenting to goe to the farms with Will Harding at night, to a venison feast, and ... for dalliance with said Harding was whiped.” How it fared with Will we are not told, but presume there was safety for him only in exile, as there was no marked discrimination in favor of his sex at that time. As connected with this case it is further recorded that “Goodman Hunt and his wife for keeping the councells of said William Harding, baking him a pastry and plum cakes, and keeping company with him on the Lord’s day, and she suffering Harding to kisse her, they being only admitted to sojourn in this plantation on their good behavior, ordered to be sent out of this towne within one month after the date hereof; yea, in a shorter time, if any miscarriage be found in them.—December 3, 1651.” On another page I find it recorded that “Will Harding *was* sentenced to be *severely* whipped, fined £10, and presently to depart the plantation, and not retourne under the penalty of severer punishment.”



A REMNANT OF SUMMER.

By E. O. P.

I went out in the dull autumnal day,
Around me fell the rain,
The bare trees shivered 'gainst the ashen sky,
My heart was full of pain.

High in a maple tree, upon a branch,
The tree-trunk close beside,
A little empty bird's nest, snug and neat,
My tearful eyes espied.

And straightway, for the time, from grief and care
My sad heart was beguiled,
And on this remnant of the summer gone
Through rain and tears I smiled.

Not oft has life so dull and drear a day,
But something bright appears
To speak of sunshine and the spring time flown,
And bring a smile through tears.



THE LIFE OF A PLANET.

By RICHARD PROCTOR.

The material life of a planet is beginning to be recognized as being no less real than the life of a plant or of an animal. It is a different kind of life; there is neither consciousness such as we see in one of those forms of life, nor such systematic progress as we recognize in plant-life. But it is life, all the same. It has had a beginning, like all things which exist; and like them all, it must have an end.

The lifetime of a world like our earth may be truly said to be a lifetime of cooling. Beginning in the glowing vaporous condition which we see in the sun and stars, an orb in space passes gradually to the condition of a cool, non-luminous mass, and thence, with progress depending chiefly on its size (slower for the large masses and quicker for the small ones), it passes steadily onward toward inertness and death. Regarding the state in which we find the earth to be as the stage of a planet's mid-life—viz., that in which the conditions are such that multitudinous forms of life can exist upon its surface, we may call that stage death in which these conditions have entirely disappeared.

Now, among the conditions necessary for the support of life in general are some which are unfavorable to individual life. Among these may be specially noted the action of those subterranean forces by which the earth's surface is continually modeled and remodeled. It has been remarked with great justice, by Sir John Herschel, that since the continents of the earth were formed, forces have been at work which would long since have sufficed to have destroyed every trace of land, and to have left the surface of our globe one vast limitless ocean. But against these forces counteracting forces have been at work, constantly disturbing the earth's crust, and, by keeping it irregular, leaving room for ocean in the depressions, and leaving the higher parts as continents and islands above the ocean's surface. If these disturbing forces ceased to work, the work of disintegrating, wearing away,

and washing off the land would go on unresisted. In periods of time such as to us seem long, no very great effect would be produced; but such periods as belong to the past of our earth, even to that comparatively short part of the past during which she has been the abode of life, would suffice to produce effects utterly inconsistent with the existence of life on land. Only by the action of her vulcanian energies can the earth maintain her position as an abode of life. She is, then, manifesting her fitness to support life in those very throes by which, too often, many lives are lost. The upheavals and downsinkings, the rushing of ocean in great waves over islands and seaports, by which tens of thousands of human beings, and still greater numbers of animals, lose their lives, are part of the evidence which the earth gives that within her frame there still remains enough of vitality for the support of life during hundreds of thousands of years to come.

This vitality is not due, as seems commonly imagined, to the earth's internal heat. Rather the earth's internal heat is due to the vitality with which her frame is instinct. The earth's vitality is in reality due to the power of attraction which resides in every particle of her mass—that wonderful force of gravitation, omnipresent, infinite in extent, the property whose range throughout all space should have taught long since what science is teaching now (and has been foolishly blamed for teaching), the equally infinite range of God's laws in time also. By virtue of the force of gravity pervading her whole frame, the crust of the earth is continually undergoing changes, as the loss of heat and consequent contraction, or chemical changes beneath the surface, leave room for the movement inward of the rock-substances of the crust, with crushing, grinding action, and the generation of intense heat. If the earth's energy of gravity were lost, the internal fires would die out—not, indeed, quickly, but in a period of time very short compared with that during which, maintained as they constantly are by the effects of internal movements, they will doubtless continue. They are, in a sense, the cause of earthquakes, volcanoes, and so forth, because they prepare the earth's interior for the action of her energies of attraction. But it is to these energies and the material which as yet they have on which to work, that the earth's vitality is due. She will not, indeed, retain her vitality as long as she retains her gravitating power. That power must have something to work on. When the whole frame of the earth has been compressed to a condition of the greatest density which her attractive

energies can produce, then terrestrial gravity will have nothing left to work on within the earth, and the earth's globe will be to all intents and purposes dead. She will continue to exercise her attractive force on bodies outside of her. She will rotate on her axis, revolve around the sun, and reflect his rays of light and heat. But she will have no more life of her own than has the moon, which still discharges all those planetary functions.

But such disturbances as the recent earthquakes, while disastrous in their effects to those living near the shaken regions, assure us that as yet the earth is not near death. She is still full of vitality. Thousands—nay, tens, hundreds of thousands of years will still pass before even the beginning of the end is seen, in the steady disintegration and removal of the land without renovation or renewal by the action of subterranean forces.—*The Contemporary Review*.



DISRAELI'S LONDON.

One of Disraeli's favorite ideas was that London ought to be made the most magnificent city in the world—a real *Kaiserstadt*, or imperial town, a model to all other cities in the character of its public buildings, the sanitary perfection and outer picturesqueness of its private houses, the width of its streets, etc. When Napoleon III. commenced the re-edification of Paris he used to say: “Is it not pitiful that the emperor should be doing by force what we could do so much better of our own free will, if we had a proper pride, to say nothing of good sense in the matter?” Once when he was staying at Knole, he launched out into a parody of Macaulay's idea of the New Zealander meditating over the ruins of London Bridge. He imagined this personage reconstructing in fancy a row of villas at Brixton: “What picture he would make of it! he would naturally suppose that knowing how to build, and having just awoken to a knowledge of sanitation, we had built according to the best ideas in our heads.” Then he took his New Zealander among the ruins of the stately commercial palaces crowded in narrow lanes all round the Bank, and the Exchange: “He would conclude that there must after all have been some tyrannical laws which prevented our merchants from combining their resources to make their streets spacious and effective, for it would seem absurd to him that intelligent men should, at a great cost, have built palaces for themselves in holes and corners where nobody could admire them properly, when by acting in concert, they might at much less expense have set much finer palaces in noble avenues, courts and squares.” Then Disraeli broke out into an animated description of his regenerate London with Wren's four grand approaches to St. Paul's, boulevards transecting the metropolis in all directions; and the palace of Whitehall rebuilt after Inigo Jones's designs to make new government offices. He would have covered the embankment pedestals with statues of admirals set in colossal groups recalling great naval achievements, and he thought Stepney ought to have its cathedral of St. Peter, and containing memorials to all the humble heroes, sailors or fishermen who lost their lives performing acts of courage on the water. When he had finished speaking

somebody observed that his plan would cost £200,000,000, and convert every ratepayer into a porcupine. “We may have to pay £500,000,000 in the end for doing things in the present way,” he answered; “and as to the porcupine, he is manageable enough if you handle him in the right way.”—*Temple Bar*.



TEMPERATURE.

By J. MORTIMER-GRANVILLE.

Such expressions as a “cool head,” “hot-headed,” and the like, commonly relate to temperament rather than temperature; but it is essential to a full comprehension of the subject before us that the *rationale* of animal heat should be stated, and the laws that govern the phenomenon of temperature actual and subjective, at least cursorily, explained.

Heat and the sensation of heat are two widely different states. When, on a chilly day or after washing in cold water, a man rubs his hands until a glow of heat seems to suffuse them, there is a very slight rise of actual temperature caused by the friction; the feeling is principally due to nerve-excitement, produced mechanically by the rubbing. The blood flows more freely into, and through, the parts excited immediately afterward, as shown by the redness, but the first impression of heat is mainly one of sensation. The feeling and the fact are not even constantly related. A person may feel hot when not only the surrounding temperature but that of his body is low; or, he may feel cold when really overheated. These perverted sensations are occasionally morbid—that is to say, form part of a state of disease—or they may arise from individual peculiarities which, perhaps, render perceptions of a particular class especially acute. On the other hand, there are conditions of the body, and special sensibilities, in which the sense of heat is dulled, and even considerable elevations of temperature are not perceived. It is easy to see how impossible it must be to form a correct judgment of the actual state of heat either around or within us by simple sensation.

Throughout the world, whether man be placed in tropical heat or arctic cold, the temperature of his body must, to maintain health, be preserved at the same point—about 98.4 to .6 degrees of Fahrenheit. A very small departure from this universal mean standard constitutes or indicates disease. The external heat is comparatively unimportant, or only of secondary moment, in the economy of nature; we can not rely upon it for the

compensation of differences in the heat generated within the body by the organism. Except for the production of a temporary effect, such as to give time for the reestablishment of the normal temperature in a body chilled, as by submersion, external heat is useless for vital purposes. The only way in which it can act is by preventing the loss of more heat, and giving a slight aid to recovery by warming the surface of the body.

If when a person is cold he goes into a heated apartment, or sits before a large fire, he receives with advantage just as much heat as will bring the skin of his body up to the normal standard; as soon as that point is reached, the organism will begin to labor to get rid of the superfluous caloric, and by sweating the heat must be kept from rising above the standard. All the heat thrust upon the body above 98.6 degrees is waste and mischievous except in so far as it may promote perspiration, which probably helps to work off some of the useless and burdensome, possibly morbid and poisonous, materials that oppress the system. This is how Turkish baths, and “sweatings” generally, do good, by exciting increased activity of the skin, and, as it were, opening up new ways of egress for matters which, if retained, might offend.

So far as the heat of the body is concerned, whether in health or disease, every degree of external heat which is above the complement to form 98.6 degrees Fahrenheit with the heat of the body itself at the time, is useless and may do harm. It follows that in fever the surrounding atmosphere should be kept cool; in depressing disease, when the heat-producing powers of the organism are small, the air around should be warm. These are precisely the conclusions to which experience and observations conduct us; and the facts now briefly stated explain the reason why.

There is no warmth in clothes; the heat comes from the body itself, generated within, or the surrounding atmosphere, or from substances with which the body may be in contact. Of course clothes, like any other materials, can be charged with heat, and will take up as much thermic or heating property as their specific capacity allows. It is this capability of receiving heat which constitutes the first condition of warmth in the comparative value of different materials of dress. The second condition consists in the physical power of any fabric to hold the heat with which the article has been charged. For example, some materials will become warmer

in a given time and retain their heat longer than others under the same conditions of exposure, first to heating and then to cooling influences. The principle of clothing should be to protect the body from external conditions which tend to abstract heat, when the surrounding temperature is lower than that of the body; and to strike heat into the organism, when the temperature of the outside air and of the substances with which the skin may be brought into contact is higher than that of the animal body itself.

Local temperature, that is, the heat in the several regions of the body is determined by conditions which control the circulation of the blood, and the function of nutrition or food appropriation. If the circulation is free in a part, its temperature is maintained; if, from any cause, the flow of blood is retarded, the local heat will be reduced. Any one may put this to the test by encasing the hands in somewhat tight gloves when the weather is cold. The pressure prevents the free passage of the blood through the vessels, and the temperature falls. There is no warmth of any kind in the gloves; they act simply as non-conductors of heat, and prevent the heat generated within the body from passing off. For example—if a piece of lint or rag be dipped in cold water and laid on the skin, and a sheet of impervious or non-conducting material, such as india-rubber or thick flannel, is wrapped closely round, the heat of the body will raise the cold water to a temperature at which it will be given off as steam the moment the covering is removed. When the extremities are enclosed in thick or dense coverings, their temperature will depend on the amount of heat generated within them, and if the flow of blood through the vessels is arrested or retarded, nothing is gained, but everything lost, by the measures taken to protect them from the external cold.

This is a matter of the highest practical moment, and needs to be thoroughly understood. The feet can not be kept warm unless the blood circulates freely in the extremities, and that will not be the case if the boot, shoes, or stockings are tight. These last-named articles of clothing are practically the worst offenders. A stocking encircling the foot and leg closely and enveloping every part, with special pressure at the instep, around the ankle, and above or below the knees, must inevitably tend to oppose the circulation and so reduce the natural heat. The arteries which bring the blood to the extremity are set deeper than the veins that carry it back, and, as the latter are provided with valves which open toward the

heart, it is too commonly supposed that the “support” afforded by the stocking will favor the return of blood more than it can impede the deeper supply-currents, and so help the circulation; but practically we know this is not the fact, for a tight stocking ensures a cold foot, and the chilliness of which many persons complain is mainly caused by the practice of gartering, and wearing stockings which constrict somewhere or everywhere.

There is a popular notion that if the feet are cold the head must be hot, and by keeping the extremities warm with wraps the “blood is drawn from the head,” and its temperature reduced. Those who have on the one hand studied the phenomenon of fever, and on the other noted the physical condition of races and individuals who habitually leave the extremities unclothed, will know that this theory of the distribution of heat is only partially true. Heat depends on the due supply of nutrient elements to the tissues. It is the expression or result of the process of local feeding. If a part is active it will be heated. When the feet are left bare the complex muscular apparatus of the extremity, which in a stiff shoe scarcely works, is called into vigorous action, the arch of the foot plays with every step, and each toe performs its share in the act of progression. This promotes growth and calls for nutrition, whereby the heat is maintained; whereas if it be simply packed away as a useless piece of organism, no amount of external heat will warm it. Work is the cause and counterpart of heat throughout the body.

The same principle applies to the head. No amount of external cooling will reduce the temperature, no drawing away of the blood by artificial expedients will permanently relieve the sense or obviate the fact of heat if the organ within the cranium is excessively or morbidly active. The brain is a peculiarly delicate and complicated organ, requiring more prompt and constant nutrition than any other part of the body, because the constituent elements of its tissue change more rapidly than those of any other in proportion to the amount of exercise. Moreover, the brain is always acting during consciousness, and even in sleep it is seldom wholly at rest, as we know from the occurrence of dreams. The faculty of nutrition is highly developed in the organ or it could not so continuously, and on the whole healthily, discharge its functions, even when other parts of the body, or the system as a whole, are suffering from disease. When the head is heated there is nearly always a local cause for it, and the remedy must be addressed to the seat of the malady. The temporary expedient of “drawing away the

blood” by applying heat to the extremities is useful as far as it goes, and may suffice to enable the organ to rid itself by the contraction of its blood-vessels from a surplus charge of this fluid, but in the absence of special causes the *reason* of the “heat of head” is undue exercise or disturbance of nutrition in the brain itself. Perhaps the seat of the over-work and consequent heating may have been limited to a particular part of the head; for example, the apparatus of sight, or hearing, as when the head becomes heated by reading too long or in a strong light. The point to understand is that when the head is physically hot it is the seat of too much or disorderly nutrition, and either the amount of brain or sense-power exercised must be reduced or the mode of action changed, and the particular part of the apparatus of perception or thought which has been too severely taxed relieved.

The true condition of health is that in which the temperature of the body as a whole and of its several parts is not disturbed by surroundings either of heat or cold. The preservation of a natural and healthy temperature is mainly to be secured by the maintenance of a regular and well distributed circulation of blood charged with the materials of nutrition.

The first condition of a free and continuous flow of blood is a healthy heart, not hampered by irritants, mental or physical. Sudden grief or fright produces cold by arresting the circulation, and the flow may be permanently retarded by anxiety. The mind has a wondrously direct influence on the heart and blood-vessels—on the latter through the nerves, which increase or reduce the calibre of the minute arteries, as in blushing or blanching at a thought. Instead of loading the body with clothes, the “chilly” should search out the physical cause of their coldness. The blood must not only circulate freely; it must be rich in nourishing materials, and not charged with poison. An excess of any one element may destroy the value of the whole. It is too much the habit of valetudinarians and unhealthy people of all kinds, to charge the blood with substances supposed to be “heating” or “cooling” as they think the system requires them. This is a mistake. The body does not need to be pampered with cordials, or refrigerated with cunningly devised potions. If it be well nourished it will be healthy.



There is something fearful in seeing a man of high character being under an obligation to a fool.—*Goethe*.



SKATING AND SKATERS.

By ROBERT MACGREGOR.

Though it appears to be impossible to fix on the time when skating first took root in this country, there can be no doubt that it was introduced to us from more northern climates, where it originated more from the necessities of the inhabitants than as a pastime. When snow covered their land, and ice bound up their rivers, imperious necessity would soon suggest to the Scands or the Germans some ready means of winter locomotion. This first took the form of snow-shoes, with two long runners of wood, like those still used by the inhabitants of the northerly parts of Norway and Sweden in their journeys over the immense snowfields.

When used on ice, one runner would soon have been found more convenient than the widely-separated two, and harder materials used than wood; first bone was substituted; then it, in turn, gave place to iron; and thus the present form of skate was developed in the North at a period set down by Scandinavian archæologists as about A. D. 200.

Frequent allusions occur in the old Northern poetry which prove that proficiency in skating was one of the most highly esteemed accomplishments of the Northern heroes. One of them, named Kolson, boasts that he is master of nine accomplishments, skating being one; while the hero Harold bitterly complains that though he could fight, ride, swim, glide along the ice on skates, dart the lance, and row, "yet a Russian maid disdains me."

Eight arts are mine: to wield the steel,
To curb the warlike horse,
To swim the lake, or skate on heel
To urge my rapid course.
To hurl, well aimed, the martial spear,
To brush with oar the main—
All these are mine, though doomed to bear
A Russian maid's disdain.

Specimens of old bone skates are occasionally dug up in fenny parts of the country. There are some in the British Museum, in the Museum of the Scottish Antiquaries, and probably in other collections. There seems to be good evidence that even in London the primitive bone skate was not entirely superseded by implements of steel until the latter part of last century.

Mr. Roach Smith, F.S.A., describing one found about 1839, says that "it is formed of the bone of some animal, made smooth on one side, with a hole at one extremity for a cord to fasten it to the shoe. At the other end a hole is also drilled horizontally to the depth of three inches, which might have received a plug, with another cord to secure it more effectually."

There is hardly a greater difference between these old bone skates and the "acmés" and club skates of to-day, than there is between the skating of the middle ages and the artistic and graceful movements of good performers of to-day. Indeed, skating as a fine art is entirely a thing of modern growth. So little thought of was the exercise that up to the Restoration days it appears to have been an amusement confined chiefly to the lower classes, among whom it never reached any very high pitch of art. "It was looked upon," says a writer in the *Saturday Review* in 1865, "much with the same view that the boys on the Serpentine even now seem to adopt, as an accomplishment, the acmé of which was reached when the performer could succeed in running along quickly on his skates and finishing off with a long and triumphant slide on two feet in a straight line forward. A gentleman would probably then have no more thought of trying to execute different

figures on the ice than he would at the present day of dancing in a drawing-room on the tips of his toes.”

During all this time, when skating was struggling into notice in Britain, in its birth-place it continued to be cultivated as the one great winter amusement. In Holland, too, where it is looked upon less as a pastime than a necessity, nothing has so frequently struck travelers as the wonderful change the advent of ice brings about on the bearing of the inhabitants. “Heavy, massive, stiff creatures during the rest of the year,” says Pilati, in his “Letters on Holland,” “become suddenly active, ready and agile, as soon as the canals are frozen,” and they are able to glide along the frozen surface with the speed and endurance for which their skating has been so long renowned, though these very qualities are bought at the expense of the elegance and grace we nowadays look for in the accomplished skater. Thomson thus graphically describes the enlivening effects of frost on the Dutch:

Now in the Netherlands, and where the Rhine
Branched out in many a long canal, extends,
From every province swarming, void of care,
Batavia rushes forth; and as they sweep,
On sounding skates, a thousand different ways
In circling poise, swift as the winds along,
The then gay land is maddened all to joy.
Nor less the northern courts, wide o’er the snow,
Pour a new pomp. Eager on rapid sleds,
Their vigorous youth in bold contention wheel
The long resounding course. Meantime to raise
The manly strife, with highly-blooming charms
Flushed by the season, Scandinavia’s dames
Or Russia’s buxom daughters glow around.

Though the poet of the “Seasons” speaks of Russia here, it is curious to note that skating is not a national amusement of the Russians, but is entirely of foreign and quite recent introduction. It is quite unknown in the interior, and no Russian—except a few who have picked up the art in St. Petersburg

—ever thinks of availing himself of the many pieces of water annually frozen hard in so cold a country.

Perhaps it is in Friesland that the skate is most especially a necessary of life. What stilts are to the peasant of the Landes, skates are to the Frisian. The watercourses of the summer are his highways when winter sets in. “He goes to market on skates; he goes to church on skates,” we are told; “he goes love-making on skates.” Indeed, it may be doubted if this province could be inhabited if the art of skating were unknown, for without it the inhabitants would be confined to home for several months of each year. Frisians of both sexes actually skate more than they walk, says M. Depping; no sooner is an infant able to stand upright than the irons are fastened on his feet; his parents lead him on to the ice, and teach him how to move along. At six years most of the young skaters have attained great proficiency, but in Frisian opinion even the best performers improve up to thirty.

Here, as elsewhere in Holland, ice races are of frequent occurrence during the winter. “The races on the ice,” says Pilati, “are the carnivals of the Dutch: they are their fêtes, their operas, their dissipations;” naturally, therefore, the people manifest the greatest interest in them; skate long distances to be present, and cherish the names of distinguished winners in a way we should never expect from such an unemotional people as the Hollanders appear when the ice is gone and when most travelers see them.

The women have races of their own; but most interesting of all the contests are those in which the sturdy dames, whom their own painters delight in depicting as gliding along to market with baskets on their heads and knitting-needles in their busy fingers, are matched against the best of the other sex. Though, as a rule, these “Atalantas of the North” excel the men rather in beauty of style than in speed, yet the prize often enough goes to one of them.

Frequently on the Continent skates have proved themselves excellent engines of war, both in actual fighting—as when a Dutch army on skates once repulsed a force of Frenchmen on the Scheldt—and as a rapid means of communication. During the winter of 1806, Napoleon, after the battle of Jena, wished to send an order with the utmost dispatch, to Marshal Mortier, directing him to make himself master, without delay, of the Hanseatic

towns. The officer charged with this order found himself at the mouth of the Elbe at a point where it was seven and a half miles from bank to bank. To cross in a boat was impossible, as the river was coated with a surface of newly-frozen ice; to get over by a bridge would necessitate a detour of more than twenty miles. The officer, knowing how precious time was, determined to skate over the thin ice; and though it was too weak to bear a man walking, he skimmed along so rapidly that he got across in safety, gaining great honor for the ingenuity and boldness that enabled him to deliver his despatch six hours sooner than he possibly could have done by the ordinary route.

In Holland, regiments have regular parades on the ice; but Norway is probably the only country where it has been considered necessary to embody a special corps of skaters. In this regiment, "the men are furnished," says Mr. Russell, in his translation of Guillaume Depping's book, "with the skates in ordinary use in the North, that fixed on the right foot being somewhat longer than that on the left. Furnished with these, the soldiers descend steep slopes with incredible rapidity, re-ascend them as quickly, cross rivers and lakes, and halt at the slightest signal, even while moving at the highest speed."

Skating has had many enthusiastic votaries, but probably none more so than the two illustrious names that continental skaters are so proud to reckon in their guild.

Klopstock, even in his old age, was so ardent a lover of it that, after skimming over the ice of Altona for hours, "to call back that warmth of blood which age and inactivity had chilled," he retired to his study and wrote fiery lyrics in its praise. His friend and great successor, Goethe, took to skating under peculiar circumstances. He sought relief in violent exercise from embittered memories of a broken-off love affair. He tried in vain riding and long journeys on foot; at length he found relief when he went to the ice and learned to skate, an exercise of which he was devotedly fond to the last. "It is with good reason," he writes, "that Klopstock has praised this employment of our physical powers which brings us in contact with the happy activity of childhood, which urges youth to exert all its suppleness and agility, and which tends to drive away the inertia of age. We seem, when skating, to lose entirely any consciousness of the most serious objects

that claim our attention. It was while abandoning myself to these aimless movements that the most noble aspirations, which had too long lain dormant within me, were reawakened; and I owe to these hours, which seemed lost, the most rapid and successful development of my poetical projects.”

That skating has been in certain circumstances something more than mere elegant accomplishment, is well illustrated by two anecdotes, told by the author of some entertaining “Reminiscences of Quebec,” of two settlers in the far West, who saved their lives by the aid of their skates. In one case the backwoodsman had been captured by Indians, who intended soon after to torture him to death. Among his baggage there happened to be a pair of skates, and the Indians’ curiosity was so excited that their captive was told to explain their use. He led his captors to the edge of a wide lake, where the smooth ice stretched away as far as the eye could see, and put on the skates. Exciting the laughter of the Indians by tumbling about in a clumsy manner, he gradually increased his distance from the shore, till he at length contrived to get a hundred yards from them without arousing their suspicion, when he skated away as fast as he could, and finally escaped.

“The other settler is said to have been skating alone one moonlight night, and, while contemplating the reflection of the firmament in the clear ice, and the vast dark mass of forest surrounding the lake and stretching away in the background, he suddenly discovered, to his horror, that the adjacent bank was lined with a pack of wolves. He at once ‘made tracks’ for home, followed by these animals; but the skater kept ahead, and one by one the pack tailed off; two or three of the foremost, however, kept up the chase, but when they attempted to close with the skater, by adroitly turning aside, he allowed them to pass him. And after a few unsuccessful and vicious attempts on the part of the wolves, he succeeded in reaching his log hut in safety.”



BOOK KNOWLEDGE AND MANNERS.

By LORD CHESTERFIELD.

I have this evening been tired, jaded, nay, tormented, by the company of a most worthy, sensible and learned man, a near relation of mine, who dined and passed the evening with me. This seems a paradox, but is a plain truth, he has no knowledge of the world, no manners, no address; far from talking without book, as is commonly said of people who talk sillily, he only talks by book; which in general conversation is ten times worse. He has formed in his own closet, from books, certain systems of everything, argues tenaciously upon those principles, and is both surprised and angry at whatever deviates from them. His theories are good, but, unfortunately, are all impracticable. Why? because he has only read, and not conversed. He is acquainted with books, and an absolute stranger to men. Laboring with his matter, he is delivered of it with pangs; he hesitates, stops in his utterance, and always expresses himself inelegantly. His actions are all ungraceful; so that, with all his merit and knowledge, I would rather converse six hours with the most frivolous tittle-tattle woman, who knew something of the world, than with him. The preposterous notions of a systematical man, who does not know the world, tire the patience of a man who does. It would be endless to correct his mistakes, nor would he take it kindly; for he has considered everything deliberately, and is very sure that he is in the right. Impropropriety is a characteristic, and a never-failing one, of these people. Regardless, because ignorant, of customs and manners, they violate them every moment. They often shock, though they never mean to offend; never attending either to the general character, nor the particular distinguishing circumstances of the people to whom, or before whom they talk; whereas the knowledge of the world teaches one, that the very same things which are exceedingly right and proper in one company, time and place, are exceedingly absurd in others. In short, a man who has great knowledge, from experience and observation, of the characters, customs, and manners of mankind, is a being as different from, and as superior to, a man of mere

book and systematical knowledge, as a well-managed horse is to an ass. Study, therefore, cultivate and frequent, men and women; not only in their outward, and consequently guarded, but in their interior, domestic, and consequently less disguised, characters and manners. Take your notions of things as by observation and experience you find they really are, and not as you read that they are or should be; for they never are quite what they should be.



A man of the best parts, and the greatest learning, if he does not know the world by his own experience and observation, will be very absurd; and consequently very unwelcome in company. He may say very good things; but they will probably be so ill-timed, misplaced, or improperly addressed, that he had much better hold his tongue. Full of his own matter and uninformed of, or inattentive to, the particular circumstances and situations of the company, he vents it indiscriminately; he puts some people out of countenance; he shocks others; and frightens all, who dread what may come out next. The most general rule that I can give you for the world, and which your experience will convince you of the truth of is, never to give the tone to the company, but to take it from them; and labor more to put them in conceit with themselves, than to make them admire you. Those whom you can make like themselves better, will, I promise you, like you very well.

A system-monger, who, without knowing any thing of the world by experience, has formed a system of it in his dusty cell, lays it down, for example, that (from the general nature of mankind) flattery is pleasing. He will therefore flatter. But how? Why, indiscriminately. And instead of repairing and heightening the piece judiciously, with soft colors and a delicate pencil; with a coarse brush, and a great deal of white-wash, he daubs and besmears the piece he means to adorn. His flattery offends even his patron; and is almost too gross for his mistress. A man of the world knows the force of flattery as well as he does; but then he knows how, when, and where to give it; he proportions his dose to the constitution of the patient. He flatters by application, by inference, by comparison, by hint; and seldom directly. In the course of the world there is the same difference, in everything, between system and practice.



UNDER THE AUTUMN SKIES.

By MRS. EMILY J. BUGBEE.

The clouds hung loose and gray,
 Across the autumn sky,
And at my feet in golden piles,
 The dead leaves, drifting lie.
No voice of summer song,
 I hear from copse or tree,
The perfume of no summer flower,
 Comes floating up to me.

Death's silence over all,
 Where music was, and bloom,
Enfolded all the sun-kissed hills,
 In drapery of gloom.
I walk as in a dream,
 Beneath the brooding sky,
While faded, as these autumn leaves,
 Life's hopes around me lie.

The keen and cruel frost
 Has touched my world with blight,
And dark on all its splendors lie,
 The shadows of the night.
The memory of its joy,
 Like billows of the sea,
Come surging up the silver strand,
 Then backward moaning flee.

Amid this sombre calm,
 Beneath these skies of gray,
And drifting of the yellow leaves
 I walk alone to-day,
And scarce can look beyond
 The shadows cold and drear,
That fold, away from mortal sight,
 The summer of my year.

In the eternal spring,

Beyond time's changing skies,
Beyond the chilling frost of death,
A resurrection lies.
I can not tell how long,
The snow shall wrap their tomb,
But sometime, shall life's blighted flowers
Burst into splendid bloom.



EIGHT CENTURIES WITH WALTER SCOTT.

By WALLACE BRUCE.

III.

One hundred years have passed away since Richard the Lion-hearted, Ivanhoe and Robin Hood met at the “Joyous passage of arms at Ashby.” Our next story, “Castle Dangerous,” opens upon days even more bitter and warlike; Scotland is rent with bitter feuds. The daughter of King Alexander the Third died in 1291, and no fewer than twelve persons claimed the throne. King Edward of England was chosen arbiter. He took advantage of sectional discord and endeavored to make Scotland subject to the English crown. He found a willing instrument in the person of John Baliol, who basely acknowledged himself vassal and subject. King Edward further demanded the surrender of three powerful castles, Berwick, Roxburgh and Jedburgh; but the people murmured and Baliol was compelled to do battle with Edward. Under this weak and treacherous leader the Scottish army was defeated in a great battle near Dunbar in 1296. Edward marched through Scotland at the head of a powerful army. He removed to London the records of the Scottish Kingdom, carried the great stone of Scone, upon which the Scottish Kings had been crowned for centuries, to Westminster Abbey, and placed the government of Scotland in the hands of John de Warrene, Earl of Surrey.

At this juncture a leader arose in the person of Sir William Wallace, the son of a private gentleman, and in no way related to the nobility of the kingdom. His glorious struggle kept alive the spark of Scottish liberty. He gathered to himself a band of brave men, and defeated the English army near Stirling. The Scottish people, as they had no king, chose him Protector, and he was titled Sir William Wallace, Governor of the Scottish Nation. He was defeated, captured by a traitor, brought to trial in the great hall of

William Rufus in Westminster, sentenced to death as an outlaw, his body divided into four quarters and placed on London bridge.

Among the followers of William Wallace were two powerful barons, Robert Bruce and John Comyn, whose claims were about equal, by descent, to the Scottish throne. They met before the high altar in the Church of Dumfries. What passed betwixt them is not known; but they quarrelled and Bruce slew him with his dagger. Scott puts a defence of this high-handed deed in the mouth of Robert Bruce which we will quote later. Having committed an act which would bring down upon his head the fierce anathema of the Romish Church, which would moreover arouse the King of England and the powerful family of Comyn, Bruce determined to put them all to defiance, and was crowned King of Scotland at the Abbey of Scone the 29th of May, 1306. Among his devoted friends was James, Lord of Douglas. His castle was on the border of Scotland, and it is in the vicinity of this castle, known as Castle Dangerous, that the scene of our romance is laid. So much for the historical preface which may be of service to the reader in connection with the incidents under our consideration.

In the old chronicles and poems of Scottish history, notably that of Barbour, considerable space is devoted to the adventures of Douglas. His castle was captured again and again by the English; but the victors held it at such hazard against the attacks of the adventurous Douglas, that it was considered a perilous and uncertain piece of property. With a romantic enthusiasm, in keeping with those chivalrous times, Lady Augusta, a wealthy English heiress, distinguished for her beauty, promised her hand and fortune to the knight, who would show his courage by defending the castle against the Scots "for a year and a day." A brave knight, John de Walton, started up and said "that for the love of that lady he was willing to keep the Perilous Castle for a year and a day if the King pleased to give him leave." The King gladly gave his consent, being well pleased to get so brave a knight for such an important fortress.

There was an old prophecy of Thomas the Rhymer, that as often as the Castle of Douglas should be destroyed it would arise grander and stronger than ever from its ruins. The prophecy had already been fulfilled and its great walls seemed able to withstand the most powerful siege. Some manuscripts of Thomas the Rhymer were also preserved in the Castle, and

our first chapter opens with a description of two travelers, showily dressed in the fashion of the wandering minstrels of the day, apparently father and son, making a pilgrimage to the castle with the avowed purpose of finding some of the papers or books of the old poet. They are lodged at the house of one Thomas Dickson. They arouse the suspicion of two English soldiers who are quartered at the Dickson farm-house. The elder minstrel is conducted to the castle and imprisoned; the younger is placed in a neighboring convent. By this time the reader begins to suspect that the younger minstrel is no other than the fair Lady Augusta, making a trip under disguise of a minstrel-boy to see how her knight is prospering. Attended by her father's minstrel she reminds one of Rosalind in "As You Like It," under the guidance of the faithful Touchstone. During her detention at the convent she confessed her secret to Sister Ursula, and they escape by night through a trap-door and subterraneous passage, although the convent is strongly guarded. They separate, and by rather an unnatural process again meet at the Douglas Kirk, where the services of Palm Sunday are converted into a warlike controversy. A hand-to-hand conflict, worthy of the Homeric heroes, is recorded between Lord Douglas and De Walton. In the midst of the fray a herald arrives, announcing the defeat of the English army, and the first triumph of Robert Bruce. De Walton surrenders to Douglas, who allows him without ransom to return to England with the Lady Augusta, and unlike the seven years' toil of Jacob for Rachel, the daughter of Laban, which was lengthened to fourteen years, the one year and a day was shortened, no doubt to the great delight of the interested parties.

The most dramatic incident in the story is the midnight interview between the English knight, De Valence, and the old sexton in the ruined burial-place of the Douglas Kirk. The story throughout is chivalrous and romantic; but "Castle Dangerous" does not rank with other stories of the Waverley series in power, incident or dramatic unity. I have already alluded to "Count Robert of Paris" as the last of the Waverley Novels written by the great magician, and it is so regarded, as "Castle Dangerous" was never really completed by the author; but it serves as a connecting link in the great chain, and, in spite of its incompleteness, gives a graphic description of years eloquent with prowess and manly courage.

“Streamered with silk and tricked with gold,
Manned with the noble and the bold
Of Island chivalry.”

I saw the solitary skiff, bearing the hope and pride of Scotland, making slow and toilsome progress, with rent sail and gaping planks, and heard above the roar of the tempest the calm reply of King Robert to his impatient brother:

“In man’s most dark extremity
Oft succor dawns from heaven.”

I saw the lights of the castle again gleam over the dark billows as the door opened to the regal wanderer asking shelter. I saw the haughty look of the proud Lorn, his lifelong enemy. I saw the bridal feast changed into warlike debate, and Scott’s lines came to my mind with pictured force:

“Wild was the scene; each sword was bare,
Back streamed each chieftain’s shaggy hair
In gloomy opposition set,
Eyes, hands, and brandished weapons met;
Blue gleaming o’er the social board,
Flashed to the torches many a sword;
And soon those bridal lights may shine
On purple blood for rosy wine.”

I saw the Abbott, with hoodless head and withered cheek stop upon the threshold, while

“Threat and murmur died away,
Till on the crowded hall there lay
Such silence as the deadly still,
Ere bursts the thunder on the hill;
With blade advanced, each chieftain bold
Showed like the sworder’s form of old,
As wanting still the torch of life
To wake the marble into strife.”

I heard the haughty words of Argentine demanding Bruce, as England’s prisoner, and the loud turmoil of fiercer chiefs demanding his life, while the brave Ronald cries:

“Forbear!
Not in my sight while brand I wear,
O’ermatched by odds, shall warrior fall,
Or blood of stranger stain my hall!
This ancient fortress of my race
Shall be misfortune’s resting-place,
Shelter and shield of the distressed,
No slaughterhouse for shipwrecked guest.”

I heard the Abbott’s stern charge asking the heroic King if he knew reason aught, why his curse should not be pronounced in requital of that rash deed at the high altar of the Church of Dumfries. I heard the eloquent defense of the King, and the unexpected and sublime blessing of the Abbott.

“Abbott!” the Bruce replied, “thy charge
It boots not to dispute at large.
This much, howe’er, I bid thee know,
No selfish vengeance dealt the blow,
For Comyn died his country’s foe.
Nor blame I friends whose ill-timed speed
Fulfilled my soon-repent’d deed,
Nor censure those from whose stern tongue
The dire anathema has rung.
I only blame my own wild ire,
By Scotland’s wrongs incensed to fire.
Heaven knows my purpose to atone,
Far as I may, the evil done,
And hears a penitent’s appeal
From papal curse and prelate’s zeal.
My first and dearest task achieved,
Fair Scotland from her thrall relieved,
Shall many a priest in cope and stole
Say requiem for Red Comyn’s soul.
While I the blessed cross advance,
And expiate this unhappy chance
In Palestine, with sword and lance.
But, while content the Church should know
My conscience owns the debt I owe,
Unto De Argentine and Lorn
The name of traitor I return,
Bid them defiance stern and high,
And give them in their throats the lie;
These brief words spoke, I speak no more,
Do what thou wilt; my shrift is o’er.”
Like man by prodigy amazed,
Upon the king the abbott gazed;
Then o’er his pallid features glance
Convulsions of ecstatic trance,
And undistinguished accents broke
The awful silence ere he spoke.
“De Bruce! I rose with numose dread

De Bruce, thou with purpose great
To speak my curse upon thy head,
To give thee as an outcast o'er
To him who burns to shed thy gore;
But, like the Midianite of old,
Who stood on Zophim, heaven-controlled,
I feel within my aged breast
A power that will not be repress'd.
It prompts my voice, it swells my veins,
It burns, it maddens, it constrains!—
De Bruce, thy sacrilegious blow
Hath at God's altar slain thy foe:
O'er-mastered yet by high behest,
I bless thee, and thou shalt be blest!
Blessed in the hall and in the field,
Under the mantle as the shield.
Avenger of thy country's shame,
Restorer of her injured fame,
Blessed in thy scepter and thy sword,
De Bruce, fair Scotland's rightful lord,
Blessed in thy deeds and in thy fame,
What lengthened honors wait thy name!
In distant ages sire to son
Shall tell thy tale of freedom won,
And teach his infants in the use
Of earliest speech to falter Bruce."

There is nothing, to my mind, in any poem more dramatic than this unexpected prayer of the abbott; and the reader does not wonder that

"O'er the astonished throng
Was silence, awful, deep and long."

The scene of the poem now changes to the stormy island of Skye, where Sir Walter pauses to give one of his beautiful descriptions in the fourteenth and fifteenth divisions of canto third.

The fourth canto takes the king *en route* past the island of Staffa, with its Fingal's Cave, and Iona, with its sainted shrine—the cradle of Christianity in Britain, now in ruin. His description of Staffa is one of the most beautiful in English verse:

“Where, as to shame the temples decked
By skill of earthly architect,
Nature herself, it seemed would raise
A minster to her Maker’s praise!
Not for a meaner use ascend
Her columns, or her arches bend;
Nor of a theme less solemn tells
That mighty surge that ebbs and swells,
And still, between each awful pause,
From the high vault an answer draws,
In varied tones prolonged and high,
That mocks the organ’s melody.
Nor doth its entrance front in vain
To old Iona’s holy fame,
That nature’s voice might seem to say,
‘Well hast thou done, frail child of clay!
Thy humble powers that stately shrine
Tasked high and hard—but witness mine!’”

In canto fifth the king returns to Scotland. He rallies his adherents, and the sixth canto closes with a graphic description of the battle of Bannockburn. The incidents are so stirring that we almost forget the fate of fair Edith and her brave Roland, but the last line of the poem assures us that they are at last happily wedded.

“The Lord of the Isles” does not possess the pleasing qualities of the “Lady of the Lake,” or the sustained vigor of “Marmion;” but it is a noble poem throughout, and abounds with passages revealing the deep reverence and exalted character of the author. The reader will note the heart-spoken prayer and God-speed of the priest as King Robert embarks upon his uncertain mission:

“O heaven! when swords for freedom shine
And monarch’s right, the cause is thine!
Edge doubly every patriot blow!
Beat down the banners of the foe!
And be it to the nations known,
That victory is from God alone.”

In connection with the “Lord of the Isles” and “Castle Dangerous,” it is well to read carefully the sixth, seventh, eighth, ninth and tenth chapters of Scott’s “Tales of a Grandfather.” It is also pleasant to note that the friendship between Robert Bruce and James Douglas was constant and unchanging; in fact, their unwavering trust and fidelity are emphasized by the dying wish of the king, who desired his heart to be carried to Jerusalem after his death, and requested Douglas to take charge of it. It was in fulfillment of a vow which he had been unable to perform—to go to Palestine and fight for the Holy Sepulchre. “Douglas wept bitterly as he accepted the office, the last mark of the Bruce’s friendship and confidence. He caused a case of silver to be made, into which he put the heart, and wore it around his neck, by a string of silk and gold.” He set off with a gallant train of the bravest men in Scotland. But the doughty James found an opportunity in Spain for a skirmish with the infidels, which he could not let pass; he was overpowered by numbers, and, seeing no chance for escape, he took from his neck the Bruce’s heart, and throwing it before him, exclaimed, “Pass first in fight, as thou wert wont to do, and Douglas will follow thee or die.” His body was found after the battle lying upon the silver case, and the heart of the Scottish king was returned to his native country, and interred beside the high altar under the east window of Melrose Abbey.



PLANT NUTRITION.

By MAXWELL T. MASTERS, M.D.

A living plant feeds, breathes, grows, develops, multiplies, decays, and ultimately dies. In so doing it receives, it spends, it accumulates, it changes. Some of these processes are always in operation, very generally more than one is going on at the same time, and the action of one is modified by and controlled by that of another. Some circumstances and conditions favor these operations, others hinder them.

The nutritive process has to be entered on the creditor side as a receipt. The plant will indeed feed upon itself for a time, or rather it will feed upon what its predecessor left it as an inheritance for this very purpose, or upon the stores accumulated in the plant itself during the preceding season; thus, when a seed, or rather the young plant within the seed, begins to grow, it is at first unable to forage for itself, but it depends for its sustenance on the materials laid up for its use during the preceding season by the parent plant. So the bud of a tree awakening into life, and beginning its career as a shoot which is to bear leaves and flowers, derives its first meals from the reserves accumulated the autumn previously in the parent branch. Very generally a little water, supplied from without, is required before the plant can avail itself of these stored-up provisions, but this is not always indispensable. Potatoes begin to sprout in their cellars or pits, as growers know to their cost, before they can have obtained a drop of water from without. In this latter case there is water enough already in the tuber to allow of food being utilized.

A certain degree of useful heat is, of course, quite indispensable. Practically, no plant will feed when its temperature is reduced as low as the freezing point, and in most cases the heat requires to be considerably greater. Each kind of plant, each individual plant, and indeed each part of a plant, feeds, and performs each item of its life-work best at a certain temperature, and ceases to work at all when the temperature falls below or

risers above a certain point. The particular degree, whether most or least favorable, varies according to the plant, its age, stage of growth and various external circumstances, which we need only mention, as their effects will be readily understood without the necessity of explanation.

Leaving, however, on one side, the temperature, we have to consider the water which is so essential, not only in the feeding processes with which we are now concerned, but with every other action of plant life. Fortunately there is, in general, no lack of it; the earth and the air contain their shares of this elementary compound in varying proportions and varying modifications as liquid or gaseous. Besides, the plant itself has so much of it that even at the driest condition compatible with life, it still constitutes a very large proportion of the entire weight. Now, it is as a rule when the plant, the seedling, or the bud is at its driest that growth begins, the necessity for food first manifests itself, and the demand for a further supply of water becomes imperative. How is the demand supplied? We have seen that there is no lack of that fluid. How is it to get into the plant?

When one liquid, say spirit, is poured into another, say water, the two gradually mix. If we suppose these liquids to consist of a number of molecules, then, mixture may be taken to be the result of the displacement say of one molecule of water by one molecule of spirit, and so, throughout the whole quantity of liquid, there is displacement and replacement till at length equilibrium is restored and a thorough diffusion results. This power of diffusion does not always exist. The molecules of water and of oil will not mix or diffuse freely through each other. Water containing carbonic acid gas will not mix, in this sense of the term, with water containing acetate of lead.

It may be a truism to say, that for the process of diffusion the liquids must be diffusible, but the fact must be carefully borne in mind in all questions relating to the feeding of plants. In the case of plants, the phenomenon of diffusion, or the gradual admixture of two liquids of different natures, is complicated by the presence of a membrane in the shape of the cell-wall. The water from the outside has to pass through the membrane to reach the protoplasm on the other side. Speaking broadly, there are no holes in the membrane through which the water can pass. Ingress is secured by that process of diffusion to which reference has just

been made, and by virtue of which the molecules of the membrane and the molecules of the water shift and change places; the space that was occupied by a molecule of membrane is now occupied by a molecule of water, and *vice versa*. The access, therefore, of water into the interior of a closed cell is the result of the process of diffusion. Where two liquids mix without any intervening membrane, the mixture is called diffusion simply; where there is an intervening membrane, the diffusion process is known as “osmosis.”

The raw material (the term is not quite accurate, but for illustration sake it may pass) is that very marvelous substance now called “protoplasm.” We must leave it to chemists and microscopists to explain its composition and indicate its appearance.

Diffusion is not equal or alike in all cases; it depends upon the extent to which the two liquids are diffusible, upon their different densities, upon temperature, and a variety of other conditions. So, in the case of osmosis, we have not only the nature of the two fluids to consider, but their relation to the membrane that separates them. The membrane may be much more permeable to one of the two fluids than to the other. Thus, in the case of a living cell, the membrane or wall is much more permeable to water than it is to protoplasm; and so it happens that, while water readily penetrates the membrane and diffuses itself in the protoplasm, protoplasm does not nearly so readily permeate the membrane as the water. Ingress of water is easy and of constant occurrence, egress of protoplasm is rare and exceptional.

Pure water or weak saline solutions, such as are generated in the soil under certain circumstances, pass readily through membrane—that is, the molecules of the one shift and change places with those of the other—while those of gummy or albuminous substances like protoplasm do not. After a time, if there is no outlet for the water absorbed, or if it is not utilized within the plant in some way, absorption and diffusion cease, the cell becomes saturated with water, and until something happens to disarrange the balance, no more is absorbed. But, even in the case where the cell is saturated with water, it may still take up other liquids, because the diffusive power of those other liquids, in relation to the cell-wall and to the protoplasm, is different from that of water, and this absorption may go on in its way till saturation point is reached for each one of them, just as in the case of water. On the other hand, it may happen that the plant may be

saturated with other substances, and incapable of taking up more of them, while at the same time pure water may be freely taken up.

Just so much and no more of each particular substance is absorbed, the exact quantity of each being regulated in all cases by the condition and requirements of the cells, their membranous walls, and their contents. Thus it happens that some particular substances may be found by the chemist to exist in large relative proportions in the plant, while the quantity in any given sample of the soil from which it must be derived is sometimes so small as to elude detection. The plant in this case, or some part of it, is so greedy, if we may so say, for this particular substance, that it absorbs all within its reach, and stores it up in its tissues or uses it in some way, the demand ensuring supply. On the other hand, the soil may contain a large quantity of some particular ingredient which is incapable of being absorbed, or which the plant does not or can not make use of, and, in consequence, none is found within the plant. The supply is present, but there is no demand.

The different physical requirements of the plant supply also the explanation of the fact that different plants, grown in the same soil, supplied with the same food, yet vary so greatly in chemical composition. Thus, when wheat and clover are grown together, and afterwards analyzed, it is found that while lime is abundant in the clover, it is relatively in small quantity in the wheat; and silica, which is abundant in the wheat, is absent from the clover. Poisonous substances even may be absorbed, if they are of such a nature as to be capable of absorption; and so the plant may be killed by its own action—by suicide, as it were.

The entrance of water into the plant and the entrance of those soluble materials which a plant derives from the soil are therefore illustrations of the process of osmosis, and are subjected to all the conditions under which osmosis becomes possible, or under which it ceases to act.

One thing we must strive to impress forcibly on the reader, because, if the notion is well grasped, it will enable him to understand plant life so much more vividly. We allude to the continual changes that are going on throughout the whole living fabric of the plant while in its active condition. Cell membrane, the protoplasm, the entire mass of liquid and solid

constituents of which the plant consists, are, as we have seen, made up of molecules, each, as it were, with a life of its own, undergoing continual changes according to different circumstances, acting and reacting one upon another so long as any active life remains.



C. L. S. C. WORK.

By REV. J. H. VINCENT, D.D., SUPERINTENDENT OF INSTRUCTION C. L. S. C.

Readings for the month: "Easy Lessons in Vegetable Biology," by Dr. J. H. Wythe; "Canadian History;" Chautauqua Text-Book No. 24; "Biographical Stories," by Nathaniel Hawthorne; Required Readings, in THE CHAUTAUQUAN.

Memorial Day, Sunday, December 9, "Milton's Day." See "Memorial Days," Chautauqua Text-Book, No. 7. Monday, December 10, may be used if preferred.

Remember the 5 p. m. Sunday "Chautauqua Vesper Service." Observe the hour personally or as local circles. Now and then a brief public service at this hour may be very profitable.

There can be no substitute accepted for the "Preparatory Latin course in English."

One of our faithful members—a member of the class of '84—on the first day of October sent this pleasant greeting to the Superintendent of Instruction: "My Dear Doctor—This is opening day. I must send you a line just to keep it—and the Lord keep you!"

The Sacramento Circle last year answered in writing over 1,000 questions, besides having prepared sixty-two original papers.

A young lady who has charge of a Young Ladies' Seminary in Washington, D. C., recently remarked that she had adopted the Chautauqua

Text-Books on History as an auxiliary in her school, as they are so condensed and so carefully arranged. She said that at the last examination of her graduating class the influence of the little Text-Books was visible in the remarkable proficiency of the pupils.

Each C. L. S. C. Local Circle in the study of Biology should secure the services of a local microscopist, if possible. Without the microscope, Biology is like Hamlet with Hamlet left out.

In one of the leading churches of one of the leading denominations in one of the leading cities of the United States, a strange thing has happened. The president of the local circle of the C. L. S. C. made application for the privilege of holding bi-monthly meetings in a room in the basement of the church, so many of the members of the circle being members of the church. The matter was referred to the president of the board, a leading lawyer, who refused the application. When asked why he should exclude such an auxiliary of the church, and especially a circle containing so much of the religious element, he responded that it "could not be a religious organization, because they were studying biology." This is very hard to believe if it were not well vouched for. If the church had been a Methodist Episcopal Church, the editor of this column would have felt at liberty to make a few direct remarks; but, as it refers to another very respectable and very orthodox branch of the Holy Catholic Church, he must content himself with this general announcement. What would this leading lawyer have said to the wise man who said: "Go to the ant, thou sluggard, consider her ways and be wise;" or to that wiser teacher who commanded his disciples to "consider the lilies." Biology zoölogical and biology botanical being commended in the Bible, and the study being necessary to the fullest interpretation of the Bible, we commend our legal friend to a little more biblical study.

Apropos to the above item is the following communication from an earnest New England member: "At a certain Sunday-school convention this question was given me to answer: 'What is the effect of the Chautauqua course of reading on Christian zeal? Does it tend to increase one's interest in Christian and Church work?' I answered in substance as follows: 'I am

very glad of the opportunity for saying, and saying confidently, that, judging from what experience and observation I have had, as also from the nature of the case, just as whatever is calculated to enlighten and invigorate the mind, deepen, broaden, elevate and strengthen character, to enlarge the soul and warm and ennoble the heart, must tend to intensify Christian zeal, so the Chautauqua course of reading and study, when conducted or pursued in accordance with the projector's idea, can not but tend to have this effect—to deepen and to invigorate, by enlightening, piety. Precisely what we need in our day is a more intelligent piety—a broader and stronger Christian manhood. Our piety generally is too narrow, or superficial, or feeble. We are apt to build up too much on some one side. We are one-sided, unsymmetrical, sanctified in spots only, as it were. We want to be built out more on all sides, that we may be *thoroughly* furnished unto *every* good work—that we may be fruitful in all directions. Meantime, if the cultivation of such a thoughtful, intelligent, reverent piety as the prayerful study of the works and Word of God is calculated to promote, is not calculated to intensify one's interest in and zeal for Christ's cause, it would be interesting to know what could.' Subsequently I dropped a line to one of the members of a circle which I organized a year ago—a bright, Christian young woman, who, though an operative in the mill, yet clearly grasped the Chautauqua Idea, and who, together with as fine a company of young men and women as were ever grouped together for any cause, has most enthusiastically and successfully pursued that idea for a year:—to this young lady I dropped a line, submitting the question: 'Do you find the C. L. S. C. helpful, or otherwise, to Christian piety?' Permit me to quote from her reply: 'Do I love my Savior, or his church, any less for what I have learned the past year? No. A thousand times no. Jesus seems ever so dear to me, as I look up into the starry heavens, and try to recall something I have learned about those wonderful worlds. And when I think of him who created, and who, by his almighty power and wisdom controls and keeps them all in place; when I think of him as my own kind Heavenly Father, though I am poor, and lowly, and ignorant, and weak, and sinful, my heart throbs with gratitude, love and praise—for he owns me as his child! O! I wish I could tell you how happy I feel to-night, my Savior seems so near and dear to me. My heart is full of love to him and to his people; and I do want to do something to help on his glorious cause. I am praying day by day that he will show me my duty, and help me to do it; and I know you will pray for

me that I may be faithful and true.' Does not this testimony have the true ring in it? Does this look much as though the C. L. S. C. had secularized the writer's mind, or diverted her energies from church channels? This lady, together with several other members of that circle, is a devoted Sunday-school worker. What is more, not a little of the glowing, enthusiastic zeal expressed above, has been kindled and developed during this very past year of C. L. S. C. reading and study."



OUTLINE OF C. L. S. C. READINGS.

DECEMBER, 1883.

The required readings for December include "Vegetable Biology," Hawthorne's "Biographical Stories," Chautauqua Text-Book, No. 24, "Canadian History," and the required readings in THE CHAUTAUQUAN.

First Week (ending December 8)—1. "Vegetable Biology," to chapter v, page 27.

2. "Biographical Stories," to chapter iii, page 19.

3. "German History" and "German Literature," in THE CHAUTAUQUAN.

4. Sunday Readings for December 4, in THE CHAUTAUQUAN.

Second Week (ending December 16)—1. "Vegetable Biology," from chapter v, page 27, to chapter viii, page 46.

2. "Biographical Stories," from chapter iii, page 19, to chapter vi, page 40.

3. Readings on Physical Science and Political Economy, in THE CHAUTAUQUAN.

4. Sunday Readings in THE CHAUTAUQUAN, for December 11.

Third Week (ending December 24)—1. "Vegetable Biology," from chapter viii, page 46, to paragraph 10, page 66.

2. "Biographical Stories," from chapter vi, page 40, to chapter viii, page 59.

3. "Readings in Art," in THE CHAUTAUQUAN.

4. Sunday Readings for December 18, in THE CHAUTAUQUAN.

Fourth Week (ending December 31)—1. “Vegetable Biology,” from paragraph 10, page 66, to the end of volume.

2. “Biographical Stories,” from chapter viii, page 59, to end of book.

3. “Selections from American Literature,” in *THE CHAUTAUQUAN*.

4. Sunday Readings for December 25, in *THE CHAUTAUQUAN*.



A REUNION AT MILWAUKEE.

The five local circles in Milwaukee named Alpha, Beta, Grand Avenue, Delta and Iota, recently had a grand reunion on the occasion of the visit of the Superintendent of Instruction. It having been announced in the papers that Dr. Vincent would attend the Wisconsin conference and preach Saturday afternoon, the circles decided to give him a reception. The committee on invitation sent out about two hundred invitations gotten up in a very tasteful and unique manner. The envelopes were covered with autumn leaves of most delicate tints, and contained each a square gilt-edged card, also covered with leaves, bearing the monogram C. L. S. C. and the following invitation: "You are kindly invited to meet Rev. J. H. Vincent, D.D., Superintendent of Instruction of the Chautauqua Literary and Scientific Circle on the evening of Saturday, October 6, in the parlors of the Grand Avenue Congregational Church. Reception, 6 to 9. Refreshments, 6:30."

There was also a smaller card containing a neat little device in a circle in the center, representing a hand holding a handkerchief, and a request in the corner that this card be shown at the door.

Besides these two cards there was a green leaf (artificial) in each envelope for a badge, and a little printed slip of instructions, directing each member to wear the leaf as a badge, and explaining the Chautauqua salute to be given Dr. Vincent when he entered the room.

The committees on decorations and on supper made diligent preparations, so that when the time arrived parlor No. 1 was tastefully arranged with vines and flowers, while through the open doors could be seen twelve tables in parlor No. 2 arranged for an inviting feast. Among the decorations was the banner of the class of '86, made of maroon velvet bordered with cream colored fringe, and bearing the class motto, in letters cut from white felt, "We study for light to bless with light."

Above the platform was a diploma granted to Mrs. William Millard of the class of '83, which arrived from Plainfield a few hours before the reception, and was used as an object lesson by the Doctor in his address. The entire event was most joyous, this being the first union meeting of the circles, and the first time many of them had ever met their revered leader. Of his address, what can be said but that it was *like him*; full of uplifting thoughts and helpful ideas of inestimable value to all Chautauquans, and delivered in his delightful manner.

On Sabbath Dr. Vincent conducted a vesper service in the Immanuel Presbyterian Church, the largest in the city. The Chautauquans gathered in an adjoining room, and forming two columns, headed by Dr. Vincent, marched into the audience room where the central seats were reserved for them. Short addresses were given by Bishop Hurst and Dr. Buckley, which, with the impressive vesper service, made the occasion one long to be remembered.



A C. L. S. C. EXPERIENCE.

From an able speech by Prof. H. A. Strong, before the local circle of Erie, Pa., we clip the following: "Says one of the workers and leaders around the C. L. S. C. camp-fire at Chautauqua: 'I was in Missouri, March last, and was compelled to take a freight train to make connection. As I entered the caboose I noticed a little candle on a cracker-box on the side of the car. There was a door on hinges made out of bits of leather, and a rough button held in its place by a screw, closed the door. After the train started, the conductor came in, and, after attending to his duties, stepped to the box, turned the button, opened the door, and took out a package of C. L. S. C. books, recognizable as such anywhere, sat down on a bench and began working with one of the Chautauqua text-books. Of course it was an absolute necessity that I should make his acquaintance. I approached him and asked him what he was doing. He said: "A friend of mine in St. Louis called my attention to this Chautauqua course of reading. I did not know what it meant, but I knew I ought to read. So, finally, I joined the circle, bought the books, and put them in this box. My brakemen read with me. One of us keeps watch and the others read. Sometimes we are switched off on a side-track, and then we make good progress. Sometimes it is pretty hard work when we have an unusually long run and much freight; but for the sake of the help it is, I am going to hold on to it." I felt like giving the fellow a round of applause, all alone as I was in the car.' Such an experience of the C. L. S. C. can be duplicated over and over again in the history of any class, and the simple truth is the realization of the vision."



THE C. L. S. C. IN TORONTO.

In Canada the course of study for 1883-4 opened with a meeting of C. L. S. C. workers and their friends in the lecture-room of the Metropolitan Methodist church, Toronto, on the evening of the 29th of September. After a few words of greeting from Mr. Edward Gurney, jr., president of the Toronto Central Circle, Rev. Dr. Thomas, pastor of the Jarvis Street Baptist Church, spoke for a short time on the importance and value of a systematic course of reading. The too common habit of desultory reading, with its waste of time and mental enervation, came in for a vigorous denunciation from the doctor. All reading, he said, should be purposeful and systematic, and no reading can be of any real profit that is not of that character. The C. L. S. C. course answered two supreme questions that can not fail to arise in the mind of any young man who is desirous to rise: "What shall I read?" and "How shall I read?" In this age of great intellectual power it was important that we should avail ourselves of every opportunity for the better equipment of our minds, so that we can use with precision the implements of our profession or calling, whatever that may be. The multitudes that are treading upon each other in the lower levels of life, are the incompetent; no first-class worker in any line need remain idle. The doctor also pointed out that this is a skeptical age, and that we should be prepared to answer, if necessary, the reflections that are being cast upon the foundations of our faith. Before closing he said: "I want to declare my entire sympathy with the work and purposes of this rapidly-spreading Chautauqua tree, from the branchings of which thousands and tens of thousands are gathering with delight and gratitude the most luscious fruit. I thank God for this course of study, by means of which the mind is led into the green pastures and beside the still waters of literature. My mind has been stirred in the matter as it would not have been if I had not examined into it closely, and if I had not been profoundly impressed by the fact that multitudes of our young people spend their spare moments in reading pernicious literature in which the serpent has left his slimy trail. I am going to join this class to-night for myself."

Rev. Mr. Milligan, of old St. Andrew's Presbyterian church, followed with an earnest, practical address on "How to Read Books." He impressed upon his hearers the fact that books are made for man, and not man for books, and that it is possible for us to abuse the gift of books by becoming literary ceremonialists, just as we can abuse any other good thing. Every investment we make in relation to books should be made with a definite purpose, and should make us richer. In our reading, too, we should have something more ultimate in view than the mere book; we should endeavor to ponder and reflect on the subject which it treats. In this way we become thinkers, and thinking becomes a necessity, and the mind and memory are enriched and strengthened. Mr. Milligan expressed his hearty coöperation and sympathy with the Chautauqua scheme, and his pleasure that it is associated with the churches. A brief round-table conference followed the addresses, in which thought and experience were interchanged, and inquiries as to the methods and progress of the Chautauqua Idea were answered by the president, and by the Canadian secretary, Mr. Peake. The local press is doing good work in bringing the advantages of the scheme before its readers, and public interest is awakening in all directions in regard to it.



SUNBEAMS FROM THE CIRCLE.

C. L. S. C. class of '87 sends out the following circular to its members:

Beloved Friends and Co-Laborers:—We greet you with joy and gladness as we enter upon our four years' college course of reading and study. It is wisely selected and admirably prepared for us in our home life. We that toil ten hours in the shop, office, and store, with the never ending farm life and detail of housekeeping, will know not a little struggle to command forty minutes per day; but we need it and will do it. Mary A. Livermore was forty-five years of age before ever attempting public speaking, and in a decade was queen of the American rostrum. Some of you at Chautauqua, last August, remember the determined earnestness of Louise R. F. Jones. She writes: "Aiken, S. C., Oct. 6. Have formed a local circle of thirteen; first meeting last night at our house, two men, eleven women; sent for our books yesterday. Have persuaded two persons in Augusta, Ga., to join the C. L. S. C. In Langley, a small town eight miles out, my 'Hall in the Grove' has been read, and a circle is the promise. In Spartanburg, S. C., a circle is formed, which, with Aiken, are the only two in the Palmetto State, so far as known." This Pansy Class of '87 ought to graduate at least 10,000, and with five hundred members like our South Carolina friend, it would be accomplished. One of our class travels, and in forty days visited over thirty newspaper offices, begging editors to publish the C. L. S. C. leaflets, and securing their sympathy. Another one, (just completing his three score years) when on trains, goes from car to car, and politely and quietly seating himself in front or back of the passenger, introduces the "People's College." Our motto, "Neglect not the gift that is in thee." Inspired words! Let each one of us make them ours in the best sense. Let us introduce circles as rapidly as possible. Every day that passes now is forty minutes behind, and it is difficult to catch up. It can be done up to the first of January. Class writing paper has been prepared. Communicate with either of the undersigned concerning it. Rev. Frank Russell, Mansfield, Ohio, president Class of '87. K. A. Burnell, 150 Madison St., Chicago, secretary Class of '87.

The Rev. C. S. Woodruff, of Bayonne, N. J., class of '82, was present at Chautauqua this year, and passed under the Arches with the class of '83. On his return home he took occasion to preach upon the subject of education, and mention the Chautauqua plan particularly. As a result he has organized a local circle of over sixty, and it is still growing. He says: "Every pastor ought to visit Chautauqua. After being inspired he should spread his enthusiasm among all his people. Let us cast out the devil of bad literature by giving the people good reading."

The Johnstown, N. Y., local circle, includes among its officers a critic and an orthoepist—two excellent officers. Much exact knowledge of pronunciation, spelling, use of words, and forms of expression may be obtained at evening sessions, if critical and wise persons are selected.

There is an energetic circle of twelve members at Shushan, N. Y., the outgrowth of one member who began the readings two years ago. There is something contagious in the C. L. S. C.

Nothing could show better the peculiar work of the C. L. S. C. than the following suggestive toasts offered at the "Opening Day Exercises" at Meriden, Conn.; they were: "The C. L. S. C., a beneficial force in the life of a business man; as a coöperative with the duties of a school teacher; for young working people, establishing an alliance between labor and culture; as promoting Christian growth and culture; for the wife and mother at home."

A very pleasant and inspiring piece of news comes from the same circle. A young printer belonging to the circle became so much interested in his studies, and so anxious for further development that leaving his trade he has undertaken a college course. The circle did a kindly act when they presented to him that most necessary book for a student—Webster's Unabridged Dictionary.

A local circle whose work is done by correspondence has been started in the province of Ontario. There are but two members, but a dozen might carry on the work with equal success. For readers who are remote from the regular societies this plan may be utilized and made a means both of culture and of sociability.

A novel and exceedingly useful idea comes to us from Union City, Indiana. On a neatly printed program there is given the outline of the exercises for four months. The circle meets fortnightly, and the date, place of meeting, exercises and participants are given for eight sessions, so that there can be no mistake or misunderstanding about the work to be done. The plan is to be commended to all circles.

The class of '85, C. L. S. C., held a meeting at the grounds of the New England Assembly at Framingham and organized by the election of the following officers: President, Rev. J. E. Fullerton, of Hopkinton, Mass.; Vice Presidents, Miss Lena A. Chubbuck, New Bedford, Mass., Alice C. Earle, Newport, R. I., Miss Marcia E. Smith, Swanton, Vt., J. B. Underwood, Meriden, Conn.; Secretary and Treasurer, Albert B. Comey, South Framingham, Mass. Plans were suggested looking to the social and other interests of the New England members; said plans to be perfected as soon as the details can be arranged by the executive committee. It is earnestly desired that all persons in the New England States belonging to class '85 will send their name and address to the Secretary. The President cordially solicits correspondence from members of the class upon matters pertaining to its interests.

From the Silver Creek, N. J., *Local* we learn that the meeting for re-organization of the C. L. S. C. has been held, and that on October 8 the first regular meeting took place. Several new members have joined the circle.

At Spring Mills, N. J., though several members have moved from the village, and a few have dropped the course, they report a prospect of doubling their numbers.

A circle of eleven members is reported at Greencastle, Pa.

Osceola, Iowa, has a circle of seventeen members, class of '87.

Some one inquires for a copy of "the rules of the C. L. S. C. to guide in their meetings." There are no rules to guide in the meetings of the local circles. The wide diversity of circumstances under which they exist would make a fixed organization impracticable. What would fit the great circles of Troy, N. Y., and Pittsburgh, Pa., would be of little use to the small circle of the village. The many plans and outlines of work in this department are presented especially to guide new circles to the plan best suited to their needs.

Montana has sent us reports of several energetic circles; the one at Bozeman, of fifteen members, is the last reported.

A circle has been organized at Hood River, Oregon.

The Summer Assembly at Monteagle, Tenn., did some excellent work in the interest of the C. L. S. C. Many circles are being formed as a result of the efforts made there to spread information concerning the methods and object of the organization.

One zealous C. L. S. C. worker writes us that while traveling through the west in search of health she has succeeded in making many think about the course, and has persuaded ten to enroll for '84. It is such individual effort that extends the boundaries of our work.

At Mountain Lake Park, Md., Assembly there was formed last summer a very interesting circle. The members are widely scattered. They come from West Virginia, Maryland and Pennsylvania, but have formed the "Mt. Lake Park C. L. S. C.," which they are keeping up while belonging to the local circles at their homes. There are some forty-three members.

As an example of how the attention of your community may be drawn to the C. L. S. C., we quote the following from the Petersburg Va., *Mail*: “Everybody has heard of Chautauqua, and the readers of *The Mail* have enjoyed several excellent papers concerning that institution from the pen of Mrs. C. D. Tinsley, of this city, who spent the summer there. But there are many people who do not know that the C. L. S. C. is spreading out its branches in all directions and offering very fine advantages to people who desire to undertake a systematic course of reading. The course extends over a period of four years, and embraces religious, scientific, and general literature of a substantial character. The books are cheap, and it is said that one may cover the whole course by reading for forty minutes each day. At the end of the fourth year, if the student has gone over the ground, a diploma is given, bearing the seal of the C. L. S. C. A number of ladies and gentlemen of this city have handed in their names. The writer is favorably impressed with what he has seen of it, and cheerfully commends it to the public. All information required may be had of Mr. C. D. Tinsley, of this city.”

A member from Canada writes: “As one of the class of ’84—the ‘Irrepressibles’—and having caught the inspiration at Chautauqua, I can hardly write or say anything strong enough to express my admiration of the movement. I wish the officers could do something for Palestine. When visiting it a year ago I induced my dragoman, Herbert C. Clark, of Joppa, to subscribe then and there for THE CHAUTAUQUAN, as we were sitting on the ruins of the old wall of Mount Zion, above the valley of Hinnom. Mr. Clark writes me that he enjoys it exceedingly. I was much of the time for ten days with Dr. Selah Merrill, the U. S. Consul, and his lady, who worthily represents the women of America. There are many fine people speaking the English language in Jerusalem and other points, who are cut off from many of the advantages of our Christian civilization. Nothing prospers under the administration of the stupid Turk, and literature especially is discouraged. I believe the C. L. S. C. is just what these good people need.”

LOCAL CIRCLES.

Vermont (West Brattleboro).—For the past three years, there have been at West Brattleboro, Vt., informal gatherings of from eight to twelve members of the C. L. S. C., who called themselves a “circle.” But until this year there has been no formal organization. At the meetings subjects were taken up in the way decided upon at the previous meeting. These meetings were found to be of much profit and interest to ourselves. As we learned to know and appreciate the “Chautauqua Idea,” we began to see growth could not be expected without organization. Accordingly a meeting was called for the evening of September 20, to which all were invited, whether they proposed to join or not. At this meeting the aims and methods of the C. L. S. C. were fully discussed, and an organization was effected under the name of the “Vincent Circle,” with a president and secretary. There was also a committee for program chosen, to serve for two months. It was decided to hold meetings once in two weeks, at houses of the members. We are happy to say that we start upon our new year with a membership of thirty-nine, eighteen of whom are regular members, representing classes ’84, ’86 and ’87. Nor is this all the gain. Another circle has been organized, under the name of “Pansy.” This circle is composed wholly of new members, who thought they could work to better advantage separately. It has twenty-two members. No doubt the secretary will report to you, as the circle is very wide-awake, and means to profit by the experience of the ’84s. It has been thought best to devote the time at the meetings during October and November to Grecian history, as that is the principal subject for study during those two months. The following was the program for the first meeting, October 4. The second meeting was similar in character:

1. Responsive Reading from “Assembly Hymnal.”
2. C. L. S. C. Song No. 6, from “Assembly Hymnal.”
3. Report of Secretary.

4. Introduction of the subject of "Grecian History," by Prof. H. H. Shaw.

5. Paper on "The Advantages of the Study of Grecian History."

6. Instrumental Music.

7. Reading of Selections pertaining to Greece, from the Second Canto of "Childe Harold."

8. Blackboard Drill on "Outlines of Grecian History," by Rev. C. H. Merrill.

9. Question Box, Questions to be answered at next meeting.

10. C. L. S. C. Song No. 19.

11. Closing Prayer.

After the regular exercises, which began at 7:30, closing at 9, an hour was spent in a social way.

Massachusetts (Lawrence).—Immediately after the Assembly at Framingham, a meeting was called in the interests of the Circle in one of our city churches, at which its purpose and method of working were fully explained. Two circles were already in existence, and with these as a basis we put in some hard work during the month of September, securing to date forty-two new members, with more to come. Two additional circles have been formed, so that we now have four, with a total membership of about one hundred. October 1, Opening Day, was duly observed by a union Round-Table of all the circles, and a large number of invited friends. A program consisting of music and readings was given, all appropriate to the occasion. We have engaged Prof. W. C. Richards for a course of lectures in November, and shall have others from time to time through the winter. We have also arranged for a monthly union meeting, each local circle in turn conducting the exercises for the evening.

Massachusetts (Franklin).—As the Bryant Bell at Chautauqua rang out its call to study on October 1, the members of our local circle assembled to celebrate the first anniversary of their existence as a local circle. Complimentary tickets were issued to their friends, and at the hour of opening the chapel was filled, the audience numbering not far from five hundred. Promptly on the hour, the new members of Class of '87 (the Pansy class) marched into the room, and taking position in open ranks allowed the Class of '86 to pass through; they taking position on the right, opened ranks, and allowed the president of the circle and the speaker of the evening to pass through, receiving as a greeting the Chautauqua salute. The program consisted of instrumental music, singing of selections from Chautauqua Songs, an address of greeting from the president, Rev. G. E. Lovejoy, the commencement address by Rev. A. E. Winship, of Boston, and the recital of the anniversary poem by Miss Laura Pond. The whole affair was a helpful and enjoyable opening of the Chautauqua work for 1883 and 1884. The circle starts upon its work with increased membership and enthusiasm, and one and all are ready to say God bless the originator of the C. L. S. C., and God speed the work in the days to come!

Connecticut (Meriden).—The Meriden branch of the C. L. S. C., held Opening Day exercises; nearly every member was present, together with a few invited guests, mostly those who have especially assisted them in their work during the past three years. The exercises were opened by the circle singing from Chautauqua Songs a song of welcome, after which an address of introduction of the several classes to the guests and a synopsis of the work of the Circle, was delivered by the president, who also took occasion to speak encouragingly to each class, and referred to their several colors and what they symbolized. At the close of the address a prayer of thanksgiving was offered, when the company sat down to a banquet of good things. After supper several toasts were offered and responded to, and several testimonials of interest in the success of the organization offered. The C. L. S. C. feel justly proud of their success, and all who have taken time to examine into its aim and the results accomplished, commend them highly.

Connecticut (Hartford).—Within a few days a general interest has been manifested with regard to a C. L. S. C. circle in Hartford—more than fifty

having expressed their desire to become members of the Class of '87. Last year, however, Hartford had but a few Chautauqua readers. Among them was a little circle of five young ladies not long out of school. They found the Chautauqua course just what they needed to give form and direction to their studies, and they sat down to the table of good things spread before them as to a mental banquet. A severe bereavement met the circle in the loss of one of their members, a young lady who had been an eager student and whose enthusiasm had done much to help the circle. By her suggestion Greek had been introduced into the course, and the Iliad was being read in connection with the Greek literature. At the last meeting before her death when it was proposed to omit some of the less interesting portions, she said, earnestly: "Don't let us skip any. Let us do our duty." The shock of her death was such that at first it seemed that they could not go on with their work, but the words of their departed friend came back to them with peculiar meaning: "Let us do our duty," and with chastened hearts they took up their work again. They did not find the course too laborious, but were able to add to it the White Seal course and some valuable supplementary reading upon the topics in question. Now, with undiminished interest, they are ready to go on with the second year, hoping that a large band will be ready to accompany them.

New York (Johnstown).—A meeting for the reorganization of Johnstown C. L. S. C. was held September 26, 1883. The names of fifteen new members were enrolled; so we launch our little craft of twenty-two members, with a prospect of taking an occasional recruit as we journey on. We have decided to meet every alternate week. At our next meeting, October 10, we read an outline on Greek History, Vol. ii., Part vii., a paper on American literature, and selections from THE CHAUTAUQUAN.

Pennsylvania (Erie).—The officers and members of the Erie local circle of the C. L. S. C. presented a very elaborate program at its opening session Monday evening, October 8, at the Y. M. C. A. Hall in this city. The hall was densely crowded, and the interest steadily increased to the close. The organization was completed, and its roll bids fair, this season, to be seventy-five strong.

Delaware (Wilmington).—Through the zealous efforts of the pastor of Asbury M. E. Church, the Asbury local circle was organized during September last. It has about thirty members. Among the members is one graduate of the class of '83. Considerable interest in the course has been aroused through the city, and there are more persons to join.

District of Columbia (Washington).—A meeting of the Banneker Circle was held September 17 for reorganization and general talk concerning the work for 1883-84. Quite a large number of our members of last year attended, and from the number of applicants for admission, it seems that we will be compelled to abandon our idea of meeting from house to house of the several members and meet at the church. It is exceedingly gratifying to note the continued interest in the work. Our meetings are held every Monday night. The pastor of one of the churches in another section of our city, attended our last meeting, in order that he might learn enough about the C. L. S. C. to organize a circle among many of his members, who seem anxious to join. We spent many pleasant and instructive evenings last year over our work, and hope to realize as much benefit from the studies of this year. Knowing of the benefits of the C. L. S. C. we are always glad to help others to join. One of our members has been influenced, through last year's work, to attend college.

Ohio (Cincinnati).—The reception to the Class of 1883, of Cincinnati and vicinity, took place on Friday evening, September 28. The spacious parlors where the reunion was held were fragrant with flowers. A beautiful piece of crayon work—"Welcome, 1883," with C. L. S. C. monogram—prepared by the superintendent of penmanship of Cincinnati public schools, together with a fine portrait of Dr. Vincent, held conspicuous places. The following was the program:

Piano solo—Miss Clara Looker.

Address of welcome to the Class of 1883—Mr. John G. O'Connell.

Class song of 1882.

Toast—"The Class of 1882." Response by Mrs. M. J. Pyle.

Class song of 1883.

Toast—"The Class of 1883." Response by Mr. Clifford Lakeman.

Vocal solo—"The Flower Girl." Miss Clara Looker.

Toast—"The Cincinnati Circles." Response by Miss Bessie Hicks.

Song—"Join O Friends in a Memory Song."

Toast—"Chautauqua." Response by Mr. M. S. Turrell.

Song—"C. L. S. C. Commencement Carol."

Toast—"Our Chancellor, Dr. J. H. Vincent." Response by Miss Harriet Wilson.

Song—"Sing Pæans over the Past."

Letters of regret were then read from unavoidable absentees. Time and space will only permit of the publication of the following letter, which is an embodiment of the sentiment contained in the others:

HOT SPRINGS, ARK., September 24, 1883.

Rev. J. G. O'Connell, President C. L. S. C. Alumni Association of Cincinnati, Ohio:—Please accept my thanks for your very kind invitation to attend the C. L. S. C. reception, Friday evening, September 28. The intervening 700 miles will prevent. But does not the Chancellor of the Out-of-Doors University say that, "When the bell at Chautauqua rings on memorial days, all true Chautauquans hear its echo?" And as this same Chancellor teaches so diligently the superiority of mind over matter, why may I not apply this teaching to my own case and say to you that I will be with you in some sort of soul-telephonic manner, and hear your speeches and join in your songs, and enjoy with you the feast of reason and the flow of soul?

I am sorry I said I couldn't go. I think you may expect me. I read most carefully the report of Commencement Day, and welcomed (in my heart) all the '83s.

A popular writer in a most popular magazine says: "There are in this life three stages of existence. The first, when we believe every thing is white. The second, when one is sure every thing is black; the third, when one knows that the majority of things are simply gray."

Members of the C. L. S. C. have gone a step further than that. To us, all the world has a *golden* hue. How *can* one fully understand the meaning of the terms, "communion of Saints," and "brotherly kindness," unless he has spent a season at Chautauqua *as a student*, in full sympathy with the great work being done there? What grand opportunities are there afforded for growth and symmetrical development of character.

Please tell your Alumni Association how glad I am to be counted one of its members. I thank you again for your kind remembrance of me.

Wishing you a most joyous reunion, and uniting with you in warmest love for our Alma Mater, I am yours sincerely,

HATTIE N. YOUNG.

The officers were elected for the coming year, and after a handsome collation bountifully served, the society parted for the evening, filled with additional enthusiasm for the success of their Alma Mater. President, Mr. John G. O'Connell; Vice Presidents, Mr. M. S. Turrell, Mrs. M. J. Pyle, Miss Mary E. Dunaway; Corresponding Secretary, Mr. Clifford Lakeman; Recording Secretary, Miss Julia Kolbe; Treasurer, Miss Selina Wood.

Illinois (Mattoon).—This is the first year of the C. L. S. C. of Mattoon. We organized the last of September, and have an enthusiastic membership of over twenty. We take the lessons as given in THE CHAUTAUQUAN, sometimes assigning the work to individuals, and again we have general recitations. During the winter we had an afternoon with Longfellow; also a

lecture upon the History of Greece, and one upon the Sun, with diagrams. Most of us have completed the work for the year, and have written the memoranda. Our meetings have been both profitable and interesting.

Iowa (Anamoso).—Our C. L. S. C. circle was organized in January, 1883, with a membership of nine ladies, all of whom have taken up the four years' course of study. The order of exercises varies somewhat, but is always exceedingly interesting, each study receiving due investigation and research. Generally, however, our president assigns the different subjects to the members on the preceding meeting, thus giving each leader time to prepare questions which will bring out all the points of interest in the lesson. Amid crowding duties we are glad to note in our membership an increasing enthusiasm over the C. L. S. C. work.

Iowa (Quasqueton).—We are a struggling little company of two regular members of the C. L. S. C. We have not been lacking in interest ourselves and are heartily in sympathy with the C. L. S. C.; think it is a grand, good thing.

Missouri (Kansas City).—The Kansas City local circle was reorganized on September 25, and was ready to begin work promptly the first week in October. We have at present forty-four members. Our circle has propagated the Chautauqua Idea, and sent off branches until now there are at least six circles in the city, and about three hundred of our citizens are reading the course.

Missouri (Independence).—A local circle was organized here in September with forty-seven regular members. We have a president, vice president, secretary, treasurer, and board of managers. We meet every Friday evening, and thus far have followed the conversational plan. All are interested, and the Chautauqua enthusiasm has the true ring. Already the '87s are looking forward to the day when they will pass through the Arches.



C. L. S. C. ROUND-TABLE.

From the record of '82. Held in the Hall of Philosophy in August, 1882, at 5 p. m. [This report had been overlooked, and as it contains much that will be interesting, is here published.]

DR. VINCENT: What are the advantages of the C. L. S. C.? What are the advantages to our homes?

A VOICE: Unity in the family, in study and spirit.

A VOICE: System of reading at home.

A VOICE: It brings good literature into the house.

A VOICE: It trains intelligent citizens in the house.

A VOICE: It saves time that would be otherwise wasted.

A VOICE: It gives pleasant subjects of thought while we are about our daily work.

A VOICE: It promotes conversation.

A VOICE: It leads us into new lines of work.

A VOICE: It makes us more attractive to each other.

A VOICE: It keeps husbands at home in the evening. [Laughter.]

MR. MARTIN: It keeps wives home in the evening.

A VOICE: It crowds out unprofitable occupation.

A VOICE: It leads to farther investigation.

A VOICE: It cultivates the conversational powers.

DR. VINCENT: It not merely brings subjects of conversation, it brings the power of conversation.

A VOICE: It makes the Southern people love the Northern people.

A VOICE: It lifts the home up a little higher.

A VOICE: It crowds out gossip.

A VOICE: It cultivates a missionary spirit.

DR. VINCENT: In what respect?

A VOICE: In getting people into the circle and into all kinds of work.

A VOICE: A lady says it makes the evening hearth exceedingly pleasant.

A VOICE: It inspires us to want to help others.

A VOICE: It has in one instance made a Christian of an Infidel.

A VOICE: In more than one instance.

A VOICE: There is a book in the course that will do that every time it is attentively read.

DR. VINCENT: What is that?

A VOICE: "The Philosophy of the Plan of Salvation."

A VOICE: And the "Tongue of Fire."

A VOICE: And "The Outline Study of Man" is a wonderful book.

A VOICE: It helps fathers and mothers to grow up with their children.

A VOICE: It helps them cultivate their memory.

A VOICE: I found that I could remember dates much better than before.

REV. W. D. BRIDGE: It brings the old into sympathy with the young.

A VOICE: It gives even old men books they would not have read.

A VOICE: It creates a spirit of union among all kinds of people that belong to it.

A VOICE: It brings the grown people into sympathy with the public school and its work.

A VOICE: It makes us better Christians and workers in the church.

A VOICE: It was suggested a moment ago that it brings the older people into sympathy with the young: I think it brings the young people into sympathy with the old.

A VOICE: It increases the respect of the young for the old also.

A VOICE: It teaches old people to become younger.

A VOICE: It makes old people wish that the thing had been thought of earlier.

A VOICE: It brings us to Chautauqua.

DR. VINCENT: That is a great thing for Chautauqua as well as for us.

A VOICE: It teaches us never to be discouraged.

A VOICE: It teaches us the spirit of propriety.

A VOICE: The first of the Chautauqua mottoes has been noticed; the other two should come in for their share.

DR. VINCENT: The other two mottoes should be recognized. It helps us to “keep our Heavenly Father in the midst.”

A VOICE: It shows in the class of '82 the proof of the third motto, “Never be discouraged.”

A VOICE: It teaches us to “look up, and not down.”

DR. VINCENT: To “look forward and not backward,” to “look out and not in,” and “to lend a hand.”

A VOICE: It leads to an investigation of science by people who had never thought of it before.

MR. INGHAM: It teaches all classes to find a book store.

DR. VINCENT: Brother Ingham is in the book trade. [Laughter.]

A VOICE: It teaches people that no one is too old to study.

A VOICE: It gives a higher idea of the responsibility of life.

A VOICE: It makes the bookseller keep good books. [Applause.]

DR. VINCENT: It makes the bookseller keep the books at a lower figure.

A VOICE: It develops the habit of systematic thought and work.

A VOICE: It discovers people to themselves, showing themselves their natural bent and power.

A VOICE: It breaks down the deep seated denominational prejudices.

DR. VINCENT: Without in the slightest degree diminishing our loyalty to them.

A VOICE: It fits the mind for its eternal mission and home.

A VOICE: It makes one see what a wonderful thing a book is.

A VOICE: It puts the divine idea into all the study: "We study the words and works of God," and this promotes unity of scientific and religious pursuits.

A VOICE: It selects a course of reading that we would not ourselves select.

A VOICE: It teaches us the value of time.

A VOICE: It teaches us to recognize God in everything.

A VOICE: It furnishes a good channel for the expenditure of money in connection with young people.

DR. VINCENT: We ought to say in connection with that, it builds up an individual library that acquires an individual preciousness; when a man looks at it he is rich, for he owns books bought himself. The square yards of books are not worth much. The books that are mine are worth much to me.

A VOICE: It makes it plain that the world is going forward and not back.

A VOICE: It helps the world to go forward, and helps others to acquire knowledge.

A VOICE: It gives us a hint as to the powers and possibilities of the mind.

A VOICE: It teaches me how very little I know myself.

A VOICE: I think it teaches old and young to appreciate art in its different forms.

DR. VINCENT: It enables people to distinguish between good preaching and poor preaching.

A VOICE: It teaches that faithful labor, though in a very limited degree, will be rewarded here and hereafter.

A VOICE: And that it will accomplish a great deal of good in addition to the reward.

A VOICE: It awakens latent energies in the mind.

A VOICE: It makes the common people better critics.

DR. VINCENT: It makes what they would call where caste prevails "common people" better critics. We have no common people in this country. We are all kings.

A VOICE: It makes us understand better the Chautauqua Idea.

A VOICE: It makes us patient in weakness and suffering.

A VOICE: It helps us bear the burdens of life.

DR. VINCENT: In many places there is no social enjoyment for those who do not dance. The C. L. S. C. gives us congenial society. I have known many people where the habit of dancing and card playing prevailed, to justify these indulgencies on the ground that there was nothing else to do. In a few such places the C. L. S. C. has turned the dance and the card table out of doors. Of course some of you do not look at that matter as I do. There may be some of you who dance or allow your children to attend dancing

school, and some of you allow your children to play cards. I have avoided dogmatism on all subjects where the Word of God does not come in as the final authority. I never like to dogmatize about these things. But I do believe that such is the condition of society to-day, and such are the unseen perils of the day—perils always present—that the family that can enjoy itself thoroughly in an intellectual way, so as not to create a taste for the stimulating power of the dance and the card table and of the theater is a safer, and in the long run, a happier family than the family otherwise controlled by so-called worldly tastes. [Applause.] It becomes us to be very free from dogmatism about these things, because we do not want to lay down laws that have not been laid down for us; but if we can, let us substitute the influences of the C. L. S. C. for these things.

WRITTEN PAPER: The C. L. S. C. gives new hope and courage to those who have thought that the days for personal improvement had gone by.

DR. VINCENT: Dr. Wilkinson, in his address the other day, made reference to the fact that I myself had never enjoyed college opportunities. I did enjoy the very best academic opportunities up to the time that I should have entered college, but circumstances, which seemed very much like Providence, interposed at that crisis in my life, where the question was settled by three contingencies. I suffered from a bronchial affection, and my friends regarded me in great peril physically. I submitted three questions to three men after serious thought and earnest prayer, and resolved to be governed by the decision of the three men if they should decide in the same line. To one, an able scholar and a most efficient preacher, and a man occupying a high position in the church, I submitted the question of my intellectual fitness, and gave him a long account of my intellectual history. To another man, my father, I submitted the financial part of the business. That was a question that he alone could settle. To a distinguished physician, one of the ablest in New York City, I submitted the question of my physical health. Now, said I, if these three men combine in their decision, I shall consider the question settled in that way. If they differ, I shall consider it still open. The decision of all three was quite in a given line, and I entered very soon into the active ministry.

The fact that I lacked the *prestige* of the college was humiliating to me to the last degree. It made me morbid for years. I was too honest to impose

on people, and therefore too likely to betray myself where no good could come of it, and where there was no necessity of it. But my humiliation led me to do this thing: To turn my theological studies and the preparation of sermons into means of mental discipline; to acquire the habit of laying hold of a subject, and of holding on to it, and persisting in holding on to it until I could master it, so that if I did not have more than a smattering (and I did have a smattering of Greek and Latin and Hebrew to begin with), I would have the discipline of thinking on subjects and of tearing them open on my own account. I tried to do that through all the years of my active ministry.

I drew up for myself a sort of C. L. S. C. thirty years ago, and took glimpses of all that the boy examines in college, so that the C. L. S. C. of to-day developed out of it, and different as it may be, it is the result of bitter experience and immense effort, so far as I was personally concerned.

I really ought not to have mentioned these things to you. I have never done so anywhere except to a limited circle of friends. When I watch boys in college, their pleasures and struggles; when I look at the buildings, at the bronze statue of the first president of Yale, the libraries, the art department, the scientific department; when I hear that old bell ring from day to day, when I look on the *campus* and see the boys marching or lounging, singing the college songs; when I see them striving for preëminence in the athletic arena; when I remember that certain prerogatives depend upon victory on this side or the other; when I see old men who were students fifty or sixty years ago, the oldest that are left, and see the joy that comes from the inspiration of such memories, then I see that it is a great thing to be able to give old people and every-day people a touch of the joy and hope and memory that colleges alone can give, and no one unless identified with such an institution can feel.

It is for that purpose that we have the “Hall in the Grove,” and the “Arches,” the “Memorial Days,” the “Badges,” the “Diplomas,” etc. Privileges heretofore limited to college life are thus and now guaranteed to the old and the young. This is another benefit that comes from the C. L. S. C. [Applause.] I should have taken a shorter time to tell it, but I could not.

WRITTEN PAPER: In accordance with your request for the members of the Circle to remember each other at the throne of grace each Sabbath

afternoon, would it not be well to have a set hour, say five o'clock, Sunday afternoon?

DR. VINCENT: The suggestion is a good one. We will call five o'clock Sunday afternoon "Our Sacred Hour." Mr. Bridge, make an item for the columns of THE CHAUTAUQUAN, that it may reach all the members of the Circle.

As I said the other night, we are not all of the same way of thinking, but we may all think upward, and whatever the degree of our thought and the kind of our faith, if the look be upward, there will be an uplift. If with sincere desire we pray for others and seek God's glory, he will lead us into all truth. Let us appoint with your approval five o'clock Sabbath afternoon for the uplook in order to uplift. Those who approve lift your hands.

My friends, while the formal worship—the going aside and kneeling down, and observing the form of worship—is very useful, the idea of prayer is not limited to the place or particular mode, or to the words you speak. Prayer is sometimes the mightiest that leaps without words out of the inmost heart to the highest heaven. Let us think a prayer wherever we may be. Sometimes when people are too busy with their hands and under the pressure of every-day labor to retire, and have not words or place for the specific act of prayer, the uplift of the soul, the upreach, is prayer that brings down abundant blessings. Let it be so with us. Let us not be bound too much by times and circumstances and words. Let us have the heart, and let forms and words come as they will, and let us not neglect times and forms and words.



QUESTIONS AND ANSWERS.

ONE HUNDRED QUESTIONS AND ANSWERS ON “EASY LESSONS IN VEGETABLE BIOLOGY.”

By A. M. MARTIN, GENERAL SECRETARY C. L. S. C.

1. Q. How is the word Biology made up, and what does it mean? A. It is made up of two Greek words—*bios*, life, and *logos*, a discourse. It means the study of living things.

2. Q. What does Biology include in its survey? A. Both animals and vegetables, and considers their forms and peculiarities, the parts of which they are composed, their relations to each other, and the uses which they serve.

3. Q. What are the subjects of Physics and Chemistry? A. The general forces of nature and the changes in non-living matter.

4. Q. What is the teaching of the Bible and of all the religions of mankind, the belief of the most eminent philosophers, the doctrine held by the early Christian fathers, and maintained by the majority of scientific and unscientific men as to the difference between a living body and the same body after death? A. That it arises from the union of matter and spirit.

5. Q. What is it that entitles any thing to be called a living being? A. The presence of little particles of living matter scattered through it.

6. Q. What does this living matter look like when seen through the microscope? A. Like a little bit of jelly or albumen. It is generally transparent; is neither quite solid or fluid.

7. Q. What is it called? A. It is often called protoplasm, or first formation. It is also called by the better term bioplasm, or living formation.

8. Q. What is said as to the resemblance of the particles of bioplasm to one another, no matter where they belong? A. They always look alike. There is no difference under the microscope between the bioplasm of a blade of grass or a whale, or an oak, a rose, a dog, or a man.

9. Q. What does chemical examination show as to all living matter? A. That it is composed of the same elementary materials. Oxygen, hydrogen, carbon, and nitrogen enter into the construction of every piece of bioplasm.

10. Q. In what three different states do we recognize matter in every bioplast, or living particle? A. Matter not yet alive, but about to become so, called pabulum, or nutriment. Living matter in the strictest sense, or bioplasm. Formed material, or matter which was alive, but is so no longer.

11. Q. What peculiarity has living matter as to motion? A. All bioplasm has spontaneous motion. Non-living matter has inertia.

12. Q. What are the three kinds of motion of bioplasm? A. Inherent motions of individual particles among themselves. Constant change of shape. Wandering movements.

13. Q. What is the peculiarity of living matter as to the power of nutrition and growth? A. The non-living increases in size by external additions; but bioplasm selects appropriate material from its food, or pabulum, changes the chemical relations of this material, and appropriates it to its own structure in such a way that it grows from within.

14. Q. What is the peculiarity of bioplasm as to reproduction A. Bioplasm can generate or reproduce its own kind of living matter.

15. Q. What power has a living thing to preserve its own identity? A. A living being preserves its identity amid all the material changes which take place.

16. Q. In the grouping together of living things according to their real relationships, what do types represent? A. General plans of structure.

17. Q. How are classes formed? A. By the special modification of a type.

18. Q. What are orders? A. They are groups of the same class related by a common structure.

19. Q. What is a family or genus? A. A still smaller group having generally the same essential structure.

20. Q. What is a species? A. It is the smallest group whose structure is constant.

21. Q. What are individuals? A. They are the units of organic life, forming a complete animated existence.

22. Q. What are peculiarities of races or breeds called? A. Varieties.

23. Q. How are vegetables and animals distinguished from each other? A. By the term kingdom, and the types in each kingdom are called sub-kingdoms.

24. Q. Under what five types or plans of structure can all the multitude of plants which clothe the earth or dwell in the sea be arranged? A. Protophytes, Thallogens, Acrogens, Endogens and Exogens.

25. Q. What are the elementary masses of bioplasm usually called? A. They are usually called cells, even if they are merely pieces of animated jelly, uninclosed by an outside shell or membrane.

26. Q. What is the principal difference between animals and plants? A. The latter can be nourished by simple mineral or chemical (that is unorganized) matter, while animal nutrition requires material which has been organized, or made part of a living being.

27. Q. What do most vegetable cells produce on the outside? A. A membrane or cell wall, within which the living matter is, as it were, imprisoned.

28. Q. What concentrations of living matter are there within a cell? A. A concentration called a *nucleus*, and sometimes a still further concentration within the nucleus, called *nucleolus*, or little nucleus.

29. Q. Of what substance is the cell wall composed? A. A substance somewhat like starch, called cellulose.

30. Q. When it becomes solid how is it known? A. As woody tissue.

31. Q. How is common wood made up? A. Of a number of these cells arranged side by side.

32. Q. Of what shape may vegetable cells be? A. They may be globular, oval, conical, prismatic, cylindrical, branched, or of any other form.

33. Q. What are some of the varieties of formed material into which the bioplasm within the cell wall may be transformed? A. They may be solid, as coloring matter, starch, crystals, and resin; or fluid, as oil and gum, or solutions of sugar or tannin.

34. Q. What is the most important of these substances called? A. Chlorophyll, the source of the green color of plants.

35. Q. What other product of vegetable cells is even more widely distributed than chlorophyll? A. Starch.

36. Q. How do cells generate? A. By self-multiplication.

37. Q. What are the simplest forms of plant life? A. Those that consist of a single cell.

38. Q. In the higher classes of plants what is the character of the union of cells which forms tissues and organs? A. It is permanent.

39. Q. What are made by the union of cells into groups? A. The woody fibers of plants, and the cellular tissue which makes the softer, fleshy and pithy parts.

40. Q. What has observation shown as to the production of new cells in the highest plants? A. That they are not produced everywhere uniformly, but in particular spots.

41. Q. What terms have been applied to places of this kind? A. Growing-point, and growing or formative layer.

42. Q. Where may growing-points and formative layers be seen? A. Growing-points may be seen in the tips of buds, and formative layers between the wood and bark of trees.

43. Q. What names have been given to the tissue which is here formed by the division and union of cells? A. Formative or generating tissue.

44. Q. What are in direct contrast to the generation tissues? A. The healing tissues, or cork tissues.

45. Q. How are vessels made? A. By the union of several cells, the partition-walls disappearing, while the union continues at the margin.

46. Q. What are bast-tubes or bast-fibers? A. They are long, pointed, thick-walled tubes, commonly united into bundles.

47. Q. To what part of the flower is the term nectaries, or honey-glands, given? A. To any part of a flower which secretes honey or sugary fluids.

48. Q. What is the first independent tissue formed in flowering plants by the union of cells? A. The epidermis or skin.

49. Q. What is each of the pores found among the epidermic cells called? A. A stoma, or mouth.

50. Q. What are hairs? A. They are epidermal structures, composed of one or more cells.

51. Q. What do we find next to the epidermis? A. The cortex, or bark, often composed of cells containing starch or chlorophyll.

52. Q. What is beneath the bark? A. The formative layer or cambium, in which thin-walled cells become transformed into vascular or bast-cells, and thence are changed into permanent cells.

53. Q. What do groups of cells thus formed, united into bundles and penetrating the rest of the tissue, form? A. The fibro-vascular bundles.

54. Q. What are the simpler types of plants that have no fibro-vascular bundles, called? A. Cellular plants.

55. Q. What are the rest termed? A. Vascular plants.

56. Q. Of what does the fundamental tissue generally consist? A. Of thin-walled cells containing starch, although other forms of cells may be present.

57. Q. What is the simplest form of individual plant life? A. A particle of living matter inclosed in a membrane or cell-wall.

58. Q. What are plants of this type of structure called? A. Protophytes.

59. Q. Where are many of these one-celled plants found? A. In the green slime which grows on stones and on boards in damp places.

60. Q. What is one of the simplest forms, often found in rain-water casks, called? A. The protococcus.

61. Q. What are the unicellular plants most interesting to those who study with the microscope? A. Diatoms.

62. Q. In the living state where are diatoms found abundantly? A. In every pond, rivulet, ocean and rock-pool.

63. Q. What do they form in a fossil state? A. Large strata of rock material.

64. Q. What are thallogens? A. Plants composed of a tissue of cells, or bioplasts, but with no clear distinction of stem, root and leaves.

65. Q. What three classes are included under this type? A. Algæ, or sea-weeds; Lichens, or the dry, leafy, or mossy patches on trees, stones, etc.; and Fungi, or mushrooms, molds, and their allies.

66. Q. Into what three orders have Algæ, or sea-weeds, been divided? A. The red, the olive and the green sea-weeds.

67. Q. How are Fungi regarded by some scientists? A. As neither animal nor vegetable, but forming a sort of third kingdom.

68. Q. What seems to be the principal business of the Fungi? A. The removal of the waste material of both animal and vegetable life.

69. Q. What are Acrogens? A. Plants which grow at the summit only, and not in diameter.

70. Q. What plants do we find in fresh-water ponds and rivers, growing in tangled masses of dull green color that illustrate the manner of growth in

the type of Acrogens? A. Stone-worts, consisting of two genera, Chara and Nitella.

71. Q. What are the nodes, and what the internodes in the stone-worts? A. The points on the axis, or stem, from which the branchlets spring, are called nodes, and the intervening parts are internodes.

72. Q. How is each internode formed? A. By the growth and elongation of single cells.

73. Q. How are the branchlets produced? A. By the sub-division of single cells.

74. Q. What other families of plants are examples of Acrogens? A. Ferns and Mosses.

75. Q. What are Endogens? A. Plants whose vessels and woody fibers first grow within the stem. The seed has but a single lobe, or cotyledon.

76. Q. What families of plants are found in the type of Endogens? A. Grasses, Rushes, Lilies, and Palms, with similar families.

77. Q. In the growing plant what part grows from the axis upward, and what part from the axis downward? A. The stem grows from the axis upward, and the root downward.

78. Q. What is the root formed by the downward elongation of the axis called? A. It is called the primary root.

79. Q. What is the stem of a plant? A. That part which bears the leaves, flowers, and fruit.

80. Q. What is the length of life of the stem and roots? A. It may be only a single year, or annual; two years, or biennial; or a number of years, or perennial.

81. Q. What are thorns? A. Undeveloped branches, and many plants which are thorny when wild are not so under cultivation.

82. Q. Of what are leaves constituted? A. Cells, with cavities, fibro-vascular bundles and epidermis.

83. Q. How do the veins in the leaves of Endogens differ from those in the leaves of Exogens? A. They are generally parallel or straight in Endogens, and do not form a network as in Exogens.

84. Q. What are five of the names given to leaves according to their shapes? A. Lanceolate, or narrow and tapering; oblong, or narrow and not tapering; cordate, or heart-shaped; sagittate, or arrow-shaped; and ovate, or egg-shaped.

85. Q. What is the function or use of leaves? A. To expose the juices of the plant to light and air, and thus aid in forming the woody matter of the stem and the various secretions.

86. Q. What constitute a plant's organs of nutrition? A. The root, stem and leaves.

87. Q. What is the flower of a plant? A. It is the organ, or assemblage of organs, for the production of the seed.

88. Q. What are the four whorls in which the parts of a flower are usually arranged called? A. The outer whorl is the calyx, the next the corolla, the third the stamens, and the innermost the pistil.

89. Q. To what is the term fruit applied in botanical language? A. To the mature, perfect pistil, whether dry or succulent.

90. Q. What nutritious grains are classed among the family of Endogens called grasses? A. Wheat, barley, oats, rice and Indian corn.

91. Q. What other families are noted members of the type of Endogens? A. Palms and bananas.

92. Q. What are some of the other families of the type of Endogens? A. The orchid, the lily and the bulrushes.

93. Q. What are Exogens? A. Plants whose woody fibres grow in outer layers. The seed has two lobes, or cotyledons.

94. Q. How many different species are included in this type? A. About seventy thousand.

95. Q. What are Incomplete Exogens? A. Those whose flowers have no corolla. They are of two kinds.

96. Q. What are the first kind? A. Those whose seeds are naked, as in the cone-bearing family, consisting of the fir and spruce tribe, the cypress tribe, and similar plants.

97. Q. What are the second kind? A. Those whose seeds are contained in the ovary, as the amaranth, buckwheat, laurel, nettle, fig, and the catkin-bearing family.

98. Q. What are some of the plants in the next sub-division of the type of Endogens, those whose flowers have both calyx and corolla? A. The honeysuckle, teasel, lobelia, convolvulus, primrose, and labiate and composite families.

99. Q. What are some of the families of plants found in another class of Exogens that also have calyx and corolla, but the corolla has distinct petals, and the stamens are attached to the calyx? A. The umbelliferous, the leguminous, and the cactus families.

100. Q. What are the distinguishing characteristics of the highest class, or the most perfect Exogens? A. The calyx and the corolla are present, the petals are distinct and inserted into the receptacle, and the stamens grow from beneath the ovary.



POPULAR EDUCATION.

CHAUTAUQUA LITERARY AND SCIENTIFIC CIRCLE.

President—Lewis Miller.

Superintendent of Instruction—J. H. Vincent, D.D.

Counselors—Lyman Abbott, D.D.; J. M. Gibson, D.D.; Bishop H. W. Warren, D.D.; W. C. Wilkinson, D.D.

Office Secretary—Miss Kate F. Kimball.

General Secretary—A. M. Martin.

1.—AIM.

This new organization aims to promote habits of reading and study in nature, art, science, and in secular and sacred literature, in connection with the routine of daily life (especially among those whose educational advantages have been limited), so as to secure to them the college student's general outlook upon the world and life, and to develop the habit of close, connected, persistent thinking.

2.—METHODS.

It proposes to encourage individual study in lines and by text-books which shall be indicated; by local circles for mutual help and encouragement in such studies; by summer courses of lectures and "students' sessions" at Chautauqua, and by written reports and examinations.

3.—COURSE OF STUDY.

The course of study prescribed by the C. L. S. C. shall cover a period of four years.

4.—ARRANGEMENT OF CLASSES.

Each year's Course of Study will be considered the "First Year" for new pupils whether it be the first, second, third, or fourth of the four years' course. For example, "the class of 1887," instead of beginning October, 1883, with the same studies which were pursued in 1882-83 by "the class of 1886," will fall in with "the class of '86," and take for their first year the second year's course of the '86 class. The first year for "the class of 1886" will thus in due time become the fourth year for "the class of 1887."

5.—C. L. S. C. COURSE OF READING, 1883-84.

I. REQUIRED.

History of Greece.[\[1\]](#) By Prof. T. T. Timayenis. Vol. 2; parts 7, 8, 10 and 11. Price, \$1.15.

Stories in English History by the Great Historians. Edited by C. E. Bishop, Esq. Price, \$1.

Chautauqua Text-Books.—No. 16, Roman History; No. 24, Canadian History; No. 21, American History; No. 5, Greek History. Price, 10 cents each.

Preparatory Latin Course in English. By Dr. W. C. Wilkinson. Price, \$1.

Chautauqua Text-Books.—No. 23, English Literature. By Prof. J. H. Gilmore. Price, 10 cents.

Primer of American Literature. By C. F. Richardson. Price, 30 cents.

Biographical Stories by Hawthorne. Price, 15 cents.

How to Get Strong and How to Stay So. By W. Blaikie. Price, cloth, 80 cents; paper, 50 cents.

Easy Lessons in Vegetable Biology. By Dr. J. H. Wythe. Price, cloth, 40 cents; paper, 25 cents.

Philosophy of the Plan of Salvation. By J. B. Walker. Price, cloth, \$1; paper, 50 cts.

Chautauqua Text-Books.—No. 18, Christian Evidences; No. 39, Sunday-School Normal Class Work; No. 43, Good Manners; No. 4, English History. Price, 10 cents each.

THE CHAUTAUQUAN, price, \$1.50, in which will be published:

Sunday Readings. Selected by Dr. J. H. Vincent.

Readings in Commercial Law. By Edwin C. Reynolds, Esq.

Readings in Political Economy. By Prof. George M. Steele, D.D.

Readings in French History and Literature. By Dr. J. H. Vincent.

Studies in American History and Literature. By A. M. Martin, Esq.

THE CHAUTAUQUAN will also contain, in the department of Required Readings, brief papers, as follows:

Readings in German History and Literature.

Readings in Roman History.

Readings in American Literature.

Readings about the Arts, Artists, and their Masterpieces.

Readings in Physical Science.

ADDITIONAL READINGS FOR STUDENTS OF THE CLASS OF 1884.

Hints for Home Reading. By Dr. Lyman Abbott. Price, cloth, \$1; boards, 75 cts.

The Hall in the Grove. By Mrs. Alden. (A Story of Chautauqua and the C. L. S. C.) Price, \$1.50.

Outline Study of Man. By Dr. Mark Hopkins. Price, \$1.50.

II. FOR THE WHITE SEAL.

Persons who pursue the “White Seal Course” of each year, in addition to the regular course, will receive at the time of their graduation a white seal for each year, to be attached to the regular diploma.

History of Greece.[\[1\]](#) By Prof. T. T. Timayenis. Vol. 2. Completed. Price, \$1.15.

Chautauqua Library of English History and Literature. Vol. 2. Price, cloth, 50 cents; paper, 35 cents.

Church History. By Dr. Blackburn. Price, \$2.25.

Bacon’s Essays. Price, \$1.25.

III. REQUIRED.—FOR THE WHITE (CRYSTAL) SEAL FOR GRADUATES OF ’82 AND ’83.

For the benefit of graduates of the C. L. S. C. who, being members of local circles, wish to continue in the same general line of reading as undergraduate members, a White Crystal Seal Course is prepared. This consists mainly of books belonging to the current year’s study, but not previously read by the graduates. An additional white seal is also offered to the graduates, the books for which are specified under paragraph 4. Some of these books were in the first four year’s course, and are therefore to be *re-read*. The payment of one dollar at one time entitles a graduate to the White Crystal and White Seals for four years. If only fifty cents is paid, it will be credited for but one year.

THE CHAUTAUQUAN. Required Reading.

History of Greece.[\[1\]](#) By Prof. T. T. Timayenis. Vol. 2. Completed. Price, \$1.15.

Preparatory Latin Course in English. By Dr. W. C. Wilkinson. Price, \$1.

Credo. By Dr. L. T. Townsend. Price, \$1.

Bacon’s Essays. Price, \$1.25.

IV. REQUIRED.—FOR ADDITIONAL WHITE SEAL FOR GRADUATES OF '82 AND '83.

Brief History of Greece. By J. Dorman Steele. Price, 60 cents.

Stories in English History by the Great Historians. Edited by C. E. Bishop. Price, \$1.

Easy Lessons in Vegetable Biology. By Dr. J. H. Wythe. Price, cloth, 40 cents; paper, 25 cents.

Biographical Stories. By Nathaniel Hawthorne. Price, 15 cents.

How to Get Strong and How to Stay So. By W. Blaikie. Price, cloth, 80 cents; paper, 50 cents.

Philosophy of the Plan of Salvation. By J. B. Walker. Price, cloth, \$1; paper, 50 cts.

Primer of American Literature. By C. F. Richardson. Price, 30 cents.

Chautauqua Text-Books, Nos. 4, 5, 16, 18, 21, 23, 39 and 43. Price, each, 10 cents.

The following is the distribution of the books and readings through the year:

October.

History of Greece.[\[I\]](#) Vol. 2. By Prof. T. T. Timayenis. Parts 7 and 8.

Chautauqua Text-Books.—No. 5, Greek History. By Dr. J. H. Vincent.

Primer of American Literature. By C. F. Richardson.

Required Readings in THE CHAUTAUQUAN

November.

History of Greece.[\[I\]](#) Vol. 2. By Prof. T. T. Timayenis. Parts 10 and 11.

Chautauqua Text-Books.—No. 5, Greek History. By Dr. J. H. Vincent.

Required Readings in THE CHAUTAUQUAN

December.

Easy Lessons in Vegetable Biology. By Dr. J. H. Wythe.

Biographical Stories. By Nathaniel Hawthorne.

Chautauqua Text-Books.—No. 24, Canadian History.

Required Readings in THE CHAUTAUQUAN

January.

Philosophy of the Plan of Salvation. By J. B. Walker. 14 chapters.

Chautauqua Text-Books.—No. 18, Christian Evidences. By Dr. J. H. Vincent.

Chautauqua Text-Books.—No. 39, Sunday School Normal Class Work.

Required Readings in THE CHAUTAUQUAN

February.

Philosophy of the Plan of Salvation. By J. B. Walker. Completed.

Chautauqua Text-Books.—No. 21, American History; No. 24, Canadian History.

How to Get Strong and How to Stay So. By W. Blaikie.

Required Readings in THE CHAUTAUQUAN

March.

Preparatory Latin Course in English. By Dr. W. C. Wilkinson. Half of book.

Required Readings in THE CHAUTAUQUAN

April.

Preparatory Latin Course in English. By Dr. W. C. Wilkinson. Completed.

Chautauqua Text-Books.—No. 16, Roman History. By Dr. J. H. Vincent.

Required Readings in THE CHAUTAUQUAN

May.

Stories in English History by the Great Historians. By C. E. Bishop. Half of book.

Chautauqua Text-Books.—No. 4, English History. By Dr. J. H. Vincent.

Chautauqua Text-Books.—No. 23, English Literature. By Prof. J. H. Gilmore.

Required Readings in THE CHAUTAUQUAN

June.

Stories in English History by the Great Historians. Completed.

Chautauqua Text-Books.—No. 4, English History. By Dr. J. H. Vincent.

Chautauqua Text-Books.—No. 43, Good Manners. By J. P. .

Required Readings in THE CHAUTAUQUAN

6.—SPECIAL COURSES.

Members of the C. L. S. C. may take, in addition to the regular course above prescribed, one or more special courses, and pass an examination upon them. Pupils will receive credit and testimonial seals to be appended to the regular diploma, according to the merit of examinations on these supplemental courses.

7.—THE PREPARATORY COURSE.

Persons who are too young, or not sufficiently advanced in their studies to take the regular C. L. S. C. course, may adopt certain *preparatory lessons* for one or more years.

For circulars of the preparatory course, address Miss K. F. KIMBALL, Plainfield, New Jersey.

8.—INITIATION FEE.

To defray the expenses of correspondence, memoranda, etc., an annual fee of fifty cents is required. This amount should be forwarded to Miss K. F. Kimball, Plainfield, N. J., (by New York or Philadelphia draft, Post-office order on Plainfield, N. J., or the new Postal Note, to be ready about September 1.) Do not send postage-stamps if you can possibly avoid it. *Three-cent stamps will not be received.*

N. B.—In sending your fee, be sure to state to which class you belong, whether 1884, 1885, 1886, or 1887.

9.—APPLICATION FOR MEMBERSHIP.

Persons desiring to unite with the C. L. S. C. should forward answers to the following questions to Miss K. F. KIMBALL, PLAINFIELD, N. J. The class graduating in 1887 should begin the study of the lessons required October, 1883. They *may* begin as late as January 1, 1884.

1. Give your name in full.
2. Your post-office address, with county and State.
3. Are you married or single?
4. What is your age? Are you between twenty and thirty, or thirty and forty, or forty and fifty, or fifty and sixty, etc.?
5. If married, how many children living under the age of sixteen years?[\[U\]](#)
6. What is your occupation?
7. With what religious denomination are you connected?
8. Do you, after mature deliberation, resolve, if able, to prosecute the four years' course of study presented by the C. L. S. C.?

9. Do you promise, if practicable, to give an average of four hours a week to the reading and study required by this course?

10. How much more than the time specified do you hope to give to this course of study?

10.—TIME REQUIRED.

An average of forty minutes' reading each week-day will enable the student in nine months to complete the books required for the year. More time than this will probably be spent by many persons, and for their accommodation a special course of reading on the same subjects has been indicated. The habit of thinking steadily upon worthy themes during one's secular toil will lighten labor, brighten life, and develop power.

11.—MEMORANDA.

The annual 'examinations' will be held at the homes of the members, and in writing. Duplicate Memoranda are forwarded, one copy being retained by each student and the other filled out and forwarded to the office at Plainfield, N. J.

12.—ATTENDANCE AT CHAUTAUQUA.

Persons should be present to enjoy the annual meetings at Chautauqua, but attendance there is not necessary to graduation in the C. L. S. C. Persons who have never visited Chautauqua may enjoy the advantages, diploma, and honors of the "Circle."

13.—MISCELLANEOUS.

For the history of the C. L. S. C., an explanation of the LOCAL CIRCLES, the MEMORIAL DAYS to be observed by all true C. L. S. C. members, ST. PAUL'S GROVE at Chautauqua, etc., etc., address (inclose two-cent stamp) Miss K. F. KIMBALL, Plainfield, N. J., who will forward the "Chautauqua Hand-Book, No. 2," sixty-four pages. Blank forms, containing the ten questions given in paragraph 9, will also be sent on application.

14.—CHAUTAUQUA PERIODICALS.

THE CHAUTAUQUAN, organ of the C. L. S. C.; 76 pages; ten numbers; \$1.50 per year. CHAUTAUQUA ASSEMBLY DAILY HERALD, organ of Chautauqua meetings; 8 pages; 48 columns. Daily in August; 19 numbers. Contains the lectures delivered at Chautauqua; \$1 per volume. Both periodicals one year, \$2.50. Address Dr. Theodore L. Flood, Editor and Proprietor, Meadville, Pa.

15.—BOOKS OF THE C. L. S. C.

For all the books address Phillips & Hunt, New York, or Walden & Stowe, Cincinnati or Chicago.

[I] Students of the new class (1887) to be organized this fall, and graduates of the classes of 1882 and 1883, not having read volume 1 of Timayenis's History of Greece, will not be required to read volume 2, but instead of volume 2 of Timayenis's, will read "Brief History of Greece." Price, paper, 60 cts.

[J] We ask this question to ascertain the possible future intellectual and moral influence of this "Circle" on your homes.



CHAUTAUQUA NORMAL COURSE.

Season of 1884.

LESSON II.—BIBLE SECTION. *The Bible from God Through Man.*

By J. L. HURLBUT, D.D.

The Sunday-school teacher in his work uses one book, and one only. To that one book he appeals as an authority; the doctrines contained in that book he asserts as truth; the moral system of that book he insists upon as the standard for man's obedience. It is therefore necessary to know concerning the Bible:

I. The claims of the Bible believer.

II. The evidences supporting those claims.

I. There are *four claims* made on behalf of the Bible by those who believe in it.

1. *Its Genuineness.* By this we mean that we possess the book substantially as it was written. Not that we have an absolutely perfect text, or that the translations represent precisely the original, or that we know just when or by whom all the books were written, but that the work has come into our possession without serious mutilation or interpolation. We can accept it as the Bible.

2. *Its Authenticity.* By this we mean that the book contains the truth. Its records are trustworthy history; its reports of discourses or parables or conversations give the substance of their thoughts; its statements upon every subject can be depended upon as honest and truthful.

3. *Its Inspiration.* By this we mean simply that this book came from God. “Divine inspiration we understand to be an extraordinary divine agency upon teachers while giving instruction, whether oral or written, by which they were taught what and how they should write or speak.” (Dr. Knapp, quoted by McClintock and Strong.)

4. *Its Authority.* By this we mean that the Bible contains God’s law, and was given to us as the standard in life. It contains “the only rule, and the sufficient rule, for our faith and practice.” No doctrine is to be accepted unless it is in accordance with the teachings of the Bible, and no law is binding which conflicts with the higher law of the Scriptures.

II. *The Evidences Supporting these Claims.* It is not necessary to present the proofs of each claim apart from the others. Those attesting the genuineness of the Bible will be given with Lesson iv, “The Canon of Scripture;” but the other claims are so linked together that the proofs of one are the proofs of all. If the Bible can be proven *true*, its truth is of such a nature as to show a divine original; and if it proceeds from God, it comes as God’s law. Hence we present together the *Ten Evidences* of its Authenticity, Inspiration and Authority.

1. *Its Adaptation to Human Need.* (1) We start with the proposition that *there is a God*; a person who governs the universe; not a mere personification of law or force, but a spiritual existence. (2) *God has a Law.* If God has no law for man, then for man there is practically no God. (3) *We have a right to know that law.* What would be thought of a law-maker with absolute power, who concealed his decrees, yet expected his subjects to obey them, and punished them for disobedience? (4) *We find just such a law as we need in the Bible*, and we find it nowhere else, for it is not stamped into our consciousness, nor is it written in nature. (5) We conclude then that *the Bible contains the Divine Revelation.*

2. *Its General Acceptance.* The common consent of intelligent society has accredited this book as authentic and divine. (1) We find an *early acceptance* among those best acquainted with its facts, and nearest to them; the Old Testament regarded as divine among the Jews; the New Testament among the Christians. (2) We find a *continuous acceptance* through all the centuries since; at no time the chain of belief being broken. (3) We find a

present acceptance now; in this age of searching investigation, when nothing is accepted on ground of tradition only, the Bible has more readers, more students, more believers in the intelligent classes than at any previous period of its history.

3. *Its Characteristics.* The Bible contains four traits which, taken together, distinguish it from all other books. (1) *Its Variety.* Written at intervals through 1,600 years, by more than thirty authors, in different lands and different languages, it contains history, poetry, genealogy, biography, ethics, epistles, doctrine, and many other classes of composition. (2) *Its Harmony.* Underneath its variety of the surface there is a harmony, so that its statements and its principles are nowhere discordant. Contrast with this the discords of scientists. Could we place on one shelf sixty-six books on astronomy, written during sixteen centuries, by thirty writers, and find them harmonious? (3) *Its Unity.* Amid all the different subjects of the Bible there is one unifying purpose. It presents as its theme *Redemption*, and every chapter in every book falls into line in relation to that central thought. (4) *Its Progressiveness.* There is a steady development of truth in Scripture, a growing light through its centuries. We see the revelation beginning with Adam, taking a step upward with Noah, another with Abraham, again with Jacob, and so mounting higher in turn with Moses, Samuel, David, Isaiah, Malachi, Peter and Paul, each on a loftier platform of spiritual knowledge than the age before him, until John crowns the pyramid of truth in his gospel and the Apocalypse. Not all the earth can show another book besides the Bible with all these four traits, which show the work divine.

4. *The Harmony of its Relations.* The statements of the Bible come into relation with facts ascertained in various departments of knowledge; yet in none of these do we find contradiction, in all an ever increasing harmony as our knowledge grows. (1) *With Localities.* The Bible names more than two thousand places in the ancient world; lands, rivers, seas, mountains, towns, villages, brooks, etc., yet not a single locality has been placed wrongly by the Scripture. (2) *With Existing Institutions.* We find in the world such bodies of people as the Jews, the Samaritans, the Christian church; such services as the passover, baptism, the Lord's Supper, etc. Take away the Bible and none of these can be accounted for; open the Bible, their origin is plain. (3) *With Historical Monuments.* During the present century thousands of ancient inscriptions have been brought forth and deciphered, and the

history of great empires has been written, bearing close relation to the history of the Bible. But not a line of the Bible annals has been discredited by these explorations, and many Bible statements have been placed in clearer light. (4) *With Science*. Though “the conflict of science and the Bible” has been often referred to, yet the testimony of the best scientists is that the opening chapters of Genesis are in substantial and growing accord with geology; that the tenth chapter of Genesis tallies with the latest conclusions of comparative philology; and that modern astronomy furnishes the best illustrations of the attributes of God as revealed in Scripture.

5. *The Fulfillment of its Prophecies*.—It is very evident that no man, unaided by Divine wisdom, can know the future and make prediction of coming events. Yet there is a book containing many prophecies, which have been fulfilled to the letter. (1) There are *predictions concerning places*, as Babylon, Nineveh, Jerusalem, Tyre, Egypt, all differing in their statements, yet all brought to pass. (2) There is a series of predictions concerning Christ, beginning in Eden and extending through the Old Testament, growing in definiteness as the hour of fulfillment drew near, and all accomplished. Thus the New Testament and the Old mutually prove each other.

6. *The Person of Christ*.—We find in the gospels four accounts, by different writers, of one Person. They tell us that he was at once God and man; that he grew up in a country village, yet surpassed all the wisdom of the philosophers; that he could create food, yet suffered hunger; that he could raise the dead, yet submitted to be tortured and crucified; that he was free from worldly ambition, yet became the founder of the greatest kingdom earth has seen. The life, the character, the personality, is so unique and original that no one could have invented it. Hence the writers of the gospels must have drawn their sketch from the life.

7. *The Candor of its Writers*.—The authors of these documents write like honest men, telling their story plainly, without partisan bias. They relate the sins of their heroes, Abraham’s deception, Jacob’s double-dealing, Moses’ anger, David’s crime, Peter’s denial, Paul’s quarrel with Barnabas. Their tone of sincerity shows the truthfulness of the narration.

8. *The Elevation of its Teachings.*—Here is a book, written in an age when even the most cultured nations worshiped idols and held the grossest conceptions of God, with correspondingly low ideals of morals for men. Yet in such ages, the Bible presents a view of God to which the world has been slowly broadening its vision; and a standard of character which rises far above that of Plato, Cicero, or Confucius, and is now adopted as the ideal manhood by ethical philosophers. Whence, but from a divine source, came those lofty teachings of the Scriptures?

9. *Its Influence Upon the World.*—What the Bible has done shows the hiding of its power. (1) *See its effects upon nations.* The lands where it is honored, America, England, North Germany, are the three lands of most advanced civilization and largest hope for the race. The lands where it is forbidden, as Spain, or where it is unknown, as China, are those whose condition is most hopeless. (2) *See its effects upon individuals.* The people who study the Bible are not the drunkards, thieves, criminal classes. Those who have the word in their minds and hearts become purer, better, higher than others. It transforms men from sinners to saints, and its influence makes earth a picture of heaven. No false book, no deceiving book could thus make the world better.

10. *Its Self Convincing Power in Experience.*—There is in the consciousness of man a conviction that the religion of the Bible rests upon a sound foundation. And he who puts the Bible to the test in his own experience, who lives its life, and follows its law, and enjoys its communings, finds an assurance to the satisfaction of his spiritual nature, that this book contains God's message to his soul. Every Christian's experience is, therefore, a testimony to the truth and the inspiration of Scripture.

[To those who wish to pursue this subject further we recommend the following works: "Credo," by L. T. Townsend; "The Logic of Christian Evidences," by Dr. Wright; Chautauqua Text Book No. 18; "Christian Evidences," by Dr. Vincent; "The Christ of History," by Principal Young; "Historical Illustrations of the Old Testament," by Rawlinson & Hackett; "The Story of Creation," by Dr. Campbell; and "Farmer Tompkins and His Bibles," by W. J. Beecher, D.D.]

CHAUTAUQUA NORMAL CLASS—S. S. SECTION.

LESSON II.—THE SUPERINTENDENT: HIS QUALIFICATIONS, DUTIES AND RESPONSIBILITY.

By R. A. HOLMES, A.M.

In Lesson I we considered the place, purpose and prerogatives of the Sunday-school. That it may keep to its place, accomplish its purpose, and enjoy its prerogatives, efficient organization is necessary. By universal consent the chief officer of such organization is called "The Superintendent." Experience has proved that the character of the school and its success or failure, as measured by the standards already given, depend very largely upon the character of the superintendent and his understanding of his work. This lesson will content itself with answering briefly three questions:

I. *What are the Qualifications of the Model Superintendent?*—The purpose of the school is the conversion and spiritual education of those who are under its influence. This, therefore, must be the purpose of the superintendent. As one can not teach what he does not know, so he can not accomplish a purpose unless he knows practically the steps which lead to its accomplishment. The superintendent therefore must be (a) *both converted and spiritually educated*. Conversion implies oneness with Christ in will and desire. Christ's will is the conversion of the world. To effect it he instituted the church on earth. The superintendent must therefore be (b) *a member of the church, and a firm believer in it and its power*.

The church in its endeavor to accomplish its holy mission has instituted the Sunday-school. Its special function is the teaching of the word. Its great need is and has been competent teachers. Their appointment and continuance in office rests with the superintendent. The superintendent should therefore be (c) *a good judge of human nature*; (d) *a person of approved teaching ability*.

The school in active operation uses as its only text-book the Holy Scriptures. The text-book is a difficult one. It deals with the deepest

problems of spiritual life and death. It is the offspring of a remote day, and is filled with allusions to a state of society and social customs entirely foreign to anything with which we are familiar. A trained teacher in secular education with no knowledge of this book may make utter failure as a teacher of it. A knowledge of it in its entirety is absolutely essential to the teacher in the Sunday-school. The superintendent must therefore be (e) *a thorough and intelligent scholar in Bible lore.*

The membership of the Sunday-school, aside from teachers and officers, is largely composed of children and youth. By nature humanity tires of monotony. Children are more restive under monotonous routine than those who have won self-control by culture. To keep in the school its children and youth, to keep them interested in its purposes while in the school, and to hold them untiringly to the true work of the school, needs fertility of brain to give proper variety to the conduct of the school, intelligence to discern the effects of all measures that are adopted, tact to change and adapt to the ever varying conditions of school life, and common sense to direct and govern the whole. The superintendent must therefore be (f) *a person fertile in expedients* and (g) *a person of intelligence, tact and common sense.*

But often in the conduct of the school infelicities occur. The different parts do not move in harmony with each other. Cases of variance between pupils and teachers arise. Often times the school suffers from financial lack. The chief officer of the school is the one to whom all such matters come for final adjudication. The superintendent therefore must be (h) *a person of good executive ability*, that with firm, strong hand he may hold each part of the system of which he is the center revolving in its own orbit, never flagging, never tiring, never ceasing to do its own part in the work, never clashing with any other. Such are some of the principal qualifications of the superintendent.

II. *What should be his personal character?*—In general, all that is suggested in the foregoing outline as to qualifications. But our requirements must not end there. A man may be a so-called Christian and yet be far from possessing the character which is an essential to the Sunday-school superintendent. He may be a church member, and be even less than a so-called Christian. He may be a good judge of human nature, and yet himself a poor illustration of it. He may be possessed of fine teaching power, and

yet misuse it. He may know the Bible as well as Erasmus, and yet be like Erasmus, the subject of Luther's keen reproach of being everything in word, and nothing in deed. He may be all we have described, and yet lack in character.

The superintendent therefore should be pious, "having reverence for God, and for religious duties." He should be devout, that is, should carry into daily life the active expression of his piety. This would forbid sudden anger, inconsiderate levity, trifling with Scriptures, by thoughtless quotations, and all outward conduct that does not comport with true consecration. He should be honest, truthful in word and act, humble, loyal, and scrupulously observant of the Sabbath.

His constant motto should be as he daily studies to build character in himself and others, "Study to show thyself approved unto God, a workman that needeth not to be ashamed." Let the student make for himself an outline of what the superintendent should be in character.

III. *What are his duties?*—They are four fold. (1) To his church. (2) To his pastor. (3) To his teachers. (4) To his pupils.

His duties to his church are plain.

1. *He should attend the regular services of the church regularly.*—This can admit of no negative. He should do it for its effect on himself. He should do it as an encouragement to his teachers. He should do it as an example to his pupils.

2. *He should impress habitual church going as a duty upon his teachers and pupils* from the desk in the Sunday-school room, and should use all means to effect the object.

3. *He should contribute regularly and uniformly to all the benevolent objects* which the church presents as worthy of Christian liberality. The reasons for this are too plain to need mention.

4. *He should urge to the same duty the teachers and pupils of the school,* that they may each do their part, no matter how small, in the work of Christian benevolence.

5. *He should contribute of his means as God prompts* him to the support of his church, and not measure himself by the standard of proportionate values. He should also teach the same duty in his school.

6. *He should be loyal to his own particular church*; should know its particular beliefs; should pray for its particular welfare; and fearlessly do whatever lies in his power to promote its purity and peace.

II. *His duties to his pastor.*

1. Is that of *Coöperation*. The pastor and superintendent should know each other's plans and purposes thoroughly. The pastor should always be able to feel that in his superintendent he has one upon whom he can depend, who will aid him in his work; share with him a certain portion of the duties devolved upon him, and in all possible ways be like Aaron and Hur, hand upholders in the fight against Amalek.

2. That of *Allegiance*. The pastor is the one man of all the church upon whom all eyes are fixed. Among his multitude of acts, some will be misunderstood. Among the multitude of tongues some will be captious and critical. A spark may kindle a conflagration. The superintendent owes it to church and pastor to be loyal to his pastor and render him the knightly service which the king could expect from the lord. He should also teach the same duty to teachers and pupils in the school.

3. *He should be his Pastor's Index Rerum*; not his mentor, but his reference, to which he can turn for information concerning affairs in that portion of the church represented by the school. Sick children to be visited, poverty to be helped with true charity, anxious souls looking for the Savior, these and many similar are within the superintendent's knowledge oft times, when unknown to the pastor. To bring them to the pastor's knowledge is an evident duty.

4. *That of Harmony*. The pastor and superintendent should agree. The school should have no plans or methods contrary to the pastor's desires. Church and school should walk the same path, and in it go hand in hand.

III. *His duties to his Teachers*. While these are many we mention but five, and these without discussion, leaving the student to fill up the outlines.



EDITOR'S OUTLOOK.

THE C. L. S. C. PLAN.

No organization that has appeared in the past fifty years has been more favored than the Chautauqua Literary and Scientific Circle. From the first the surroundings have been such as to aid its growth. Eminent educators and literary men pronounced their blessings on its head the day it was born. Thousands of people shouted its praises to the echo, in the grove at Chautauqua, as soon as they saw what it was and heard its name. Chautauqua had a history of five years to place behind the C. L. S. C.—a history of enterprising investigation in the fields of science and philosophy, Biblical literature, church and Sunday-school work, and moral reforms. It was five years of hard work to popularize useful information on all these lines of thought. This was a good beginning for the C. L. S. C., and right here it started. With the summer meetings at Chautauqua it has been associated during these first five years of its history. The C. L. S. C. Commencement exercises are held in the Hall of Philosophy, in St. Paul's grove, at Chautauqua, and from thence the diplomas are sent out to the graduates all over the world.

It never was the design of Dr. Vincent or Mr. Lewis Miller, the founders of Chautauqua, that all the work of students should be pressed into the compass of three weeks of meetings in August, but rather that Chautauqua should be carried into towns and cities, into homes and offices and workshops all over the land. When the C. L. S. C. appeared and its curriculum was announced with the promise that every person who should complete the four years course of reading in ancient and modern history and literature, the sciences, philosophy and art, would graduate and receive a diploma signed by the officers of the C. L. S. C., the idea was easily carried abroad. The press of the country was ready, as we now see, to assist. The plan was written up and philosophized upon from the beginning; but more than this was needed to insure success. To make the Chautauqua Idea as

practical in a town five hundred or a thousand miles away as it was at Chautauqua was a hard task to perform; but when it was decided that the *individual* could enroll his name in the C. L. S. C. office and pursue his studies at home, or when traveling, by devoting forty minutes a day to his books, and could fill out examination papers at the end of each year, the practicability of the plan was admitted by everybody. The organization was simple, the working of the system has been almost perfect, and each succeeding year has witnessed a marvelous growth; classes ranging from 7,000 up to 14,000 members have been enrolled from year to year until the present outlook is more encouraging than all the past.

The local circle has come to be an important factor in the working of the organization. Men are clannish, and in the work of education the world has always recognized the social element as a powerful agency. It was natural that in the C. L. S. C. men and women, who had no scruples on the question of the co-education of the sexes, should come together and effect local organizations, elect their officers and do their work methodically, under the inspiration of one another's presence. Just as in raising a building ten men are stronger than one man, so in a town or city ten persons will lift up the Chautauqua Idea in more homes and attract the attention of more people to it than one person possibly can. "In union there is strength," and while the practical working of the "local circle" is to be seen in the growing intelligence of its individual members, it is a fact that through the local circle the C. L. S. C. is taking hold of the people in all parts of our land, and thus demonstrating that the founders of Chautauqua have inaugurated an educational system which has the merit of being a "Home College," whose privileges may be enjoyed by all classes and conditions of people. While it is not sectarian or even denominational, it is Christian, and carries correct ideas of God and the Bible, of Jesus Christ and redemption, of the Holy Ghost and Christian life into every reader's mind and into every family where the course of study is received.

MARTIN LUTHER.

Our heritage of civil and religious liberty is an outgrowth of the Reformation, begun in the fifteenth century. By common consent the

Protestant churches confess indebtedness to Martin Luther, the principal agent raised up by God for the deliverance of his people. We gladly join our brethren of a free press and the heralds of a free gospel, in making some mention of this fourth centennial day. Want of space must greatly abridge the tribute we would bring, and forbids any attempt to weave such fitting chaplets as other hands will certainly bring to the altar.

Four hundred years ago to-day, November 10, 1483, Martin Luther was born in Eisleben, Saxony. The great German reformer, whose words shook the world, and whose power, after centuries, is felt by millions indebted to him, was of humble origin, his parents being peasants of the poorer class, but religious, honest, self-respecting people. He refers tenderly to them, and says: "In supporting their family they had a hard and bitter fight of it." His own privations and hardships in early life were met with something of the heroism and persistence of endeavor that marked his later years. In school, though a sprightly lad, full of fun and frolic, and often corrected for his faults by a severe master, he was yet a diligent student, eager for communion with all truth. His ambition and thirst for knowledge led him gracefully to accept what was unfavorable in his circumstances, yet not passively or without methods of improving them. The spirited youth, with some others under like pecuniary embarrassments, rather than leave school, for a time sought bread in the neighboring villages, and found way to the hearts of their benefactors by singing at their doors. The songs of the boys seem to have been offered and accepted as a remuneration for the material aid they needed, and thus the depressing sense of mendicancy was not so seriously felt. He earned his master's degree when yet young, having by his proficiency in both classical studies and philosophy attracted the attention of some scholarly men. He left school with honors, but not happy. Soon after began the great struggle of his eventful life. On a careful introspection he found in his quickened soul cravings that human knowledge could not satisfy. Educated a Catholic, and observant of all their rites and ceremonies, but finding little comfort in them, in his unrest and almost despondence, he entered a monastery, thinking by fastings, penance and prayers to find relief for a wounded conscience. The way to him was dark; the conflict terrible; the unhappy monk knew of sin, but not the Savior. The day of his deliverance was at hand, though for a time he saw but the dawn. With the Bible found in his cell as his almost only guide, he at length clearly

apprehended the way of salvation by faith alone—believing he was justified. The change was great, and the whole tenor of his after life confessed it. The strong, earnest, cultured man, rejoicing now in the gospel liberty, himself baptized with the spirit and fully consecrated to work for others, was a fit instrument for inaugurating any needed reformation. Led by the spirit and ever true to his convictions, he was soon, though wishing to avoid the issue, in open conflict with the Papal authorities. How bravely, and with what results the battle was fought, is well known. It was an open, manly fight. Any disguise with him was simply impossible. He never masked his own position, nor sought to flank that of the enemy. The warfare, on his part, was honorable, but the shafts he forged were pointed, and hurled with tremendous force. His multitudinous disquisitions, essays and replies came in quick succession, as the exigencies of the controversy called for them. He wrote, any reader will say, rapidly, from the fullness of his mind and heart; and very few authors have left on their works so strong an impress of their own personality. He is perhaps best known in his “Table Talk.” There is a freshness in these off-hand sayings that is charming, and quite disarms criticism. His greatest gift to the German people was his faithful translation of the Bible into their vernacular, and his commentaries that are still held in high esteem. The reformer’s influence, great while he lived, has increased immensely during the four centuries. As a biblical critic and expositor his ability is now recognized by the general church. He held to the spiritual and supernatural in religion, but recognized the human as well as divine factor in the books of the Bible, and in that, too, the church is in sympathy with him.

THE TEMPERANCE QUESTION.

Of this question it is the political aspect which at the present time is most prominent. It is becoming a grave, disturbing force in our politics. Viewing the temperance cause in the light of political action, it is clear that it is advancing, and that those who have the cause at heart have reason to thank God and take courage. No little chagrin was felt when it was known that the noble action of the people of Iowa a year ago, in voting for constitutional prohibition, was, owing to a technicality, of none effect. But again in that great state the battle has been fought; this time in a different way. The

Republican party there had the wisdom to champion the prohibition measure; this plank was squarely inserted in the party platform, and in the campaign recently closed it was the leading issue. We have the result of the election, and it should give the friends of temperance encouragement and hope. A second time this righteous principle has triumphed. The Republican party has won the day, and if its avowed purpose is redeemed in the State of Iowa, the sale of strong drink will soon be made a crime. We turn to the state of Ohio, and here, too, we see sure tokens that the temperance cause is moving forward. The confession comes from prominent politicians, that if, in Ohio as in Iowa, their party had adopted prohibition it might have been better. This was not done; but the question in the late election was submitted to a popular vote and the result, all things considered, is most encouraging. Some sanguine people may have had faith that the prohibitory amendment would be carried, but perhaps the number was not large. That it received the great vote it did in a state where the liquor interest is of such magnitude and so strongly intrenched, is something to cheer and make thankful the hearts of good people.

One does not need the vision of a prophet to see that the day of the triumph of prohibition in our country is coming on. The right is to win. The time is in the not-distant future when state laws and state constitutions will say that men shall not make their living by pandering to the depraved appetite of fellow men. The rum-seller's business will be made illegal and criminal. Even those who are looking forward to the prohibition of the liquor traffic by the national constitution will not long be called fanatical and visionary. But meanwhile other work for temperance besides that looking to this condition of things, so much to be desired, should not be neglected. Personal effort to preserve the youth and reclaim men is always demanded. People are clearly in error who say: "Prohibition or nothing." Laws whose aim is the curtailing of liquor selling, should be sought, enacted, sustained and enforced as better than none at all. Until we can have prohibition, let us have as stringent restrictive enactments as possible. It is a short-sighted view of things which prompts such a sentiment as this: "If we can not have prohibition, let us have free rum." The adage of the "half-loaf" and the "whole" is full of sound wisdom. We can but think there are earnest temperance men who make a grave mistake. Prohibition—unquestionably the true measure to apply to the liquor traffic, and for whose adoption we

should persistently work—fills their minds and hearts. They bend their energies to secure this. But for other legal measures, falling short of this desideratum, and aiming only to restrict the wretched traffic, they have no support. Everywhere restrictive liquor statutes are seen very imperfectly executed for want of interest and determined effort on the part of temperance people, whose rigid enforcement would work a grateful change in our communities. If the law says that the saloon shall not be opened on Sunday; that it shall be closed at a certain hour of the night; that intoxicants shall not be sold to youth under a certain age, or by any provision looks to the diminution of the great curse of our people, it should be regarded as good so far as it goes, for so much of prohibition as it contains, and should have the support of good citizens, though their hope looks and their labors are directed to the total prohibition by law of the sale of strong drink as a beverage. To make the best and most of what we have is the true policy in every issue of life. If we can not have prohibition now, we can see that our laws are enforced. When they are thoroughly enforced, we will be much nearer prohibition.

AN EXTRA DAY IN THE CALENDAR.

“There is an island off the coast of New Zealand where the day of the week changes. There Saturday is Sunday, and Sunday, Monday. When Sunday noon closes, Monday noon begins. A man sits down to his dinner Sunday noon, and it is Monday noon before he is done eating.”

A correspondent sends us the above statement and asks, is it correct? We answer: Not to the islanders, who, as ourselves, have but 365 solar days in a year. But to a stranger coming there on his voyage round the world, who has 366 at his disposal, it is true. He has one day to spare, has no name or place for it in the week, and just drops it out of his reckoning, as though it had never been. The explanation is simple enough, even for the young. The revolution of the earth on its axis, from west to east, once in 24 hours, gives the sun an apparent motion round the earth from east to west. To us the sun rises and sets. The succession of day and night is just the same as if the sun really went round the earth. As the sun’s apparent motion is from east to west, a man traveling eastward, at whatever speed, will see the sun rise,

reach the meridian, and set, a little sooner each day than the day before. So the time indicated by his watch, and that by the sun will differ more and more as he goes on; and what he gains each day in time will evidently be to a solar day, as the distance traveled is to the earth's circumference. One degree east will make a difference of four minutes, fifteen degrees an hour, one hundred and eighty degrees twelve hours. Having reached the one hundred and eightieth meridian, his chronometer and the sun are just twelve hours apart, so he changes his reckoning, to avoid confusion, and at noon Sunday calls it Monday. The correction is of course too much, but if he waits till beyond that time it amounts to more than half a day, and is constantly increasing. If the error is to be corrected all at once—and this is the only way that is found practicable—it should be done when it amounts to half a day. When he has completed the circuit of the earth a whole day will have been gained. If another man, from the same place of departure, go west, or with the sun, he will lose a day, and the two meeting would be, if neither had changed his reckoning, two whole days apart—yet each had the same number of hours and minutes. He who had the greater number of days had them just so much shorter. There is, of course, no reason in the nature of things, why the days of the week should be changed on the one hundred and eightieth meridian rather than elsewhere. There must be some point from which longitude is reckoned, and to avoid confusion English and American navigators agree on Greenwich, near London, and their nautical charts, almanacs, etc., are arranged accordingly. If they had taken as their starting point Washington, the one hundred and eightieth meridian would have been west of where it is, the number of degrees between the places.



EDITOR'S NOTE-BOOK.

The privilege of joining the new C. L. S. C. class just forming will be granted till the first day of January, 1884. This class will graduate in 1887. It begins work with genuine Chautauqua enthusiasm. Send applications for membership to Dr. J. H. Vincent, Plainfield, N. J.

The Protestant Episcopalians held their General Convention in Philadelphia in October, and during this month they consecrated an assistant Bishop for New York City, and another for the city of Baltimore. The Methodist Episcopal Church will hold their General Conference in the same city in May, 1884.

General Sherman says that he regards the Indian question as substantially eliminated from the problem of the army. The completion of the trans-continental lines of railway, and the extensive emigration into the territories have made large contributions to the settlement of the question. But for all that, we shall find many demands made upon us by the Indians in the future. Fair treatment of them will go far toward preventing trouble.

Governor Murray, of Utah, reports to the Secretary of the Interior that a secret organization among the Mormons, which has been in existence for a number of years, nullifies the laws of the United States and prevents the execution of the decrees of the Supreme Court. The Governor proposes to repeal the act giving a legislature to the territory, and to rule the people directly by the United States Government. That is a good suggestion, but why does not Governor Murray do something to prevent Mormon missionaries importing men, and especially women, from European countries to keep their ranks full? We send missionaries to foreign lands to preach the gospel, and permit the Mormons to bring agents of evil over here by the hundreds and thousands.

The lively canvass for the election of Mayor of Brooklyn, N. Y., has brought to light the fact that the cost of the Brooklyn bridge was \$21,000,000.

That fine military organization known as the Cleveland Greys has decided to purchase ten acres of land on the shores of Chautauqua Lake for a summer camping ground.

General Sheridan is now commander of the armies of the United States. His abilities as a fighter, which made his splendid reputation in the Shenandoah Valley and on other fields of battle, are not needed now, but rather the qualities which made him an excellent quartermaster as a staff officer. The nation is to be congratulated that while the great generals of the war, Grant and Sherman, are retiring, so capable and worthy an officer as Sheridan, who won a world-wide fame by his skill and heroism in battle, is promoted to this important command.

It is estimated that the German-American element in this country can not fall short of nine millions. This embraces all that were born in the Fatherland, and all that were born of German parents in this country, and that speak the German language.

Three hundred thousand voters in Ohio declared themselves in favor of constitutional prohibition at the election in October. The moral force of that vote is tremendous. Never before did the Prohibitionists, who believe in carrying their cause into politics, act more wisely than when they compelled an old and powerful political organization to take up their cause and plead for its success—"wisdom is justified of her children." If they did fail the effort was a great success, as is every action for a good cause. When the dominant political party shall adopt prohibition as one of the chief planks in its platform it will hold the Christian and temperance voters in its ranks, but when it throws this cause overboard these people will think seriously of turning their political machinery upside down.

Mr. V. C. Dibble expresses these sensible views on a live question in a recent number of the *Journal of Education*: "The objection to classical culture rests upon the assumption that it is not practical; an assumption which, although not uncommon, is nevertheless incorrect. There is no issue between classical education and that which is practical. The only education worthy of any serious advocacy is the practical—that which is adapted to the condition of its subjects, and which will prepare them for the real work which life will demand of them. Education is in fact life begun."

The paper on which the United States currency is printed is manufactured at Dalton, Mass., and the Boston *Herald*, in a recent issue, gives the following particulars: Eighteen or twenty Treasury girls, who earn \$3 a day, count the sheets, examining each one closely, and rejecting all imperfect ones. An automatic register at the end of the machine registers every sheet as it is cut off and laid down. The register man takes them away in even hundreds, and they are immediately counted in the drying room. In all the various processes of finishing every sheet is counted, and they are again counted on their receipt at the Treasury Department in Washington. The great protection of the government against counterfeiting lies in the paper here made. The distinctive feature is the introduction of colored silk threads into the body of the paper while it is in the process of manufacture. They are introduced while the paper is in the pulp, and are carried along with it to the end of the machine, where it is delivered as actual paper. This has been more fatal than anything else to the professional counterfeiters.

The political work during the past month has been a contest in several states for state officers. Massachusetts has attracted the attention of politicians everywhere, because General Butler was the most conspicuous figure in the campaign. He was a musical candidate. Editors of political papers never failed to criticise him and to praise him. He mixed up with schools, charitable institutions, moral reforms, and the industries of the state. He has been defeated by a heavy majority, and Mr. Robinson, the Republican candidate, elected over him. It is now predicted by the wise ones that this will close General Butler's race for the presidency, but this may prove to be false, because all ordinary rules fail when applied to an

abnormal character like General Butler. He rides the stormiest sea of any man in American politics.

Concerning candidates for the Presidency, all aspirants seem to be using a kind of tactics that will keep their names out of sight, while they gather all the strength possible for the coming struggle. In late years a number of eminent men have run well in the newspapers and in political street talk, but when the votes were counted in the National Convention they failed. Senator Don Cameron is in Europe, and rumor says he will remain there till late in the summer of 1884. Ex-Senator Conkling has lost his political influence, and Senator Logan is obliged to share the political fortunes of his party in Illinois with Secretary Robert Lincoln. This trio, Cameron, Conkling and Logan, who were mighty forces in the last National Republican Convention, will not be able to dominate the action of their states in the next campaign for the Presidency. Perhaps, as one result, the voice of the people will be more potential, and, in such a case, correct ideas of government will triumph.

Ready made houses is an important branch of manufacturing in some parts of the country. "A correspondent of the *Old Colony Memorial* paid a visit not long ago to Fairfield, Maine, where a large establishment is located for the production of these knock down houses, and he says that few have any idea to what extent this business has been carried in Waterville and its neighborhood, or to what perfection it has been brought. In the establishment to which we refer dwelling houses are made, like boots and shoes, in any quantity, and of any size or style, and for any market in the wide world. Not long since this concern received a single order for fifty houses for Cape May, to be delivered speedily and in complete finish. These houses were to be, not sheds nor shanties, but regularly ordered dwellings; and they were made accordingly and so delivered, and contain hundreds of occupants at this moment. An order will be received for a \$50,000 hotel, or an ornate, French-roof, cottage for a fine country estate, and these as easily and expeditiously furnished as an ordinary boarding house for a country village, or a barn for a ranch in Kansas or Colorado." This would be a good plan for persons to adopt who contemplate building cottages at Chautauqua. Try it.

“The first railroad in Palestine is being laid out, and the preliminary survey has been completed as far as the Jordan. It is to run between Acre and Damascus, and is called the Hamidié line, because it is named after his present Majesty, the Sultan Abdul Hamid. Probably one reason why the firman has been granted so easily lies in the fact that it passes through a great extent of property which he has recently acquired, to the east of the plain of Esdraelon. The concession is held by ten or twelve gentlemen, some of whom are Moslems and some Christians, but all are Ottoman subjects resident in Syria. Among the most influential are the Messrs. Sursock, bankers, who own the greater part of the plain of Esdraelon, and who have, therefore, a large interest in the success of the line.”

Several eminent Englishmen have visited this country during the past month. Lord Coleridge, representing the law, Henry Irving the stage, Matthew Arnold, literature, and Père Hyacinthe, theology. The reception of these gentlemen in our eastern cities indicates that the world has a peculiar fondness for its own. Henry Irving was received by more people, entertained more elegantly, and eulogized with more applause, than any one in the list. Yet he has not done a tithe as much for the elevation of his fellow men across the waters as any one of the others. Is it not still true, “The children of this world are wiser in their generation than the children of light?”

A Hindoo prophet, Babu Protap Chunder Mozoomdar, has come to this country from the Orient. He was educated in the religion of Brahminism. Some years ago he renounced idolatry, and in company with his cousin, Keshub Chunder Sen, joined the Brahmo Somaj, a theistic movement started in 1830 by Ram Mohun Roy. Mr. Sen and Mr. Mozoomdar have since become prominent leaders in this religious and social movement. Mr. Mozoomdar left India last spring for a tour around the world. He is about forty-two years of age, is above the average height, is of dark complexion and finely-cut features. He is the author of two books; one on “The Faith and Progress of the Brahmo Somaj,” and one just published entitled “The Oriental Christ,” which is a devout and poetic conception of Christ as seen by an Oriental mind.

The term “dude” is a very convenient nickname for the over-nice or simpering individuals who are found in considerable numbers on every line of the world’s work. There is the fashionable dude, scholarly dude, literary dude, artistic dude, etc. They are a useless class of persons, unless they serve as scare-crows to frighten other people from the line of life on which they move. Perhaps this is as good service as can be claimed at the hands of such a set of weaklings.

The Arctic relief expedition has proven to be a great failure. No relief for the Greeley party was provided by the expedition, and yet it has returned home. The verdict which public opinion seems to render is, that the “Arctic Relief Expedition” was *badly managed* from first to last.

It is said that one result of President Arthur’s visit to the Northwest is a determination to appoint only residents of territories to the important territorial offices. This is a concession to the people of the territories who are dissatisfied with appointments from without.

The decision of the Supreme Court on the Civil Rights bill turns the whole question over to the government of the states in which the colored people live. If they do not secure justice there, they have another high privilege in reserve, namely, the right of appeal to a higher court.

The dynamite explosions in October, on the underground railroad in London, were ineffectual attempts as movements either against the city or general government. Some Irish leaders claim that the Irish did not do the mischief, but that designing Englishmen who mean to keep up perpetual war between Ireland and England, were the guilty parties. The ways of this conflict are as dark as the railroad tunnel under London.

The Chautauqua Board of Trustees will hold their annual meeting at the Sherman House, in Jamestown, N. Y., on Wednesday and Thursday, the 9th and 10th of January, 1884.

The *Deaf-Mute Advance* comes to our table once every week, from Jacksonville, Ill. As the name indicates, it is published in the interest of deaf-mutes, and is doing much to inspire with a desire for education the class of people to whom it ministers. In a late number the editor says: "A young lady from the country came to Danville some days ago, driven by a green boy, who had his first view of town life. She had occasion to go to the Deaf and Dumb Institute, and the boy, when he went home, said he saw the people there 'winking at each other on their fingers.'"

Mr. Moody successfully opened his great mission in England on Nov. 4. Four meetings were held, each of which was attended by from 4,000 to 6,000 persons. The iron hall built for the occasion proved to be complete in all its arrangements, affording seating room for 5,800 persons. All around on the sides of the hall appropriate texts were displayed, such as "God is Love," and over the platform, "We pray you in Christ's stead be ye reconciled to God." Mr. Sankey sang with customary effect. Mr. Moody's powerful addresses showed that he had not lost his hold on the people. At the close of the evening meeting a man in the hall shouted out that Mr. Moody's last mission in London had been a failure. Mr. Moody answered by calling for volunteers to come out boldly on the Lord's side, whereupon about three thousand men arose *en masse*. The incident caused much excitement.

The first number of *The Outlook*, the paper published in the interest of the class of '84, is out. It is a stirring little sheet, brimming over with class news, class gossip and class enthusiasm. The '84s are especially fortunate in having such an editor as Mr. Bridge to lead them. This little quarterly will undoubtedly do much toward awakening the class and making their closing year even more brilliant than their beginning. Let every member subscribe.

A novel and entertaining exhibit was held in Paris in October. It was called "The Exposition of the Incoherent Arts," and was arranged by and contributed to by young artists. Such a collection of absurdities is rarely seen, this one being on a much larger scale than those in previous years, and those who attend go to laugh. It is necessary to be a Frenchman and a Parisian to thoroughly appreciate all the happy hits and plays upon words,

but even a foreigner can find food enough for laughter. The proceeds of the exhibition are for the poor of Paris, and it is expected that it will net quite a good sum. The exhibition abounds in pictures of the realistic school. For instance, where there is a figure wearing a shoe it will very likely be a genuine shoe attached; or hair will be stuck on instead of painted, suns and moons be represented by gold and silver paper pasted on, and one painting gives a ship sailing along accompanied by fishes, the fishes being two or three regular dried herrings glued to the canvas. One of the most prominent pictures is a portrait of the lecturer and critic, M. Henri de Lapommeraye. The hair and mustache, the eyeglass, the book just laid down, the letter he is reading, and the glass of sugar and water at hand, are all real objects attached to the picture, and of course, stand out most “naturally” from the canvas. No. 85 is entitled, “Poem of a Pig.” It is a very striking geometrical fantasy, the five different handlings of plain geometrical figures giving a pig drama in five acts. First act, pig strolling along seeking whom he may devour; second act, a sudden noise startles him, he scents the wind; third act, feeling he is pursued, he turns his head; fourth act, a knife shines in the air, he guesses, he flees; fifth act, fate is fate, and the beast sees heaven. No. 167 is “A Wild Pansy” (study of flowers). One forgets that “*une pensée sauvage*” can also mean a savage thought, and the surprise comes in to find the flowers of the picture are a fierce young boy and a scared-looking cat, and the boy is murdering the cat by running a spear through its neck. These are but samples of the whimsicalities.



ASTRONOMY OF THE HEAVENS FOR DECEMBER.

By PROF. M. B. GOFF.

THE SUN.

On the twenty-first day of this month, in our latitude, occurs the shortest day in the year. The sun rises at 7:20 and sets at 4:37, making the day's length 9 hours, 17 minutes. This is also the day on which properly our winter begins. The sun has reached the winter solstice, his most southerly point, and now begins his journey northward, causing the days to grow gradually longer.

THE MOON.

The moon's phases occur in the following order: First quarter, on the 7th, at 6:38 a. m.; full moon on the 13th, at 10:20 p. m.; last quarter on the 21st, at 3 a. m., and new moon on the 29th, at 7:51 a. m., Washington time (or, according to the "new reckoning," eight minutes and twelve seconds later in each case). The moon approaches nearest to the earth on the 12th, at 10:24 a. m., and is farthest away on the 24th, at 10:12 a. m. Its greatest altitude in this latitude will be on the morning of the 14th, when it will be $60^{\circ} 53'$ above the southern point of our horizon.

MERCURY

Will be evening star during the entire month. But it matters little, so far as seeing it with the naked eye is concerned, until near the close of the month, whether it be morning or evening star. On the 1st it sets at 4:35 p. m., and is then too near the sun to be seen. On the 30th it sets at 6 p. m., a few minutes before the moon, and about $5\frac{1}{2}^{\circ}$ south of the latter body. On

the 31st it sets at 6:03 in the evening, about one and one-third hours later than the sun, and $1\frac{1}{2}^{\circ}$ north of where the sun disappears. On both these evenings, and for several days both before and after these times, this planet is visible to the ordinary eye, and by its bright white light is readily recognized. Although visible several times each year, it is a remarkable fact that it has been seen by comparatively few persons. In the higher latitudes it is much more difficult to see than in the lower, and the atmosphere of some parts of Europe is very unfavorable for its observation.

VENUS

Will also be evening star throughout the month, setting on the 1st at 5:36, and on the 31st at 6:36 in the evening. It is at its greatest distance from the sun at 2 p. m. on the 12th. By far the brightest star in the west after sunset, a failure to recognize it would be almost impossible. It is the *Hesperus* of the ancients. On the evening of the 31st it is about seven degrees south and a little west of the moon.

MARS

Will be the morning star, rising at 9:45 p. m. on the 1st, and at 7:51 p. m. on the 31st. From the 1st to the 23d it will have a direct motion, that is, a motion from west to east, of 11 minutes and 18 seconds of arc; on the 23d, it will be stationary, and from the 23d till the end of the year it will have a retrograde motion, that is, from east to west, of one minute and forty-two seconds of arc. About nine o'clock on the evening of the 18th, it is north of the moon $8^{\circ} 18'$.

JUPITER,

King of the planets, will also rank as morning star. On the 1st he will rise at 8:33, and on the 31st at 6:23 in the evening, and like Mars will maintain nearly the same position in the heavens during the whole month, his motion being $9' 22''$ retrograde. On the 16th, at 9:51 p. m., he will be $5^{\circ} 43'$ north of the moon. The moons of Jupiter can be readily seen with a telescope of moderate power, or good opera glasses.

SATURN,

Though properly an evening star, shines from “dewy eve till early dawn,” rising on the 1st at 4:27 p. m., and setting next morning at 6:47; and on the 31st rising at 2:21 in the afternoon, and setting the following morning at 4:42. His motion will be 9' 9" retrograde, and on the 12th at 7:53 p. m., he will be 55 minutes north of the moon. The rings of Saturn are an object of great interest to every observer and the present is a favorable time to see them in great splendor, though the view in December 1884 and 1885 will be still finer.

URANUS

Begins the month by rising at 1:02 a. m., (thus putting himself among the morning stars), and at the close of the month at 11:11 p. m. His motion, which is direct, but only 1' 42" in thirty days, seems slow enough, but when we reflect that he actually travels an average absolute distance of over thirty million miles a day, we can but wonder at his terrific speed. He is located about two degrees southeast of *Beta Virginis*, and “can be seen with the naked eye, if one knows where to look.”

NEPTUNE,

The most distant of the planets, rises on the 1st at 3:30 p. m., and sets on the 2d at 5:26 a. m., and on the 31st rises at 1:31 in the afternoon and sets the following morning at 3:25. His motion is retrograde, and amounts to 2' 34" for the month. This planet is of no special interest to the ordinary reader, as “to recognize its disk with ease,” requires a magnifying power of three hundred or upward.



C. S. L. C. NOTES ON REQUIRED READINGS FOR DECEMBER.

EASY LESSONS IN VEGETABLE BIOLOGY.

It will be observed that many of the words and expressions used by Dr. Wythe in his “Lessons” are pronounced and explained in the “Glossary and Index.” Where such explanations and pronunciations occur, of course no notes have been prepared.

P. 12.—“Albumen,” al-bu'men. The word is derived from *albus*, the Latin word for white. Albumen forms a part of all animal fluids and solids. The white of an egg is almost pure albumen. In the vegetable world it is the food laid up for the nourishment of the embryo around which it lies.

P. 13.—“Slide.” A small slip of glass about three inches long by one wide. It is intended to place objects on which are to be examined under the microscope. The “cover” here referred to, is a square of very thin glass, which is placed over the object to hold it in place.

“Capillary attraction,” căp'il-la-ry. When a capillary (hair-like) tube is dipped into a liquid, there is an attraction between the solid and the particles or molecules of the liquid which causes the latter to rise in the tube if it wets it, or to sink if it does not.

“Heated stage.” The simplest means for heating the stage or slide upon which an object is placed, is by a small alcoholic lamp placed at the corner of the stage.

P. 15.—“32 deg. F.” 32 degrees on the Fahrenheit thermometer. This scale, invented in 1714, by Fahrenheit, is commonly used in England and America, although the centigrade scale (where the distance between the boiling point and zero is divided into one hundred parts) takes its place largely on the continent, and in most scientific works.

P. 18.—“Amœba,” a-mœ'ba. Readers who live in warm climates will have no trouble in finding the *amœba* attached to the stems of plants or floating in pools of stagnant water. To find a specimen in December in temperate latitudes will be more difficult, but by collecting grass and stems of water plants, with water from a pool, and keeping them in a warm room for a time, specimens may be obtained.

P. 22.—“Cinchona,” cin-chō'na. The tree from which the well known drug, Peruvian bark, is obtained.

“Quinine.” Kwi'nīn is Webster's preferred pronunciation, kwe-nīn', Worcester's. Quinine is an alkaloid obtained from the cinchona bark.

P. 28.—“Nucleus.” The nucleus may be easily seen in a thin section from an apple or potato, placed under a microscope.

“Mucilaginous,” mū-ci-lăg'i-noŭs. Like mucilage.

P. 29.—“Silica,” sīl'i-ca. Flint or quartz.

P. 31.—“Tannin,” tăn'nin, or tannic acid. An astringent principle found in nut-galls and the bark of many trees. If treated with ether a solid is obtained which is soluble in water. It is used in tanning leather.

“Starch.” The grains of starch are easily found. Wheat, oats, arrow-root, sage and tapioca all furnish excellent examples.

“Crystals.” Răph'i-des is the name given by botanists to the little crystals found in the tissues. A section of an onion will reveal them. Also, the juice of the hyacinth.

P. 34.—“Cochineal Cactus,” cōch'i-nēal. The cactus from which the cochineal insects, whose bodies are sold in the shops for a red dye, are obtained. The plant is also called cochineal fig, and is raised with as great care for food for the insect, as is the mulberry tree for the silk worm.

P. 35.—“Vessels.” The “dotted” vessel, or duct, is a long, open tube whose sides are marked by pores, or dots. A transverse section of a radish will furnish an example. A fern will furnish an example of “reticulated” cell; almost any common plant contains the “annular” and “spiral” cells.

P. 36.—“Epidermis.” All varieties of cells will be found in the epidermis together with the mouths, or stomata. For examination a layer should be carefully removed with a razor or knife, a little water put on it to prevent too rapid drying, and the whole covered by a thin glass.

P. 37.—“Volatile oil.” That which wastes away when exposed to the atmosphere; as musk.

P. 39.—To prepare sections of wood for slides there is a very simple instrument which may be made at home by any one possessed of a little ingenuity. Let a block of hard wood be selected, one and one-fourth inches square by two inches in length, its ends perfectly smooth. In one end drill a hole one-fourth of an inch in diameter, lengthwise, one and one-half inches. In the other end insert a common wood screw, its point filed square, until it reaches the hole. In the side of the block, one-half inch from the end in which the hole has been bored, insert another screw, with square point; this is to be used as a clamp. To cut the section take a branch the proper diameter, and which has been boiled in water, place it in the hole, and clamp solidly with the screw at the side. With a keen razor cut off the end even with the block. By turning the screw in the end of the block the branch will be pushed forward any distance desired, and the section can be cut by a sliding motion of the razor across the stem. The slices should be removed from the knife with a camel's hair brush, slightly dampened, and may be preserved in weak spirits. The work is, of course, very delicate, and requires the skill and nicety of touch which only practice brings. To mount the sections in Canada balsam, as Dr. Wythe advises, the object should be placed exactly in the center of the slide, which must be carefully cleaned from dust, and a drop of the balsam placed upon it; hold the slide over a flame until the balsam spreads over the object. Air bubbles should be broken with a needle. A glass cover, warmed, should now be placed on the object and pressed sufficiently to remove the superfluous balsam. The whole should be put in a warm place until thoroughly dry.

P. 42.—“Showers of blood,” or blood-rain. A shower of reddish dust mixed with rain, which has been known to fall in several places on the eastern coast of the Atlantic.

P. 42.—“Diatoms.” “They are found in great abundance in the mud of rivers, lakes and ponds. They are also present in those deposits of clay which once formed the beds of rivers and lakes, and which are now dry. In order to procure the diatoms from these deposits, the earth or clay should be well washed with pure water, and the deposit allowed to settle and the water poured off. This may be repeated several times. The deposit is then to be washed with hydrochloric acid, and when the effervescence is over, the acid is poured off, and a fresh portion is added. This may be repeated several times. When no action occurs by its use cold, the deposit may be transferred to a watch-glass, and kept over a spirit lamp, at a temperature of about 200° for three or four hours. The deposit must then be well washed with pure water, and will be found to consist almost entirely of diatoms.”—*Lankester*.

P. 47.—“Fungi,” fŭn'ji. No class is so easy to study in the winter. If fruit, bread and the like are allowed to mould, any number of interesting objects will be found. In the woods fungi are to be gathered from bark and old logs. Of these the peziza, or cup-moulds will be found most pleasing. Lichens also abound, and numerous sections can be made from them.

P. 62.—“Big Trees.” These Big Trees are Cedars (*sequoia gigantea*). “Calaveras,” kă-lă-vā'räs.

“Buds.” Many plants form their buds in the fall. A careful search will reveal such for examination. The lilac and trailing arbutus form their flower buds in autumn, and in vigorous plants a section of the bud will show distinctly the flower stowed away for spring.

P. 63.—“Leaves.” “Opposite” leaves are seen in the chickweed and fuchsia; “whorled,” in Prince’s pine; the “alternate,” in the rose family. The arrangement of leaves on the stem has been reduced to a science, called *phyllotaxy*.

P. 64.—“Bracts.” Seen in the camellia and strawberry. The white portion of the calla blossom is a colored bract called spathe; also, the “pulpit” of the common Indian turnip or Jack-in-the-pulpit.

“Sessile” leaves are seen in the upper leaves of the common primrose and spring beauty. All plants of the violet and the rose families bear *stipules*.

P. 65.—“Lanceolate,” as in the peach; *oblong*, the radical leaves of shepherd’s purse; *cordate*, in the blood-root; *sagittate*, in the stem leaves of shepherd’s purse; *ovate*, in chickweed and violet; *pinnate*, as in the rose; *bipinnate*, as in the sensitive plant.

P. 68.—When the stamens and pistils are on separate trees or plants, the fertilization is accomplished in various ways; insects or birds carry the pollen in many cases, in others the wind wafts it.

“*Hypogynous*,” as in the cress, radish, cabbage, and other cruciform plants.

P. 69.—*Perigynous*, as in the rose family; *epigynous*, as in the caraway, celery, and parsnip.

P. 76.—“*Labiata*.” The word means lip-shaped, and the order is named from the peculiar shape of the corolla.

P. 77.—“*Composite*,” or compounded; “*Herbaceous*,” her-bā'shus. Plants with soft stems which die every year.

“*Coriander*,” cō,ri-an'der; “*Asafoetida*,” ās'a-fēt,i-da.

P. 78.—“*Papilionaceous*,” pa-pīl'yo-nā'shus. From the Latin for butterfly.

“*Tamarind*,” tām'a-rīnd. A tree 60 to 80 feet in height, with dense foliage. A native of Africa and India. Its pods are preserved and used as a medicine, or as an article of diet.

“*Senna*,” sēn'na. A drug prepared from the dried leaves of the cassia, a shrub raised in India and Nubia. A variety of cassia is found in the United States, but its leaves are less powerful. “*Acacia*,” a-kā'shī-a, “*Mimosa*,” mī-mō'sa.

P. 79.—“*Ranunculus*,” ra-nūn'cu-lūs. The word means a little frog. Pliny is said to have so named this species because many of its members grow in water where frogs abound.

“*Aconite*,” ac'o-nite. A plant related to the Hellebores; the common wolf’s bane, or monk’s hood.

“Cruciate,” kru'shĭ-āt. The petals are arranged in the form of a cross.

P. 80.—“Chimborazo,” chim-bo-rā'zo. A peak of the Andes in Ecuador. It is the sixth in height among the lofty peaks of the range.

P. 82.—“Floras.” The whole number of plants native to any section forms its flora.

“Urticaceæ,” ur-ti-ca'ce-æ. Nettles.

P. 83.—“Rhododendrons,” rhō,do-dĕn'dron; “Azalias,” a-zā'le-as. These plants both belong to the order of heathworts or *ericaceæ*, the order to which the huckleberry, cranberry, trailing arbutus, and other well-known plants belong.

BIOGRAPHICAL STORIES.

P. 14.—This picture, West declared sixty-seven years after it was painted, contained some touches that he never surpassed.

P. 15.—“Camera-obscura,” cām'e-ra ob,scū'ra. Literally, a dark chamber.

P. 16.—“Parma.” A province in the north of Italy.

“Death of Wolfe.” This picture contained one feature which at that period was entirely new. West used costumes in his picture which were appropriate to the time and character. Before this the classical costume was used on all occasions. It is said that Sir Joshua Reynolds, the president of the royal academy before West, tried to dissuade him from this innovation, but was the first to acknowledge his success.

P. 24.—“Gulliver,” gŭl'li-ver. The hero of a satire, “Gulliver’s Travels,” by Swift. He is represented first as a surgeon, and then as captain of several ships. He takes voyages, during which he discovers many strange countries, among them the country of the Lilliputs (lĭl'li-pŭt), a race of little men.

P. 29.—“Uttoxeter,” ŭks'e-ter.

P. 34.—“Merry-Andrew.” One whose business it is to make sport for others—a clown, a court fool.

P. 68.—“Oxenstiern,” ۆks ,en-steern'.

P. 70.—“Lutzen,” loot'seen. A town of Prussia.

P. 72.—“Muscovy,” mus'co-vy. The former name of Russia.

GERMAN HISTORY.

P. 129, c. 1.—“Merovingians,” mer-o-vin'gi-ans. The first Frankish king of whom we have authentic accounts was Chlodio, who ruled about the middle of the fifth century. His successor, Merovæus, gave his name to the first *house*, or dynasty of the Franks; of him we know little more than that he fought against Attila.

“St. Remigius,” re-mij'i-us, or St. Remy, reh'me,. (439?-533.) The Apostle of the Franks. When but a young man he was made Bishop of Rheims. By his zealous work he spread Christianity widely through the Frankish kingdom. A contemporary declares him to have been the most eloquent man of his times.

“Rheims,” reemz. A city in northeastern France, whose bishops date from the fourth century, and whose cathedral is one of the finest gothic edifices in Europe.

P. 129, c. 2.—“Arian.” The religion of Arius, a bishop of the fourth century, who held that Christ, though chief of created beings, was not equal to God. A book lately issued by Appleton & Co., “Arius the Libyan,” will be found to be an interesting account of his life and doctrines.

“Paris.” The first account which we have of Paris is from Cæsar, who visited it in the last century before Christ. It was then but a collection of huts on an island in the Seine, and was called Lutetia.

“Poitiers,” poi-teerz'. A town of France, one hundred and eighty miles southwest of Paris.

“Abderrahman,” äbd-er-räh'män; “Viceroy,” vīce'roy. One who rules in the place of a king; a substitute. “Caliph,” cā'liph. The successors of Mohammed were all called caliphs.

“Damascus.” Soon after the death of Mohammed Damascus was captured by his followers and made their capital. Such it remained for ninety years. On the fall of the Ommiyades, their successors, the Abbassides chose Bagdad as their capital.

P. 130, c. 1.—“Islam,” ĭz'lam. Meaning obedience, submission, is a name given to the religion of Mohammed.

“Gregory I.” (540-604.) Born of a noble family and educated for public life, he was made prefect of Rome in 573, but his strong religious nature led him to give up his position, turn over his wealth to the Church and become a monk. Through the fourteen years of his pontificate, Gregory employed every means to purify and strengthen the Church, sending missionaries into all parts of the world, combating Arianism and rectifying many abuses.

“Augustin.” The Apostle of the English. A Benedictine monk at Rome when chosen by Gregory I. to go on a mission to the Saxons. He went to England about 597, was received kindly by King Ethelbert, and allowed to preach through Kent. After a time Ethelbert adopted Christianity and was baptized. This led to the complete triumph of the religion throughout the kingdom. Augustin was made Archbishop of Canterbury, and until his death had charge of the Church in England.

“Holy See.” A name given to the office of the pope, and also applied to the pope himself, or his court. *See* is derived from the Latin verb “to sit,” and literally means a seat, or site, hence a place where power is exercised.

P. 130, c. 2.—“Asceticism,” as-cet'i-cism. The practice common among members of the early Church of withdrawing from all business and society to devote themselves to a rigorous life of penance and self-denial.

“Vatican Hill.” The *Mons Vaticanus* of the ancient Romans, from which the palace of the Vatican takes its name.

“Dacia,” dā'ci-a. A province of the Romans north of the Danube, and comprising parts of the present countries of Hungary, Transylvania and

Roumania.

“Dalmatia,” dal-mā'ti-a. A narrow strip of country lying along the eastern shore of the Adriatic, now belonging to the Austro-Hungarian Empire.

“Istria,” is'tri-a. A peninsula, now of Austria, on the northeastern coast of the Adriatic.

“The German Crown.” In 1806, one thousand and six years after the crowning of Charlemagne, a league known as the Confederation of the Rhine, and of which Napoleon Bonaparte was protector, was formed between the central and southern German states. Its real object was to aid France against Prussia and Austria. The king of Germany resigned his crown then, and the empire came to an end.

P. 131, c. 1.—“Roland.” Roland, or Orlando, was the nephew of King Charlemagne, brought up by him and trained to be a warrior. He is the hero of one of the most famous stories of the middle ages, “The Song of Roland.” Various authors have used this tale. The *Orlando furioso* of Ariosto, and *Orlando innamorato* of Boiardo, are prominent among these. In a metrical narrative the story was sung by the minstrels of those times. According to this song Charles had been seven years in Spain warring against the heathen, until there remained but one king unsubdued, Marsilius of Saragossa. He had promised homage, and the step-father of Roland, Ganelon, was, by Roland's advice, sent on an embassy to him. Ganelon was angry because Roland had advised that he be sent, and in revenge betrayed to Marsilius the pass through which the rear guard and most valiant portion of Charles' army, under Roland would pass. Charles, with the body of his army, passed through, and when Roland appeared with his twenty thousand men, an army of four hundred thousand heathens attacked him. Roland fought until only a fragment remained before blowing his enchanted horn to summon Charles to his aid. Before his uncle could reach the pass every man was dead. The Saracens fled back to Spain, but the king pursued, completely defeating them. But the death of Roland robbed the conquest of all its glory, and threw France into mourning.

“Paladin,” pāl'a-dīn. A distinguished knight.

“Roncesvalles,” ron-thēs-väl'yēs. The pass in which Roland and his band were destroyed.

“Otto the Great.” Otto I. (936-973.)

“Ardennes,” ar'den'. The forest of Ardennes lies in the northeastern part of France, covering a portion of the department of the same name, and extending into Belgium.

P. 131, c. 2.—“Frisian,” fris'i-an. Belonging to the Frisians, a tribe formerly living beyond the Batavi, but pushed to the borders of the North Sea by the Franks.

“Aix,” āks. A town in southern France whose thermal springs were known to the Romans.

“Alcuin,” äl'kwin. (735-804.) His great reputation for learning caused Charles to invite him in 780 from England, his native country, to open a school in France. This institution is supposed to have been the germ of the present University of Paris. Alcuin afterward opened a school at Tours, which became very famous.

“Verden.” A town lying southeast of Bremen, in Prussia.

P. 132, c. 1.—“Haroun-al-Raschid.” See notes in THE CHAUTAUQUAN for November.

“Bretons.” The inhabitants of Brittany, the triangular peninsula which extends from the western coast of France into the Atlantic Ocean.

“Almayne,” al'-main.

P. 132, c. 2.—“Eginhard,” ěg'in-hart. He had been a pupil of Alcuin, and by him was introduced at court. Eginhard's history of Charles and his accounts of the Franconian kings have given him a permanent place among the writers of the middle ages. See Longfellow's poem, “Emma and Eginhard,” in “Tales of a Wayside Inn.”

NOTES ON GERMAN LITERATURE.

It will be found helpful to read the extracts from German Literature in connection with the "Outline of German Literature" in THE CHAUTAUQUAN for November, thus fixing the period to which each author belongs. Care has been taken in selecting the extracts to choose only from those who are in the first rank, and omit all minor writers. The selections are intended to show the style of each, and the lines of thought which he followed.

P. 132, c. 2.—"Würzburg," wurts'burg. A city of Bavaria on the Main, whose history dates back to the sixth century.

"Minster." The word comes from the Latin *monasterium*, and is applied to the church or chapel belonging to a monastery; also, as here, to a cathedral.

"War of Wartburg." In 1206 the landgrave, Hermann I., summoned the poets of his nation to a musical tournament in the castle of Wartburg, in the Thuringian Forest. The competition ran so high that it was called the Wartburg war, and in 1300, a poem, "The War of Wartburg," appeared, celebrating the event.

P. 134, c. 1.—"Luther." The value of Luther's literary work can not be estimated. As a poet, his hymns have won him a permanent place. Beside his original verses, he re-arranged and set to music many of the Psalms, thus really founding the church music of Germany. His translation of the Bible must be counted his most important work; but beside this he left a mass of sermons, theses, tracts and controversial writings which, at the time of their writing, wielded wide influence. These latter show most plainly the fiery spirit of their author, his clear conceptions of truths, and his pure style. Special attention should be called to his "Table Talk," which contains numberless short and pithy statements of his opinions, and to his Catechism on the Decalogue, Apostles' Creed, and the Lord's Prayer.

P. 134, c. 2.—"Nathan the Wise." This was Lessing's last drama, and was written to show his own religious views and to advocate the wide toleration in which he believed. The three chief characters, Nathan a Jew, Saladin a Mohammedan, and a Christian are thrown together, and their association causes a strong friendship in spite of their diverse views. They practice the utmost charity toward each other. The story of the "Three Rings" contains the point of the drama. It is taken from an old Italian novel.

READINGS IN ART.

P. 142, c. 1—"Nicolo Pisano," pe-sā' no. (1200?-1278?) His architectural works are also important, he having designed churches for Padua, Venice and Florence, and a campanile for Pisa.

"Pisa," pee'sā. A city of Italy, west of Florence, on the river Arno. Its cathedral contains much fine art.

"Siena," sī-en'nā. A city of Tuscany, south of Florence.

"St. Dominic," dōm'ī-nik. (1170-1221.) The founder of the order of "preaching friars."

"Bologna," bo-lon'yā. An Italian city at the foot of the Appenines, and north of Florence.

"Andrea Orcagna," or-kān'yä. (1329-1389.) The son of a Florentine sculptor who instructed him in art. His surname, Orcagna, is said to have been a corruption of L'Archagnuolo (the Archangel). His fame as a painter and architect was equal to that as a sculptor.

"San Michele," mī-ka'la. The churches of Italy are almost without exception named after the saint to which they are dedicated, as in this case, San Michele, after St. Michael.

"Giotto," jōt'o. (1276-1336.) He was born near Florence, and brought up a shepherd lad. While tending his sheep it is said that a Florentine painter found him drawing on the surface of a rock, and was so convinced of his genius that he took him to Florence to be educated. His talent was so great that he was soon employed in the decoration of the church at Assisi. The details of his life are not known except that he traveled extensively through Italy, being employed in ornamenting many prominent buildings. It is as a painter that Giotto is best known, and as such he did much to awaken art from its unnatural and stiff forms, and to introduce realism. "Campanile," kām-pa-nē'la.

"Ghiberti," gee-bēr'tee. (1378-1455.) He learned the trade of a goldsmith, but at that time the goldsmith's art included others, especially

designing and coloring. At first Ghiberti was a fresco painter, but was called to sculpture by his success with the bronze doors over which he spent most of his life. “San Giovanni,” jo-vän'nee.

P. 142, c. 2.—“Donatello” do-nä-těl'o. (1383-1466.) He was a native of Florence. One of the Medici became his patron, and he was enabled to apply himself to art. He was a painter of merit as well as a sculptor, and the only one of the age worthy to be ranked with Ghiberti.

“Brunelleschi,” broo-něl-lës'kee. (1377-1444.) Better known as an architect than as a sculptor. “Zenobius,” ze-no'bi-us.

“Lucca del Robbia,” del-rob'e-ä. (1400?-1463?) He was trained to the goldsmith's art, but took up sculpture. To him is attributed the interest in porcelain which started in Europe in the fifteenth century. His process of glazing was probably learned from the Saracens, and consisted essentially in using stanniferous (containing tin) enamel. This rendered the terra-cotta work permanent, and gave a white background.

“Uffizi,” oof-fē'tsi. “A palace of Florence whose galleries are among the best of Europe. It contains paintings of all the principal European schools, and many famous statues. There are halls devoted to sculptures of the fifteenth and sixteenth centuries, drawings of the old masters, engravings, ancient bronzes, medals, gems, cameos, intaglios, the whole forming one of the finest collections in the world.” There is also a library in the palace which is rich in manuscript and letters of Italian writers.

“Verrocchio,” vāir,rok'ke-o. (1432-1488.) Of the very little known of Verrocchio, the most interesting fact is that he was the first to take a mould of the human form to aid in designing.

“Leonardo da Vinci,” le-o-nar'do dā vīn'chee. (1452-1519.) A native of Florence. In youth he was recognized as an almost universal genius and speedily surpassed all instructors. He became attached to the court of Milan in 1483 as a musician and improvisatore, and remained there until 1499, when he removed to Florence. About this time he was employed in various cities as an architect and engineer. He went to France in 1515 with Francis I., as court painter, and there died. As a sculptor we have no remains of his work. Vinci was as remarkable a thinker as artist. Hallam says of his literary

fragments: “They are like revelations of physical truths vouchsafed to a single mind. The discoveries which made Galileo and Kepler and other names illustrious, the system of Copernicus, the very theories of recent geologies are anticipated by da Vinci, within the compass of a few pages.”

“Contucci,” con-tuk'che. “Sansovino,” sän-so-vee'no. (1460-1529.) Sansovino was a Florentine, and his early works were executed there. Thence he went to Rome and worked, and in 1513 took charge of the Holy House of Loreto, at which he worked until his death.

P. 143, c. 1.—“Loreto,” lo-rā'to. “The Holy House, in which, according to tradition, the Virgin Mary was born, the annunciation and incarnation took place, and the holy family resided on their return from Egypt. The legend is that the house was transported by angels in 1291 from Nazareth to the eastern coast of the Adriatic, and thence in 1294 to the coast of Italy. It is a rudely built brick house, 13½ feet high, with one door and one window.... The relics, treasures and offerings of different pilgrims are numerous and valuable. The house is enclosed in a marble casing, which is covered with exquisite sculptures.”

“Buonarroti,” boo-ō,nä-rot'ee. (1475-1563.) A painter, sculptor, and architect. He came from a family of high rank. His artistic genius was early displayed, and he was put under masters to study. Lorenzo de' Medici, pleased with his ability, took him to his palace, where he studied until his patron's death in 1492. He was summoned to Rome by Julius II., to design his tomb, and in 1508 he began the decoration of the Sistine chapel. When seventy years old he was called upon to take up architecture, and finish St. Peter's, then under way. For the rest of his life he was engaged upon this church, together with several other buildings of Rome. Michael Angelo was not only an artist, but a writer; his sonnets are among the best in any literature.

“Centaurs.” Fabulous creatures of mythology—half man and half horse.

“Carrara,” kār-rä'rä. A city of northern Italy, which gives its name to a chain of mountains belonging to the Appenines. They contain quarries of fine and valuable marble.

“Giuliano,” joo-le-ä'no; “Lorenzo,” lo-ren'zo; de' Medici, da mēd'e-chee. A family distinguished in Florentine history, of which Lorenzo, called The Magnificent (1448-1492), was the most famous member. Giuliano, his brother, was assassinated in 1478, an attempt being made against them both, instigated, some say, by Pope Sixtus IV.

“Modena,” mod'e-na; “Bagarelli,” bah-gah-rah'lee.

“Padua,” pad'u-a; “Riccio,” ret'cho. (1480-1532.)

“Tatti,” tah'tee. (1479-1570.) Also called Jacopo Sansovino, from his master. Tatti was a Florentine, and worked in his native city and at Rome until 1527, when he went to Venice; there he founded a school and did much work.

“Mars.” The Roman god of war, corresponding to the Greek Ares.

“Neptune,” nep'tune. In Roman mythology the god of the sea. The Poseidon of the Greeks.

P. 143, c. 2.—“Pacher,” pa'ker. He lived about 1480.

“Veit Stoss.” (1483-1533.) His early labors were in the churches of his native city. The second period of his life was spent in Nuremberg where many of his works remain. Of them it is said: “They are distinguished by a tender fervor and grace, a mild softness of form, and a clearly developed style of relief, with a great deal of life likeness.”

“Jörg Syrlin,” yŭrg seer'len. He lived in the latter half of the fifteenth century.

“Riemenschneider,” re,men-schnī'der; “Von Schrenburg,” fon schren'burg; “Vischer,” fīsh'er. He began his work in 1489, and died 1529.

“Chef d'œuvre,” shā-doovr', masterpiece.

“Regensburg,” re'gens-borg. The German name for Ratisbon, a city of Bavaria.

“Apollo,” a-pŏl'lo. One of the most popular of Greek divinities. Numerous offices were filled by him; he was the god of song and music, of

prophecy, of punishment, of protection, and of the sun. Smith says of him: "It may safely be asserted that the Greeks would never have become what they were without the worship of Apollo." And again: "In him the brightest side of the Grecian mind is asserted."

"Orpheus," or'phe-us. "Eurydice," eu-rid'i-ce. Orpheus was a poet and musician of Grecian mythology. His lyre enchanted even the beasts of the field, and the rocks were moved from their places at its sound. His wife, Eurydice, died and was carried to Hades, but Orpheus followed and by the music of his lyre won back his wife on condition that he should not look upon her until they were past the bounds of the infernal regions. His love overpowered him, he turned to see if she followed, only to see her disappear. His grief, it is said, led him to treat the Thracian women with contempt, and he was murdered by them.

"Maximilian," maks-ĩ-mil'yan; "Innsbrück," ins'brook.

"Fontainebleau," fon'tan,blō'. A suburb of Paris, famous for its splendid palace. This palace was begun in the tenth century, and has been added to, remodeled and ornamented by various monarchs since. Its architecture is of all schools, its pictures, statues and books invaluable.

P. 144, c. 1.—"Bernini," bér-nee'nee. He began work for the pope at the age of eighteen, and spent several years on the churches and palaces of Rome. His fame was so great that he was invited to Paris in 1665 to complete the Louvre; but his plans for this were never carried out. His latter life was spent in Italy.

"Alessandro Algardi," al-les-san'dro äl-gar'dee. (1598-1654.) "Puget," pü,'zha,. (1622-1694.) "Girardon," zhe,rar,don'. (1628-1715.) "Houdon," oo,don'. (1741-1828.)

"Pigalle," pe,gal'. (1714-85.) "Duquesnoy," dü,ka'nwä,. (1594-1646.) "Schlüter," schli'ter. (1662-1714.)

"Winckelman," wĩnk'el-män (1717-1768). A German archæologist. His researches and writings stimulated the interest since taken in archæology, and he is regarded as its founder. Also his theories of the beautiful and "History of Art" opened a new field in German thought.

“Canova,” kä-no'vä; “Pompeii,” pom-pe'yi; “Herculaneum,” her'cu-la-ne-um.

“Theseus,” the'se-us. A legendary hero of Attica, of whom many wonderful adventures are told. This story of the Minotaur (a monster, half man, half bull,) is that Theseus was taken to Crete along with the youths and maidens which were offered every year to the monster. The king's daughter fell in love with him and gave him a sword with which he killed the Minotaur, and then escaped from the labyrinth in which he was confined, by a thread which he had unraveled as he went in.

“Dannecker,” dän'ek-er. (1758-1841.) “Chaudet,” sho'da. (1763-1810.) “Thorwaldsen,” tor'wawld-sen; “Villa Carlotta,” vē'lyâ car-lot'a.

“Gutenberg,” goo'ten-bërg. (1400?-1468.) The reputed inventor of printing.

“Mayence,” mā'yângs. The French name for Mentz, a city of Hesse on the banks of the Rhine. “Leuchtenberg,” loik'ten-bërg.

P. 144, c. 2.—“Schadow,” shä'do; “Stettin,” stet'teen. A town of Prussia.

“Blücher,” bloo'ker; “*Naïvete*,” nä'ēv-tā, simplicity, ingenuousness. “Rauch,” rowk; “Bülow,” bü'lo; “Scharnhorst,” sharn'horst; “Charlottenburg,” shar-lut'ten-boorg. A town of Prussia.

“Dürer,” dü'rer. (1471-1528.) A German painter and engraver.

P. 145, c. 1.—“Thiergarten,” teer'gar-ten. A park in Berlin.

“Schierelbein,” shě'rel-bīn; “Dirschau,” deer'show. A town of Prussia.

“Rietschel,” reet'shel; “Friedenskirche,” frē'dens-ker'ka; “Hähnel,” hä'nel; “Brühl,” brül; “Schwanthaler,” shwän'tä-ler; “Bosio,” bo'si-o; “Duret,” dü'rā'; “Pradier,” prä'de-ā'; “Barye,” bä'rē.

P. 145, c. 2.—“Steinhäuser,” stīn'how-zer; “Carlsruhe,” karls'roo; “Hildebrand,” hil'de-brand; “Kessels,” kēs'els.

SELECTIONS FROM AMERICAN LITERATURE.

P. 146, c. 1.—“Whately,” hwāt'li. (1787-1863.) Archbishop of Dublin; author of several important works, chief among which is his “Elements of Logic.”

“Thackeray,” thāk'e-ri. (1811-1863.) An English novelist.

P. 146, c. 2.—“Steele.” (1671-1729.) An English essayist.

“Addison.” (1672-1719.) An English poet and essayist.

P. 147, c. 1.—“Benjamins,” “a smart coat.” It is said to have been so called from a tailor of that name who first made it. Perhaps also from association with the “coat of many colors.”

“Purlieus,” pūr'lūs. The outer part of the inn, here. The word means *pure place*, and was first applied to that portion of the forest around the castle which was free or pure from the forest laws; hence it came to mean the outer part of any place.

P. 147, c. 2.—“Plethoric,” ple-thor'ic; over-full. “Negus.” A drink made from water, wine, sugar, nutmeg, and lemon-juice, and said to have received its name from its first compounder, Colonel Negus.

“Sunnyside.” Irving’s home on the Hudson, near Tarrytown. The house is an old Dutch mansion. It was near here that Rip Van Winkle lived.

“Eildon Hills,” eel'dun. A group of hills in southern Scotland.



BOOKS RECEIVED.

“Putnam’s Handy Book Series of Things Worth Knowing.” Work for Women, by George J. Manson. New York: G. P. Putnam’s Sons, 27 and 29 West Twenty-third Street, 1883.

“The Primer of Politeness.” A Help to School and Home Government, by Alex. M. Gow, A.M. Philadelphia: J. B. Lippincott & Co.

“An Introduction to the History of Educational Theories,” by Oscar Browning, M.A. New York: Harper & Brothers, Franklin Square, 1882.

“The Life of Washington and the History of the American Revolution,” by Washington Irving. With illustrations. Centennial edition. New York: G. P. Putnam’s Sons, 1883.

“Mother Truth’s Melodies.” Common Sense for Children—A Kindergarten, by Mrs. E. P. Miller. Chicago and New York: Fairbanks, Palmer & Co., 1883.

“The American Girl’s Home Book of Work and Play,” by Helen Campbell. Illustrated. New York: G. P. Putnam’s Sons, 1883.

“Beyond the Gates,” by Elizabeth Stuart Phelps. Boston: Houghton, Mifflin & Co., 1883.

“Mary Lamb,” by Annie Gilchrist. Boston: Roberts Brothers, 1883.

“Bright and Happy Homes.” A Household Guide and Companion, by Peter Parley, Jr. Chicago & New York: Fairbanks, Palmer & Co., 1882.

“Sketches and Anecdotes of American Methodists of ‘The Days that Are no More,’” by Daniel Wise, D.D. New York: Phillips & Hunt; Cincinnati: Walden & Stowe, 1883.

“Handbook of Dates.” Arranged Alphabetically and Chronologically. Compiled by Henry Clinton Brown. New York: A. Lovell & Co., 1883.

“Library of Biblical and Theological Literature,” edited by George R. Crooks, D.D., and John F. Hurst, D.D. New York: Phillips & Hunt; Cincinnati: Walden & Stowe, 1883.

“Contrary Winds, and Other Sermons,” by Wm. M. Taylor, D.D., LL.D. New York: A. C. Armstrong & Son, 1883.

“Arius the Libyan.” An Idyl of the Primitive Church. New York: D. Appleton & Co., 1884.

“A Text-Book of Inorganic Chemistry,” by Prof. Victor von Richter. Authorized translation of the third German edition, by Edgar F. Smith, A.M., Ph.D. Philadelphia: P. Blakiston, Son & Co., 1883.

“Holland and Its People,” by Edmondo De Amicis. Translated from the Italian by Caroline Tilton. Fifth edition. New York: G. P. Putnam’s Sons.

“All Aboard for Sunrise Lands.” A Trip through California, Across the Pacific to Japan, China and Australia, by Edward A. Rand. Illustrated. New York and Chicago: Fairbanks, Palmer & Co., 1883.

“Summer Rambles in Europe,” by Alex. Clark. New York: Nelson & Phillips, publishers, 1879.



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The list of names belonging to the Intermediate Normal Class should have been inserted in THE CHAUTAUQUAN for November.

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Transcriber's Notes:

Obvious punctuation errors repaired.

Page 132, “Bagbad” changed to “Bagdad” (The Caliph of Bagdad)

Page 142, “Sansorino” changed to “Sansovino” (Contucci, better known as Sansovino)

Page 142, “unparalled” changed to “unparalleled” (unparalleled in beauty)

Page 143, “Würzburg” changed to “Würtzburg” (in the Würtzburg cathedral)

Page 145, “Steinhaüser” changed to “Steinhäuser” (Carl Steinhäuser)

Page 145, “unforgettable” changed to “unforgettable” (piquant and unforgettable flavors)

Page 146, “antithises” changed to “antitheses” (and pointed antitheses)

Page 151, “chipmuck” changed to “chipmunk” (the chipping squirrel, chipmunk)

Page 159, “until” added (not entirely superseded by implements of steel until the latter part of last century)

Page 174, “unicellar” changed to “unicellular” (the unicellular plants)

Page 182, “pensêe” changed to “pensée” (une pensée sauvage)

Page 187, “Posidon” changed to “Poseidon” (The Poseidon of the Greeks.)

Page 187, accents added to Innsbrück, Blücher, and Steinhäuser.

*** END OF THE PROJECT GUTENBERG EBOOK THE
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